

Lecture notes on Mathematical Foundations of Machine and Deep Learning

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May 20, 2026

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1 Basic notations and definitions

In $\mathbb{R}^n$, $n \in \mathbb{N}$, we consider the standard inner product

$$\langle x, y \rangle = \sum_{j=1}^n x_j y_j, \quad x, y \in \mathbb{R}^n, \quad (1)$$

and the Euclidean norm

$$\|x\| = \langle x, x \rangle^{1/2}, \quad x \in \mathbb{R}^n. \quad (2)$$

Recall the Cauchy-Schwarz inequality

$$|\langle x, y \rangle| \leq \|x\| \cdot \|y\|, \quad x, y \in \mathbb{R}^n. \quad (3)$$

If $A \in \mathbb{R}^{m \times n}$ is a matrix then by $\|A\|$ we mean a Frobenius norm

$$\|A\| = \left(\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2 \right)^{1/2}. \quad (4)$$

It holds for any $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n$ that

$$\|Ax\| \leq \|A\| \cdot \|x\|. \quad (5)$$

For a differentiable $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$ its gradient at $[x_1, \dots, x_n] \in \mathbb{R}^n$ is the following column vector

$$\nabla \mathcal{L}(x) = \left[\frac{\partial \mathcal{L}}{\partial x_1}(x), \dots, \frac{\partial \mathcal{L}}{\partial x_n}(x) \right]^T. \quad (6)$$

The gradient $\nabla \mathcal{L}(x_0)$ provides the direction of the steepest ascent for $\mathcal{L}$ at the point x_0 . For a twice differentiable $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$ the Hessian at the point $x \in \mathbb{R}^n$ is defined as

$$\nabla^2 \mathcal{L}(x) = \begin{bmatrix} \frac{\partial^2 \mathcal{L}}{\partial x_1^2} & \frac{\partial^2 \mathcal{L}}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 \mathcal{L}}{\partial x_1 \partial x_n} \\ \frac{\partial^2 \mathcal{L}}{\partial x_2 \partial x_1} & \frac{\partial^2 \mathcal{L}}{\partial x_2^2} & \cdots & \frac{\partial^2 \mathcal{L}}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 \mathcal{L}}{\partial x_n \partial x_1} & \frac{\partial^2 \mathcal{L}}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 \mathcal{L}}{\partial x_n^2} \end{bmatrix} (x) \in \mathbb{R}^{n \times n}.$$

For a differentiable function $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $\varphi = [\varphi_1, \dots, \varphi_m]$, we denote by $\nabla \varphi(x)$ its Jacobian matrix at the point x , i.e.:

$$\nabla \varphi(x) = \begin{bmatrix} \frac{\partial \varphi_1}{\partial x_1} & \frac{\partial \varphi_1}{\partial x_2} & \cdots & \frac{\partial \varphi_1}{\partial x_n} \\ \frac{\partial \varphi_2}{\partial x_1} & \frac{\partial \varphi_2}{\partial x_2} & \cdots & \frac{\partial \varphi_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial \varphi_m}{\partial x_1} & \frac{\partial \varphi_m}{\partial x_2} & \cdots & \frac{\partial \varphi_m}{\partial x_n} \end{bmatrix} (x) \in \mathbb{R}^{m \times n}. \quad (7)$$

Sometimes the Jacobian matrix is alternatively denoted by $\frac{\partial \varphi}{\partial x}$. Note that the Jacobian of a differentiable function $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$ is equal to the transpose of its gradient. For differentiable functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $g : \mathbb{R}^m \rightarrow \mathbb{R}^k$ the Jacobian matrix of its composition at $x \in \mathbb{R}^n$ is as follows

$$\nabla(g \circ f)(x) = \nabla g(y) \Big|_{y=f(x)} \cdot \nabla f(x), \quad (8)$$

see, for example, Theorem 5 on page 189 in [36]. The equation above is called the *chain rule*, and it is a fundamental mathematical tool for the *backpropagation procedure*, see [9].

We call $w^* \in \mathbb{R}^n$ the *critical point* of $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$ if the gradient $\nabla \mathcal{L}(w^*)$ does not exist or $\nabla \mathcal{L}(w^*) = 0$. For differentiable $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$ the point $w^* \in \mathbb{R}^n$ is a critical point iff $\nabla \mathcal{L}(w^*) = 0$. All local extrema are critical points. Not all critical points are local extrema - often, they are saddle points.

2 Models, risks and bias vs variance trade-off

2.1 Models and risks

By a model $\mathcal{M}(\cdot, w)$, parametrized by a vector $w \in \mathbb{R}^n$, we mean a measurable function from the (input) space $\mathbb{R}^M$ to the (output) space $\mathbb{R}^N$, where

$$\forall_{w \in \mathbb{R}^n} \mathcal{M}(\cdot, w) : \mathbb{R}^M \rightarrow \mathbb{R}^N \quad (9)$$

is Borel measurable. The vector w is often called *weights* or *tunable parameters*. So, for a fixed w , having the input x the model gives us an output $\mathcal{M}(x, w)$. In supervised learning, we adjust the weights w according to the training data

$$D = \{(x_1, y_1), (x_2, y_2), \dots, (x_K, y_K)\}, \quad (10)$$

where $x_j \in \mathbb{R}^M$, $y_j \in \mathbb{R}^N$, and chosen (appropriately to the problem that we want to solve) loss function $\mathcal{L}_K = \mathcal{L}_K(w)$. The loss is often defined as an empirical risk

$$\mathcal{L}_K(w) = \frac{1}{K} \sum_{j=1}^K \ell(y_j, \mathcal{M}(x_j, w)), \quad (11)$$

where $\ell : \mathbb{R}^N \times \mathbb{R}^N \rightarrow [0, +\infty)$ is a measurable function, and $\ell(y_j, \mathcal{M}(x_j, w))$ measures the error between the output of the model $\mathcal{M}(x_j, w)$ and the ground truth y_j for j th training sample. Optimal weights w^* , under given training data set D , are defined as

$$w^* \in \arg \min_{w \in \mathbb{R}^n} \mathcal{L}_K(w). \quad (12)$$

In general, the set of minimizers may be empty - in such cases one works with ε -minimizers or imposes additional assumptions such as compactness of the parameter set, coercivity, regularization, etc. The main paradigm in data science assumes that a dataset D comprises independent and identically distributed (i.i.d.) samples drawn from an (in general) unknown probability distribution Q , which is interpreted as the underlying data-generating process. Under this assumption and provided that $\mathbb{E}|\ell(Y, \mathcal{M}(X, w))| < +\infty$, the strong law of large numbers implies that for each w ,

$$\mathcal{L}_K(w) = \frac{1}{K} \sum_{j=1}^K \ell(y_j, \mathcal{M}(x_j, w)) \rightarrow \mathbb{E}(\ell(Y, \mathcal{M}(X, w))), \quad K \rightarrow +\infty, \quad a.s., \quad (13)$$

where the random pair (X, Y) is distributed according to Q . The *population (true) risk* is therefore

$$\mathcal{L}(w) = \mathbb{E}(\ell(Y, \mathcal{M}(X, w))) = \int_{\mathbb{R}^M \times \mathbb{R}^N} \ell(y, \mathcal{M}(x, w)) dQ(x, y) \quad (14)$$

and is generally unknown. For the regression setting we adopt the squared-error loss $\ell(y, z) = \|y - z\|^2$. In this case $\mathcal{L}_K$ and $\mathcal{L}$ reduce to the empirical and population *mean squared error*, respectively. Moreover, for every parameter vector w ,

$$\mathbb{E}[\ell(Y, \mathcal{M}(X, w))] \geq \mathbb{E}\|Y - \mathbb{E}[Y | X]\|^2,$$

where the right-hand side is attained when the model outputs the *regression function* $\mathbb{E}[Y | X]$. Thus, regardless of the model we choose, the irreducible noise term $\mathbb{E}\|Y - \mathbb{E}[Y | X]\|^2$ sets a fundamental lower bound on the achievable risk.

2.2 Bias vs variance trade-off from probabilistic perspective

Here, we examine the error and variance of regression models defined by conditional expectation.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, let $\mathcal{G}$ be a sub- σ -field of $\mathcal{F}$, and let $X : \Omega \rightarrow \mathbb{R}^d$ be a random vector with $\mathbb{E}\|X\|^2 < +\infty$. The best $L^2(\Omega)$ -approximation of X is the conditional expectation $\mathbb{E}(X \mid \mathcal{G})$. Formally, it means that for any $\mathcal{G}$ -measurable random vector Y with $\mathbb{E}\|Y\|^2 < +\infty$ we have

$$\mathbb{E}\|X - Y\|^2 \geq \mathbb{E}\|X - \mathbb{E}(X \mid \mathcal{G})\|^2. \quad (15)$$

Moreover, we have the following well-known variance decomposition

$$\text{Var}(X) = \text{Var}(\mathbb{E}(X \mid \mathcal{G})) + \mathbb{E}\|X - \mathbb{E}(X \mid \mathcal{G})\|^2, \quad (16)$$

where

$$\text{Var}(\mathbb{E}(X \mid \mathcal{G})) = \mathbb{E}\|\mathbb{E}(X \mid \mathcal{G}) - \mathbb{E}(\mathbb{E}(X \mid \mathcal{G}))\|^2 = \mathbb{E}\|\mathbb{E}(X \mid \mathcal{G}) - \mathbb{E}(X)\|^2.$$

Consider two sub- σ -fields $\mathcal{G}_1 \subseteq \mathcal{G}_2$ of $\mathcal{F}$. They yield two possible approximations (models) of X :

$$\mathbb{E}(X \mid \mathcal{G}_1), \quad \mathbb{E}(X \mid \mathcal{G}_2).$$

Because $\mathcal{G}_2$ contains more information than $\mathcal{G}_1$, the approximation $\mathbb{E}(X \mid \mathcal{G}_2)$ can be viewed as a more complex model than $\mathbb{E}(X \mid \mathcal{G}_1)$. Since $\mathbb{E}(X \mid \mathcal{G}_1)$ is $\mathcal{G}_2$ -measurable we have by (15) that

$$\mathbb{E}\|X - \mathbb{E}(X \mid \mathcal{G}_1)\|^2 \geq \mathbb{E}\|X - \mathbb{E}(X \mid \mathcal{G}_2)\|^2. \quad (17)$$

Furthermore, by (16)

$$\begin{aligned} & \text{Var}(\mathbb{E}(X \mid \mathcal{G}_1)) + \mathbb{E}\|X - \mathbb{E}(X \mid \mathcal{G}_1)\|^2 \\ &= \text{Var}(X) = \text{Var}(\mathbb{E}(X \mid \mathcal{G}_2)) + \mathbb{E}\|X - \mathbb{E}(X \mid \mathcal{G}_2)\|^2, \end{aligned} \quad (18)$$

which, together with (17), yields

$$\text{Var}(\mathbb{E}(X \mid \mathcal{G}_1)) \leq \text{Var}(\mathbb{E}(X \mid \mathcal{G}_2)). \quad (19)$$

Hence, the model $\mathbb{E}(X \mid \mathcal{G}_2)$, which uses more information, has a smaller $L^2(\Omega)$ error than the model $\mathbb{E}(X \mid \mathcal{G}_1)$, but it has a larger variance. This illustrates the classical *bias-variance trade-off* in statistical learning: enriching the information set reduces bias (mean-squared error) at the cost of increased variance.

3 Two classical problems and one neuron

In this section, we relate the task of training a single neuron to two classical problems:

- (i) linear least-squares regression (often called linear smoothing),
- (ii) binary classification.

We begin with the formal definition of an artificial neuron.

Definition 3.1. *Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be a Borel-measurable function. We call the function*

$$\mu : \mathbb{R}^N \times \mathbb{R} \times \mathbb{R}^N \rightarrow \mathbb{R} \quad (20)$$

an artificial neuron if for all $x, w \in \mathbb{R}^N$, $b \in \mathbb{R}$

$$\mu(w, b, x) = \varphi(\langle w, x \rangle + b) = \varphi\left(\sum_{j=1}^N w_j x_j + b\right). \quad (21)$$

We call:

- φ - the activation function,
- x - the input vector,
- w - the weights of the neuron and b is the bias term,
- the value $\mu(w, b, x)$ is the output of the neuron.

By definition, the regularity of an artificial neuron as a mathematical function depends entirely on the regularity of its activation function φ . As we shall see later, in the case of deep neural networks, this dependence is reinforced even further, because a deep network is essentially a conglomerate of many such neurons.

The common activation functions are¹:

$$\begin{aligned}
\textbf{Identity: } \quad & \varphi_{\text{id}}(z) = z \\
\textbf{Binary step: } \quad & \varphi_{\text{step}}(z) = \begin{cases} 1, & z \geq 0, \\ 0, & z < 0 \end{cases} \\
\textbf{Sign: } \quad & \varphi_{\text{sgn}}(z) = \begin{cases} +1, & z > 0, \\ 0, & z = 0, \\ -1, & z < 0 \end{cases} \\
\textbf{Logistic sigmoid: } \quad & \varphi_{\sigma}(z) = \frac{1}{1 + e^{-z}} \\
\textbf{Hyperbolic tangent: } \quad & \varphi_{\tanh}(z) = \tanh z = \frac{e^z - e^{-z}}{e^z + e^{-z}} \\
\textbf{ReLU: } \quad & \varphi_{\text{ReLU}}(z) = z^+ = \max\{0, z\} \\
\textbf{Leaky ReLU } (\alpha > 0): \quad & \varphi_{\text{LReLU}}(z) = \begin{cases} z, & z \geq 0, \\ \alpha z, & z < 0 \end{cases} \\
\textbf{Softplus: } \quad & \varphi_{\text{softplus}}(z) = \ln(1 + e^z) \\
\textbf{ELU } (\alpha > 0): \quad & \varphi_{\text{ELU}}(z) = \begin{cases} z, & z \geq 0, \\ \alpha(e^z - 1), & z < 0 \end{cases} \\
\textbf{GELU (approx.): } \quad & \varphi_{\text{GELU}}(z) = \frac{1}{2}z \left(1 + \tanh\left(\sqrt{\frac{2}{\pi}}(z + 0.044715 z^3)\right) \right) \\
\textbf{Swish } (\beta > 0): \quad & \varphi_{\text{swish}}(z) = z\varphi_{\sigma}(\beta z) = \frac{z}{1 + e^{-\beta z}} \\
\textbf{Softmax (vector } z \in \mathbb{R}^K): \quad & \text{softmax}(z)_i = \frac{e^{z_i}}{\sum_{k=1}^K e^{z_k}}, \quad i = 1, \dots, K
\end{aligned} \tag{22}$$

3.1 Linear least squares problem

For a given matrix $\mathbb{R}^{M \times N} \ni A = [a_{i,j}]_{i=1,2,\dots,M,j=1,2,\dots,N}$ and vector $y = [y_1, y_2, \dots, y_M] \in \mathbb{R}^M$, $M, N \in \mathbb{N}$, $M \geq N$, the *linear least-squares problem* seeks

$$\inf_{w \in \mathbb{R}^N} \|Aw - y\|, \tag{23}$$

together with an optimal vector $w^* \in \mathbb{R}^N$ that attains this infimum. We denote by $L(w) = \|Aw - y\|$ and $\mathcal{L}(w) = (L(w))^2$. Then

$$\mathcal{L}(w) = \langle Aw - y, Aw - y \rangle = \langle y, y \rangle - 2\langle Aw, y \rangle + \langle Aw, Aw \rangle, \tag{24}$$

and

$$\nabla \mathcal{L}(w) = 2A^T Aw - 2A^T y. \tag{25}$$

¹Especially in the ML/DL literature, the identity activation function is often referred to as a *linear activation function*, although this terminology is not mathematically precise.

Hence, the critical point of $\mathcal{L}$ is the solution of the following system of linear equations

$$A^T A w = A^T y. \quad (26)$$

We call that system the *normal equations*. We left the following fact as an exercise.

Proposition 3.1. *If columns of A are linearly independent, then the matrix $A^T A$ is invertible and*

$$w^* = (A^T A)^{-1} A^T y \quad (27)$$

is the unique solution of (26).

In practical applications, assuming that the columns of A are fully linearly independent is often too restrictive. Measurement data can be correlated, duplicated, or noisy, which leads to (near) linear dependence among the columns. As a result, the matrix $A^T A$ becomes singular or nearly singular, making its direct inversion, and thus the straightforward formula $(A^T A)^{-1} A^T y$, either impossible or numerically unstable. Therefore, in the next theorem, we will study the normal equations *without* assuming column independence, to obtain solutions even when $A^T A$ is singular. Moreover,

$$\nabla^2 \mathcal{L}(w) = 2A^T A, \quad (28)$$

and $A^T A \in \mathbb{R}^{N \times N}$ is symmetric and positive semi-definite. Therefore, any solution of (26) is a global minimum of $\mathcal{L}$ (and also L). We will show that the opposite is also true: every minimum of $\mathcal{L}$ solves (26).

Theorem 3.1 ([38]). *The linear least-squares problem (23) has at least one minimum point w^* . If $\bar{w}$ is another minimum point, then $A w^* = A \bar{w}$. The residual $r = y - A w^*$ is uniquely determined and satisfies the equation $A^T r = 0$. Every minimum point w^* is also a solution of the normal equations (26) and conversely. Moreover, for every minimizer w^* of L we have*

$$L(w^*) = (\|y\|^2 - \|A w^*\|^2)^{1/2}, \quad (29)$$

and $\|A w^*\| \leq \|y\|$.

Proof. Define the following linear subspace of $\mathbb{R}^M$

$$\begin{aligned} V &= \{A w \mid w \in \mathbb{R}^N\} = \left\{ \sum_{j=1}^N w_j k_j \mid [w_1, \dots, w_N] \in \mathbb{R}^N \right\} \\ &= \text{lin}\{k_1, \dots, k_N\}, \end{aligned} \quad (30)$$

where k_j is the j th column of A . We have that $\dim V \leq N$ (and $\dim V = N$ if the columns of A are linearly independent). By the orthogonal projection theorem we have that there exists unique $v^* \in V$ such that

$$\|y - v^*\| = \inf_{v \in V} \|y - v\|, \quad (31)$$

the vector v^* is the orthogonal projection of y on the linear space V . Since $v^* \in V$ there is (at least one, because $\dim V \leq N$) $w^* \in \mathbb{R}^N$ such that $v^* = A w^*$. Moreover, for all $w \in \mathbb{R}^N$

$$\|y - A w\| \geq \inf_{v \in V} \|y - v\|, \quad (32)$$

and hence

$$\begin{aligned} \inf_{w \in \mathbb{R}^N} \|y - A w\| &\geq \inf_{v \in V} \|y - v\| = \|y - v^*\| \\ &= \|y - A w^*\| \geq \inf_{w \in \mathbb{R}^N} \|y - A w\|, \end{aligned} \quad (33)$$

which implies the following equalities

$$\begin{aligned} \inf_{w \in \mathbb{R}^N} L(w) &= \inf_{w \in \mathbb{R}^N} \|y - Aw\| \\ &= \inf_{v \in V} \|y - v\| = \|y - v^*\| = \|y - Aw^*\| = L(w^*). \end{aligned} \quad (34)$$

Hence, w^* is the minimizer of the function $L(w) = \|Aw - y\|$.

If for $v^* \in V$ there exists another $\bar{w} \in \mathbb{R}^N$ such that $v^* = A\bar{w}$, then $v^* = Aw^* = A\bar{w}$. Therefore, the residual

$$\mathbb{R}^M \ni r = y - Aw^* = y - A\bar{w} = y - v^* \quad (35)$$

is uniquely determined. From the orthogonal projection theorem, we have

$$r = y - v^* \perp V, \quad (36)$$

and

$$\forall w \in \mathbb{R}^N w^T (A^T r) = (Aw)^T r = 0, \quad (37)$$

since $Aw \in V$. Hence, for $w := A^T r \in \mathbb{R}^N$ we obtain that $\|A^T r\|^2 = 0$, and

$$A^T r = 0. \quad (38)$$

Assume now that for w^* we have $v^* = Aw^*$, and w^* is the minimizer of L . Then, by (36)

$$\forall w \in \mathbb{R}^N (Aw)^T r = (Aw)^T (y - Aw^*) = 0, \quad (39)$$

where

$$(Aw)^T (y - Aw^*) = w^T (A^T y - A^T Aw^*). \quad (40)$$

Therefore, it must hold that

$$A^T y - A^T Aw^* = 0, \quad (41)$$

and w^* solves the normal equations.

Finally, we show that every solution w^* to (26) is also a minimizer of $L(w) = \|Aw - y\|$. Let

$$V^\perp = \{r \mid \forall v \in V \ r^T v = 0\}, \quad (42)$$

be the orthogonal complement of V . Then $r \in V^\perp$ iff

$$\forall w \in \mathbb{R}^N \ r^T Aw = 0. \quad (43)$$

Therefore, we must have $r^T A = 0$ and hence,

$$V^\perp = \{r \mid r^T A = 0\}. \quad (44)$$

Since

$$\mathbb{R}^M = V \oplus V^\perp, \quad (45)$$

the vector $y \in \mathbb{R}^M$ has a unique decomposition

$$y = s + t, \quad s \in V, \quad t \in V^\perp. \quad (46)$$

Let $s := Aw^*$ and $t := y - Aw^*$. Then $s \in V$ and $t \in V^\perp$, since

$$(y - Aw^*)^T A = y^T A - (w^*)^T A^T A = 0. \quad (47)$$

The last equality is due to the fact that w^* solves the normal equation (26) and

$$y^T A = (A^T y)^T = (A^T (Aw^*))^T = (Aw^*)^T A = (w^*)^T A^T A. \quad (48)$$

For all $w \in \mathbb{R}^N$ we therefore have

$$y - Aw = A(w^* - w) + (y - Aw^*) = t - z, \quad (49)$$

where

$$z = A(w - w^*) \in V. \quad (50)$$

Since $t \perp z$, we get for all $w \in \mathbb{R}^N$ that

$$\begin{aligned} (L(w))^2 &= \|y - Aw\|^2 = \|t - z\|^2 = \|t\|^2 + \|z\|^2 \\ &\geq \|t\|^2 = \|y - Aw^*\|^2 = (L(w^*))^2. \end{aligned} \quad (51)$$

This implies that $\inf_{w \in \mathbb{R}^N} L(w) = L(w^*)$.

For a minimizer w^* we have

$$\begin{aligned} 0 &\leq \|y - Aw^*\|^2 = (y - Aw^*)^T (y - Aw^*) \\ &= y^T y - y^T Aw^* - (w^*)^T A^T y + (w^*)^T A^T Aw^* \\ &= y^T y - y^T Aw^*, \end{aligned} \quad (52)$$

since

$$-(w^*)^T A^T y + (w^*)^T A^T Aw^* = -(w^*)^T A^T Aw^* + (w^*)^T A^T Aw^* = 0. \quad (53)$$

Moreover,

$$\begin{aligned} y^T Aw^* &= \langle y, Aw^* \rangle = \langle Aw^*, y \rangle = (Aw^*)^T y = (w^*)^T A^T y \\ &= (w^*)^T A^T Aw^* = (Aw^*)^T Aw^* = \|Aw^*\|^2. \end{aligned} \quad (54)$$

This implies the thesis. $\square$

We also relate the minimization of $L = L(w)$ to the minimization of an appropriate quadratic form $Q = Q(w)$. In particular, we have the following lemma.

Lemma 3.1. *Let $B \in \mathbb{R}^{N \times N}$ be a symmetric and positive semi-definite, and $z \in \mathbb{R}^N$. Then*

$$\forall_{w^* \in \mathbb{R}^N} \left[Bw^* = z \Leftrightarrow Q(w^*) = \inf_{w \in \mathbb{R}^N} Q(w) \right], \quad (55)$$

where

$$Q(w) = \frac{1}{2} w^T B w - w^T z, \quad w \in \mathbb{R}^N. \quad (56)$$

Proof. ($\Leftarrow$) Assume that w^* is minimizing Q . Then $0 = \nabla Q(w^*) = Bw^* - z$.

($\Rightarrow$) Assume that w^* satisfies $Bw^* = z$. Then for all $x, r \in \mathbb{R}^N$

$$\begin{aligned} Q(x + r) &= \frac{1}{2} (x + r)^T B (x + r) - (x + r)^T z \\ &= Q(x) + r^T Bx + \frac{1}{2} r^T B r - r^T z \\ &\geq Q(x) + r^T (Bx - z), \end{aligned} \quad (57)$$

since $r^T B r \geq 0$, and $\mathbb{R} \ni r^T Bx = (r^T Bx)^T = (Bx)^T r = x^T B^T r = x^T B r$ for symmetric and positive semi-definite B . Hence, for all $w \in \mathbb{R}^N$

$$\begin{aligned} Q(w) &= Q(w^* + (w - w^*)) \\ &\geq Q(w^*) + (w - w^*)^T (Bw^* - z) = Q(w^*), \end{aligned} \quad (58)$$

and this gives that w^* minimizes Q . $\square$

Corollary 3.1. *If $B \in \mathbb{R}^{N \times N}$ is symmetric and positive definite, $z \in \mathbb{R}^N$, then the unique solution of $Bw = z$ is the minimizer of Q .*

From Lemma 3.1 and Theorem 3.1 we get the following corollary.

Corollary 3.2. *There exists at least one minimizer w^* of L and*

$$\forall_{w^* \in \mathbb{R}^N} \left[\inf_{w \in \mathbb{R}^N} L(w) = L(w^*) \Leftrightarrow A^T A w^* = A^T y \Leftrightarrow \inf_{w \in \mathbb{R}^N} Q(w) = Q(w^*) \right],$$

where $L(w) = \|Aw - y\|$, $Q(w) = \frac{1}{2} w^T A^T A w - w^T A^T y$, $w \in \mathbb{R}^N$.

We left as an exercise showing that

$$Q(w) = \frac{1}{2} \left((L(w))^2 - \|y\|^2 \right), \quad w \in \mathbb{R}^N. \quad (59)$$

Let us denote by $a^{(j)} = [a_{j,1}, \dots, a_{j,N}]^T \in \mathbb{R}^N$, $j = 1, 2, \dots, M$, the j th row of the matrix A . Then for all $w \in \mathbb{R}^N$

$$\mathcal{L}(w) = \|Aw - y\|^2 = \sum_{j=1}^M \left(\langle a^{(j)}, w \rangle - y_j \right)^2. \quad (60)$$

Let us consider the following training data set

$$D = \{(a^{(1)}, y_1), \dots, (a^{(M)}, y_M)\}, \quad (61)$$

and an artificial neuron $\mu(w, 0, x)$ with the linear activation function, i.e.,

$$\mu(w, 0, x) = \langle w, x \rangle, \quad x, w \in \mathbb{R}^N. \quad (62)$$

For this artificial neuron, we know by Corollary 3.2 that there exists (at least one) optimal weight w^* for which

$$MSE(w^*) = \inf_{w \in \mathbb{R}^N} MSE(w), \quad (63)$$

where the mean square error is defined as

$$\begin{aligned} MSE(w) &= \frac{1}{M} \mathcal{L}(w) = \frac{1}{M} \sum_{j=1}^M \left(\langle a^{(j)}, w \rangle - y_j \right)^2 \\ &= \frac{1}{M} \sum_{j=1}^M \left(\mu(w, 0, a^{(j)}) - y_j \right)^2. \end{aligned} \quad (64)$$

Therefore, finding the optimal weights w^* that minimize $MSE(w)$ in the training data set D is equivalent to minimizing the loss function $\mathcal{L}(w) = \|Aw - y\|^2$, which in turn is equivalent to solving normal equations (26). Hence, solving the linear least squares problem is equivalent to training a single neuron with a linear activation function, using the mean squared error as the loss function to be minimized.

In practice, solving the normal equations (26) with direct methods, even when $A^T A$ is invertible, can be numerically inefficient, especially for large N . Therefore, in the next sections, we will show how to solve these equations iteratively, using a suitable variant of the gradient-descent method.

In what follows, we present several data-fitting problems that can be cast in the linear least-squares framework.

Examples - Scalar input data

In this example, we want to fit various functions to the following data set

$$\{(z_1, y_1), \dots, (z_M, y_M)\}, \quad (65)$$

where $z_j, y_j \in \mathbb{R}$, to minimize the mean squared error.

a) (*Classical linear regression*) In this case we take $y = [y_1, \dots, y_M]$ and

$$A = \begin{bmatrix} 1 & z_1 \\ 1 & z_2 \\ \vdots & \vdots \\ 1 & z_M \end{bmatrix}. \quad (66)$$

In this case, the mean square error (64) has the form

$$MSE(w) = \frac{1}{M} \sum_{j=1}^M (w_1 + w_2 z_j - y_j)^2, \quad w \in \mathbb{R}^2, \quad (67)$$

and its minimization give us the optimal $w^* = [w_1^*, w_2^*]$, to that the regression line is $f^*(z) = w_1^* + w_2^* z$, $z \in \mathbb{R}$. (Note that in this case it is possible to compute w^* explicitly - we left this as an exercise.)

b) (*Polynomial regression*) Assume that we want to fit to the data (75) a polynomial of degree $r \geq 1$. In this case, we need to take

$$A = \begin{bmatrix} 1 & z_1 & z_1^2 & \dots & z_1^r \\ 1 & z_2 & z_2^2 & \dots & z_2^r \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & z_M & z_M^2 & \dots & z_M^r \end{bmatrix}, \quad (68)$$

and we minimize

$$MSE(w) = \frac{1}{M} \sum_{j=1}^M (p_r(w, z_j) - y_j)^2, \quad w \in \mathbb{R}^{r+1}, \quad (69)$$

where

$$\begin{aligned} p_r(w, z) &= w_1 + w_2 z + w_3 z^2 + \dots + w_{r+1} z^r \\ &= \langle [1, z, \dots, z^r], [w_1, w_2, \dots, w_{r+1}] \rangle, \end{aligned} \quad (70)$$

to obtain the optimal weights $w^* = [w_1^*, \dots, w_{r+1}^*]$. The regression polynomial is

$$p_r(w^*, z) = w_1^* + w_2^* z + w_3^* z^2 + \dots + w_{r+1}^* z^r, \quad z \in \mathbb{R}. \quad (71)$$

In practice we chose a moderate r , since r too large may lead to overfitting.

c) (*Trigonometric polynomial regression*) In this case we want to fit to the data (75) a trigonometric polynomial of degree $r \geq 1$. We take

$$A = \begin{bmatrix} 1 & \sin(\pi z_1) & \dots & \sin(r\pi z_1) & \cos(\pi z_1) & \dots & \cos(r\pi z_1) \\ 1 & \sin(\pi z_2) & \dots & \sin(r\pi z_2) & \cos(\pi z_2) & \dots & \cos(r\pi z_2) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \sin(\pi z_M) & \dots & \sin(r\pi z_M) & \cos(\pi z_M) & \dots & \cos(r\pi z_M) \end{bmatrix}, \quad (72)$$

and we minimize the mean square error

$$MSE(w) = \frac{1}{M} \sum_{j=1}^M (t_r(w, z_j) - y_j)^2, \quad w \in \mathbb{R}^{2r+1}, \quad (73)$$

where

$$\begin{aligned} t_r(w, z) &= w_1 + \sum_{k=1}^r w_{k+1} \sin(k\pi z) + \sum_{k=r+1}^{2r} w_{k+1} \cos((k-r)\pi z) \\ &= \left\langle [1, \sin(\pi z), \dots, \sin(r\pi z), \cos(\pi z), \dots, \cos(r\pi z)], [w_1, \dots, w_{r+1}, w_{r+2}, \dots, w_{2r}] \right\rangle, \end{aligned}$$

to obtain the optimal weights $w^* = [w_1^*, \dots, w_{2r+1}^*]$. The regression polynomial is

$$t_r(w^*, z) = w_1^* + \sum_{k=1}^r w_{k+1}^* \sin(k\pi z) + \sum_{k=r+1}^{2r} w_{k+1}^* \cos((k-r)\pi z), z \in \mathbb{R}. \quad (74)$$

Examples - Vector input data

In this case, we fit various multivariable functions to the dataset

$$\{(z_1, y_1), \dots, (z_M, y_M)\}, \quad (75)$$

where $z_j = [z_j^{(1)}, \dots, z_j^{(m)}] \in \mathbb{R}^m$, $y_j \in \mathbb{R}$.

a) (*Classical multidimensional regression*) For fitting a hypersurface to the data point we need to take

$$A = \begin{bmatrix} 1 & z_1^{(1)} & z_1^{(2)} & \dots & z_1^{(m)} \\ 1 & z_2^{(1)} & z_2^{(2)} & \dots & z_2^{(m)} \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & z_M^{(1)} & z_M^{(2)} & \dots & z_M^{(m)} \end{bmatrix}, \quad (76)$$

and we minimize

$$MSE(w) = \frac{1}{M} \sum_{j=1}^M (h(w, z_j) - y_j)^2, \quad (77)$$

where

$$\begin{aligned} h(w, z) &= w_1 + \sum_{k=1}^m w_{k+1} z^{(k)} \\ &= \langle [1, z^{(1)}, \dots, z^{(m)}], [w_1, w_2, \dots, w_{m+1}] \rangle, \quad z \in \mathbb{R}^m. \end{aligned} \quad (78)$$

The optimal regression hypersurface is $h(w^*, z)$. If $m = 1$ then we obtain classical linear regression with scalar input data.

b) (*Multivariate polynomial regression*) For $m = 2$ we show how to define the matrix A if we would like to fit to the data points multivariate polynomial of degree two. For this set

$$A = \begin{bmatrix} 1 & z_1^{(1)} & z_1^{(2)} & (z_1^{(1)})^2 & (z_1^{(2)})^2 & z_1^{(1)} z_1^{(2)} \\ 1 & z_2^{(1)} & z_2^{(2)} & (z_2^{(1)})^2 & (z_2^{(2)})^2 & z_2^{(1)} z_2^{(2)} \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots \\ 1 & z_M^{(1)} & z_M^{(2)} & (z_M^{(1)})^2 & (z_M^{(2)})^2 & z_M^{(1)} z_M^{(2)} \end{bmatrix}, \quad (79)$$

and the loss function to minimize is

$$MSE(w) = \frac{1}{M} \sum_{j=1}^M (h(w, z_j) - y_j)^2, \quad w \in \mathbb{R}^6, \quad (80)$$

with

$$h(w, z) = \langle [1, z^{(1)}, z^{(2)}, (z^{(1)})^2, (z^{(2)})^2, z^{(1)} z^{(2)}], [w_1, w_2, \dots, w_6] \rangle, \quad z \in \mathbb{R}^2,$$

and the optimal multivariate polynomial is $h(w^*, z)$.

Example - Autoregressive model of order p

Let us assume that we have time series data

$$y_0, y_1, \dots, y_M, \quad (81)$$

where $y_j \in \mathbb{R}$. One of the most well-known econometric model is the autoregressive model of order p , denoted by $AR(p)$, $p \in \mathbb{N}$. In this model, we assume that the time series follows the linear recursive equation of order p , i.e.:

$$y_k = w_0 + w_1 y_{k-1} + w_2 y_{k-2} + \dots + w_p y_{k-p} + \varepsilon_k, \quad (82)$$

where the parameters $w = [w_0, w_1, \dots, w_p] \in \mathbb{R}^{p+1}$ are chosen in such a way that the following mean square error is minimized

$$\begin{aligned} MSE(w) &= \frac{1}{M-p+1} \sum_{k=p}^M \varepsilon_k^2 \\ &= \frac{1}{M-p+1} \sum_{k=p}^M \left(y_k - (w_0 + w_1 y_{k-1} + w_2 y_{k-2} + \dots + w_p y_{k-p}) \right)^2. \end{aligned} \quad (83)$$

We can settle this as the linear least squares problem with the following training data

$$D = \left\{ \left([y_{p-1}, \dots, y_0], y_p \right), \left([y_p, \dots, y_1], y_{p+1} \right), \dots, \left([y_{M-1}, \dots, y_{M-p}], y_M \right) \right\}. \quad (84)$$

Then

$$y = [y_p, y_{p+1}, \dots, y_M], \quad (85)$$

and the matrix A is built from the augmented regressors $[1, y_{k-1}, \dots, y_{k-p}]$, i.e.,

$$A = \begin{bmatrix} 1 & y_{p-1} & \dots & y_0 \\ 1 & y_p & \dots & y_1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & y_{M-1} & \dots & y_{M-p} \end{bmatrix} \in \mathbb{R}^{(M-p+1) \times (p+1)}. \quad (86)$$

Hence, $MSE(w) = \frac{1}{M-p+1} \|Aw - y\|^2$. Note that p is a *hyperparameter* that must be selected before the minimisation of $MSE(w)$ begins. The transformation of the original time-series data (81) into the vector form (84) is called *sliding window* procedure - this pre-processing step is widely used in machine- and deep-learning approaches to time-series forecasting.

Using the described approach, we can calibrate the $AR(p)$ model to the time series data without assuming anything on the law of the random residuals ε_k . (In the maximum likelihood approach, the residuals are assumed to be normally distributed.)

Example - Black-Scholes and Vasicek models

In this example, we demonstrate how to address the problem of parameter estimation for two well-known linear stochastic differential equations within the framework of a linear least-squares problem. Namely, assuming that we have observed the (financial) data

$$X(t_0), X(t_1), \dots, X(t_N) \quad (87)$$

at equidistant points $t_i = iT/N$, $i = 0, 1, \dots, N$, we aim at estimating μ, σ when we are considering the Black-Scholes model

$$dX(t) = \mu X(t)dt + \sigma X(t)dW(t), \quad t \in [0, T], \quad (88)$$

$$X(0) = X_0, \quad (89)$$

while we aim at estimating κ, μ, σ for the Vasicek model

$$dX(t) = \kappa(\mu - X(t))dt + \sigma dW(t), \quad t \in [0, T], \quad (90)$$

$$X(0) = X_0. \quad (91)$$

The Euler-Maruyama discretization of (88) yields

$$\begin{aligned} X(t_0) &= X_0, \\ X(t_{i+1}) &= X(t_i) + \mu X(t_i)h + \sigma X(t_i)\Delta W_i, \quad i = 0, 1, \dots, N-1 \end{aligned} \quad (92)$$

where $h = T/N$, $\Delta W_i = W(t_{i+1}) - W(t_i)$. (For N that is sufficiently large, we can assume that the values follow the discrete scheme (92).) Hence, after simple transformations we arrive at the following linear regression problem

$$Y_i = \bar{\mu} + \varepsilon_i, \quad i = 0, 1, \dots, N-1, \quad (93)$$

where $(\varepsilon_i)_{i \geq 0}$ is an iid sequence from the distribution $N(0, \sigma^2)$ and

$$Y_i = \frac{X(t_{i+1}) - X(t_i)}{X(t_i)\sqrt{h}}, \quad (94)$$

$$\bar{\mu} = \mu\sqrt{h}. \quad (95)$$

We obtain the estimator for $\bar{\mu}$ (and therefore also for μ) by minimizing the noise $\frac{1}{N} \sum_{i=0}^{N-1} \varepsilon_i^2$, i.e., the following mean square error

$$\mathcal{L}(x) = \frac{1}{N} \sum_{i=0}^{N-1} (Y_i - x)^2. \quad (96)$$

Since $\mathcal{L}'(x) = -\frac{2}{N} \sum_{i=0}^{N-1} (Y_i - x)$, we get that

$$\hat{\mu}_N = \frac{1}{T} \sum_{i=0}^{N-1} \frac{X(t_{i+1}) - X(t_i)}{X(t_i)}. \quad (97)$$

Additionally,

$$\mathbb{E}\left(\frac{1}{N} \sum_{i=0}^{N-1} \varepsilon_i^2\right) = \sigma^2, \quad (98)$$

so we can take

$$\widehat{\sigma}_N^2 = \mathcal{L}(\hat{\mu}_N\sqrt{h}) = \frac{1}{N} \sum_{i=0}^{N-1} (Y_i - \hat{\mu}_N\sqrt{h})^2, \quad (99)$$

as the estimator of the variance σ^2 .

In the case of the Vasicek model, the Euler-Maruyama discretization is as follows

$$\begin{aligned} X(t_0) &= X_0, \\ X(t_{i+1}) &= X(t_i) + \kappa(\mu - X(t_i))h + \sigma\Delta W_i, \end{aligned} \quad (100)$$

for $i = 0, 1, \dots, N-1$. Therefore, we get for $i = 0, 1, \dots, N-1$ that

$$\frac{X(t_{i+1}) - X(t_i)}{\sqrt{h}} = -\kappa\sqrt{h}X(t_i) + \kappa\sqrt{h}\mu + \varepsilon_i \quad (101)$$

where $(\varepsilon_i)_{i \geq 0}$ is an iid sequence of random variables that are normally distributed $N(0, \sigma^2)$. Denoting by

$$Z_i = \frac{X(t_{i+1}) - X(t_i)}{\sqrt{h}}, \quad (102)$$

$$a = -\kappa\sqrt{h}, \quad (103)$$

$$b = \kappa\sqrt{h}\mu = -a\mu, \quad (104)$$

we can write (101) as

$$Z_i = aX(t_i) + b + \varepsilon_i, \quad i = 0, 1, \dots, N-1. \quad (105)$$

Due to properties of the Wiener process, we have that the input $X(t_i)$ and the noise term ε_i are independent random variables. Hence, each Z_i is a sum of independent random variables. Moreover, the residuals $\varepsilon_i = Z_i - aX(t_i) - b$ are independent and identically normally distributed. Hence, we can treat (105) as a linear regression model in which we want to minimize the MSE

$$\mathcal{L}(a, b) = \frac{1}{N} \sum_{i=0}^{N-1} \varepsilon_i^2 = \frac{1}{N} \sum_{i=0}^{N-1} (Z_i - aX(t_i) - b)^2. \quad (106)$$

This can be seen as the linear least squares problem under the following training set

$$D = \{(X(t_0), Z_0), (X(t_1), Z_1), \dots, (X(t_{N-1}), Z_{N-1})\}, \quad (107)$$

and $y = [Z_0, Z_1, \dots, Z_{N-1}]$,

$$A = \begin{bmatrix} 1 & X(t_0) \\ 1 & X(t_1) \\ \vdots & \vdots \\ 1 & X(t_{N-1}), \end{bmatrix} \quad (108)$$

since then $\mathcal{L}(a, b) = \frac{1}{N} \|A[b, a] - y\|^2$.

3.2 Binary classification

Binary classification is the supervised-learning task of mapping each input x_j to one of two labels $y_j \in \{0, 1\}$. Given a training set

$$\{(x_1, y_1), \dots, (x_M, y_M)\}, \quad (109)$$

where $x_j \in \mathbb{R}^N$, $y_j \in \{0, 1\}$, we aim to learn a decision function $f : \mathbb{R}^N \rightarrow \{0, 1\}$, or equivalently, a probability score $p(x) = \mathbb{P}(Y = 1 \mid X = x)$. During training we choose a parametric model $\mathcal{M}(\cdot; w)$ and optimise its parameters w by minimising a suitable loss function (such as the binary cross-entropy), which penalises mismatches between predictions $\mathcal{M}(x_j; w)$ and observed labels y_j . After training, when we obtained the optimal weights w^* , the classifier assigns a new sample x to class 1 if $\mathcal{M}(x, w^*) \geq \tau$ for some threshold τ (often equal to 0.5), and to class 0 otherwise. Performance is typically assessed with metrics such as accuracy, precision-recall, or the area under the ROC curve, evaluated on a validation set or via cross-validation.

We now derive the binary cross-entropy loss, the function most commonly used for binary classification. As it was described before, we assume that the data (109) consists of realizations from the iid sequence

$$\{(X_1, Y_1), \dots, (X_M, Y_M)\}, \quad (110)$$

and we assume that

$$\mathbb{P}(Y_i = 1 \mid X_i = x_i; w) = \mathcal{M}(x_i, w), \quad (111)$$

where we consider linear models

$$\mathcal{M}(x, w) = \varphi(\langle x, w \rangle), \quad x, w \in \mathbb{R}^N \quad (112)$$

and $\varphi \in C^1(\mathbb{R}; [0, 1])$. Of course, we also have

$$\mathbb{P}(Y_i = 0 \mid X_i = x_i; w) = 1 - \mathbb{P}(Y_i = 1 \mid X_i = x_i; w) = 1 - \mathcal{M}(x_i, w). \quad (113)$$

We are choosing φ in such a way that $\mathcal{M}(x, w)$ will model the probability that Y_i belongs to the class 1, provided that we observed realization x_i for the random variable X_i . One of the most common choices is the sigmoid function

$$\varphi(x) = \frac{1}{1 + e^{-x}}, \quad x \in \mathbb{R}. \quad (114)$$

We left proof of the following lemma as an exercise.

Lemma 3.2. *The sigmoid function (114) has the following properties:*

- (i) $\varphi : \mathbb{R} \rightarrow (0, 1)$, $\inf \varphi = 0$, $\sup \varphi = 1$,
- (ii) $\lim_{x \rightarrow -\infty} \varphi(x) = 0$, $\lim_{x \rightarrow +\infty} \varphi(x) = 1$, $\varphi(0) = 1/2$,
- (iii) $\varphi'(x) = \varphi(x)(1 - \varphi(x)) \in (0, 1)$, φ is increasing in $\mathbb{R}$,
- (iv) $\varphi''(x) = \varphi'(x)(1 - 2\varphi(x))$ and $x = 0$ is the inflection point of φ ,
- (v) φ is globally Lipschitz continuous, since for all $x, y \in \mathbb{R}$

$$|\varphi(x) - \varphi(y)| \leq |x - y|. \quad (115)$$

Consistently, we can write that

$$\mathbb{P}(Y_i = y_i \mid X_i = x_i; w) = (\mathcal{M}(x_i, w))^{y_i} (1 - \mathcal{M}(x_i, w))^{1-y_i}, \quad (116)$$

for $i = 1, \dots, M$. We now adopt the maximum-likelihood (ML) principle to derive an appropriate loss function and briefly recall the rationale behind this approach.

Maximum likelihood views the observed data as an independent samples drawn from a parametric family of probability distributions $\{p_w\}_w$. It asks a single question: *for which value of w would this family assign the greatest probability to the data we actually saw?* The parameter that maximises that probability-the likelihood-is then chosen as the estimate:

$$w^* = \arg \max_w L(w \mid \text{data}) = \arg \max_w \prod_{j=1}^n p(\text{data}_j; w).$$

Philosophically, the method chooses the model, from the family of probability distributions $\{p_w\}_w$, that best explains the observations. Many familiar procedures, such as linear and logistic regression, Gaussian-mixture EM, cross-entropy training of deep networks etc., are simply instances of the principle “pick the w that makes the data most likely.”, see, for example, [19].

Since we have an iid sample, we get the following common conditional distribution

$$\begin{aligned} \mathbb{P}(Y_1 = y_1, \dots, Y_M = y_M \mid X_1 = x_1, \dots, X_M = x_M; w) \\ &= \prod_{j=1}^M \mathbb{P}(Y_j = y_j \mid X_j = x_j; w) \\ &= \prod_{j=1}^M (\mathcal{M}(x_j, w))^{y_j} (1 - \mathcal{M}(x_j, w))^{1-y_j}. \end{aligned} \quad (117)$$

We take this conditional distribution as our maximum likelihood function $L(w; D)$, where D is our dataset (109). Note that, due to Lemma 3.2, we have

$$L(w; D) = \mathbb{P}(Y_1 = y_1, \dots, Y_M = y_M \mid X_1 = x_1, \dots, X_M = x_M; w) \in (0, 1]. \quad (118)$$

We want to maximize $L(w; D)$ with respect to w , but equivalently, we minimize the following loss function

$$\mathcal{L}(w) = -\frac{1}{M} \ln L(w; D) = -\frac{1}{M} \sum_{j=1}^M \ell_j(w), \quad (119)$$

where

$$\ell_j(w) = y_j \ln(\mathcal{M}(x_j, w)) + (1 - y_j) \ln(1 - \mathcal{M}(x_j, w)). \quad (120)$$

In the literature, the function $\mathcal{L}$ is called *binary cross-entropy*. Consequently, a logistic linear binary classifier is mathematically equivalent to a single sigmoid neuron trained with binary cross-entropy. The artificial neuron's output $\mathcal{M}(x_j, w)$ is interpreted as the probability of belonging to class 1. Thus minimising $\mathcal{L}(w)$ on D is mathematically identical to training that neuron, so any optimisation routine developed in this setting for training applies directly to maximum-likelihood binary classification.

We now investigate some properties of the function (119). From (118) and Lemma 3.2 we have for all $w \in \mathbb{R}^N$ that

$$\mathcal{L}(w) \in [0, +\infty) \quad (121)$$

$$\ell_j(w) = (y_j - 1) \cdot \langle x_j, w \rangle + \ln \left[\varphi(\langle x_j, w \rangle) \right], \quad (122)$$

$$\nabla \ell_j(w) = \left(y_j - \varphi(\langle x_j, w \rangle) \right) x_j, \quad (123)$$

and

$$\nabla^2 \ell_j(w) = -\varphi(\langle x_j, w \rangle)(1 - \varphi(\langle x_j, w \rangle)) x_j x_j^T. \quad (124)$$

Therefore

$$\begin{aligned} \nabla \mathcal{L}(w) &= -\frac{1}{M} \sum_{j=1}^M \nabla \ell_j(w) = \frac{1}{M} \sum_{j=1}^M \left(\varphi(\langle x_j, w \rangle) - y_j \right) x_j \\ &= \frac{1}{M} X B(w), \end{aligned} \quad (125)$$

$$\begin{aligned} \nabla^2 \mathcal{L}(w) &= -\frac{1}{M} \sum_{j=1}^M \nabla^2 \ell_j(w) = \frac{1}{M} \sum_{j=1}^M \varphi(\langle x_j, w \rangle)(1 - \varphi(\langle x_j, w \rangle)) x_j x_j^T \\ &= \frac{1}{M} X D(w) X^T, \end{aligned} \quad (126)$$

where

$$X = [x_1 | \dots | x_M] = \begin{bmatrix} x_1^1 & x_2^1 & \dots & x_M^1 \\ x_1^2 & x_2^2 & \dots & x_M^2 \\ \vdots & \vdots & \vdots & \vdots \\ x_1^N & x_2^N & \dots & x_M^N \end{bmatrix}, \quad (127)$$

$$B(w) = [\varphi(\langle x_1, w \rangle) - y_1, \dots, \varphi(\langle x_M, w \rangle) - y_M]^T \quad (128)$$

$$D(w) = \text{diag} \left(\varphi(\langle x_j, w \rangle)(1 - \varphi(\langle x_j, w \rangle)) \right)_{j=1, \dots, M}. \quad (129)$$

In general, $B = B(w)$ depends on w in a nonlinear way, and there is no closed-form solution for the equation $X B(w) = 0$. Hence, we use gradient descent-type algorithms to minimize the loss (119).

In fact, the situation might be more complicated. Namely, depending on the geometry of the data D , the infimum of the loss (119) might not be attainable. We illustrate such a situation in the case when data are *strictly linearly separable*, i.e., there exists $v \in \mathbb{R}^N$ such that for all $j = 1, 2, \dots, M$

$$t_j \langle x_j, v \rangle > 0, \quad (130)$$

where $t_j := 2y_j - 1 \in \{-1, 1\}$. In what follows, we show that

$$\inf_{w \in \mathbb{R}^N} \mathcal{L}(w) = 0, \quad (131)$$

but there is no $w^* \in \mathbb{R}^N$ for which $\mathcal{L}(w^*) = 0 (= \inf_{w \in \mathbb{R}^N} \mathcal{L}(w))$. Notice first that (119) we can be written as

$$\mathcal{L}(w) = \frac{1}{M} \sum_{j=1}^M \ln \left(1 + e^{-t_j \langle x_j, w \rangle} \right), \quad w \in \mathbb{R}^N, \quad (132)$$

which is left as an exercise. From (132) we have that for all $w \in \mathbb{R}^N$

$$\mathcal{L}(w) > 0, \quad (133)$$

which gives

$$\inf_{w \in \mathbb{R}^N} \mathcal{L}(w) \geq 0. \quad (134)$$

Under the assumption (130)

$$m := \min_{1 \leq j \leq M} t_j \langle x_j, v \rangle > 0. \quad (135)$$

Define the sequence $\{w_k\}_{k \geq 1}$ as

$$w_k = kv, \quad (136)$$

and then for all $j = 1, \dots, M$, $k \geq 1$, it holds

$$t_j \langle x_j, w_k \rangle = kt_j \langle x_j, v \rangle \geq km, \quad (137)$$

which gives

$$\ln(1 + e^{-t_j \langle x_j, w_k \rangle}) \leq \ln(1 + e^{-km}), \quad (138)$$

and by (132)

$$0 < \mathcal{L}(w_k) \leq \ln(1 + e^{-km}). \quad (139)$$

Hence

$$\lim_{k \rightarrow +\infty} \mathcal{L}(w_k) = 0, \quad (140)$$

and, since for all $k \geq 1$

$$\mathcal{L}(w_k) \geq \inf_{w \in \mathbb{R}^N} \mathcal{L}(w) \geq 0, \quad (141)$$

we get (131). Finally, by (133) there is no w^* such that $\mathcal{L}(w^*) = 0$.

4 L^2 -regularization in linear smoothing and its probabilistic interpretation

Recall from Theorem 3.1 that the problem (23) admits at least one solution w^* , which satisfies the normal equations (26). The solution is unique provided the columns of A are linearly independent. This observation leads to the two following questions:

1. Can we slightly modify the loss function (24) (or, equivalently, the normal equations (26)) so that the resulting objective has a unique minimiser?
2. If so, can this modification be given a probabilistic interpretation?

For both questions, there are positive answers. Let us take for $\lambda > 0$

$$\mathcal{L}_\lambda(w) = (\mathcal{L}(w))^2 + \lambda \|w\|^2 = \|Aw - y\|^2 + \lambda \|w\|^2, \quad w \in \mathbb{R}^N. \quad (142)$$

Then

$$\nabla \mathcal{L}_\lambda(w) = 2[(A^T A + \lambda I)w - A^T y], \quad (143)$$

where $I \in \mathbb{R}^{N \times N}$ is the identity matrix. Hence,

$$\nabla \mathcal{L}_\lambda(w) = 0 \Leftrightarrow (A^T A + \lambda I)w = A^T y. \quad (144)$$

For any $\lambda > 0$ the matrix $A^T A + \lambda I \in \mathbb{R}^{N \times N}$ is symmetric and positive definite, since for all $z \in \mathbb{R}^N \setminus \{0\}$

$$\begin{aligned} z^T (A^T A + \lambda I)z &= z^T A^T A z + \lambda z^T z = (Az)^T A z + \lambda z^T z \\ &= \|Az\|^2 + \lambda \|z\|^2 > 0, \end{aligned} \quad (145)$$

even if there exists $z_0 \neq 0$ such that $Az_0 = 0$. Therefore, the matrix $A^T A + \lambda I$ is invertible. Moreover

$$\nabla^2 \mathcal{L}(w) = 2(A^T A + \lambda I), \quad w \in \mathbb{R}^N. \quad (146)$$

Hence, the Hessian $\nabla \mathcal{L}_\lambda$ is positive definite on the whole $\mathbb{R}^N$ and the loss function $\mathcal{L}_\lambda$ is strongly convex. Therefore, for any $\lambda > 0$ regularized system of normal equations (144) has a unique solution w^* which is also the unique minimizer of $\mathcal{L}_\lambda$.

Consider the regularized mean square error (64), defined as

$$MSE_\lambda(w) = \frac{1}{M} \|Aw - y\|^2 + \lambda \|w\|^2 = \frac{1}{M} \mathcal{L}_{\lambda M}(w). \quad (147)$$

Minimising the regularised mean-squared error MSE_λ is equivalent to solving the linear system

$$(A^T A + M\lambda I) w = A^T y,$$

where $\lambda > 0$ is the L^2 -regularisation hyper-parameter. Because the term $M\lambda I$ scales with the dataset size M , one often sets $\lambda \sim 1/M$ or $\lambda \sim 1/\sqrt{M}$, especially when M is moderate. We also have the following heuristics when choosing the regularization parameter:

- a) *Too small* λ - weights w can grow large, increasing the risk of overfitting.
- b) *Too large* λ - weights shrink excessively, which may cause underfitting.

To answer the second question consider the random perturbation matrix $E = [\varepsilon_{ij}]_{i=1,\dots,M,j=1,\dots,N}$, where each $\varepsilon_{ij} : \Omega \rightarrow \mathbb{R}$ is a random variable on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We assume that $\{\varepsilon_{ij}\}_{i=1,\dots,M,j=1,\dots,N}$ are independent and identically distributed random variables, such that $\mathbb{E}(\varepsilon_{ij}) = 0$, and $\mathbb{E}(\varepsilon_{ij})^2 = \sigma^2 < +\infty$. By $\varepsilon^{(j)} = [\varepsilon_{j1}, \dots, \varepsilon_{jN}]^T$ we denote the j th row of the random matrix E . We set

$$\tilde{A} = A + E = [a^{(j)} + \varepsilon^{(j)}]_{j=1,\dots,M}, \quad (148)$$

and then by the properties of the inner product

$$\begin{aligned}\mathbb{E}\|\tilde{A}w - y\|^2 &= \mathbb{E}\|(Aw - y) + Ew\|^2 = \mathbb{E}\left(\langle (Aw - y) + Ew, (Aw - y) + Ew \rangle\right) \\ &= \|Aw - y\|^2 + 2\mathbb{E}\left(\langle Ew, Aw - y \rangle\right) + \mathbb{E}\|Ew\|^2.\end{aligned}\tag{149}$$

We have that

$$\begin{aligned}&\mathbb{E}\left(\langle Ew, Aw - y \rangle\right) \\ &= \mathbb{E}\left(\left[\langle \varepsilon^{(1)}, w \rangle, \dots, \langle \varepsilon^{(M)}, w \rangle\right]^T \cdot \begin{bmatrix} \langle a^{(1)}, w \rangle - y_1 \\ \langle a^{(2)}, w \rangle - y_2 \\ \vdots \\ \langle a^{(M)}, w \rangle - y_M \end{bmatrix}\right) \\ &= \mathbb{E} \sum_{i=1}^M \left[\langle \varepsilon^{(i)}, w \rangle \cdot \left(\langle a^{(i)}, w \rangle - y_i \right) \right] \\ &= \sum_{i=1}^M \left(\langle a^{(i)}, w \rangle - y_i \right) \cdot \mathbb{E}\left(\langle \varepsilon^{(i)}, w \rangle\right) = 0,\end{aligned}\tag{150}$$

since

$$\mathbb{E}\left(\langle \varepsilon^{(i)}, w \rangle\right) = \mathbb{E} \sum_{j=1}^N \varepsilon_{ij} w_j = \sum_{j=1}^N \mathbb{E}(\varepsilon_{ij}) \cdot w_j = 0.\tag{151}$$

Moreover,

$$\begin{aligned}\mathbb{E}\|Ew\|^2 &= \mathbb{E}\left(\left[\langle \varepsilon^{(1)}, w \rangle, \dots, \langle \varepsilon^{(M)}, w \rangle\right]^T \cdot \begin{bmatrix} \langle \varepsilon^{(1)}, w \rangle \\ \langle \varepsilon^{(2)}, w \rangle \\ \vdots \\ \langle \varepsilon^{(M)}, w \rangle \end{bmatrix}\right) \\ &= \sum_{i=1}^M \mathbb{E}\left(\langle \varepsilon^{(i)}, w \rangle\right)^2 = M\sigma^2\|w\|^2,\end{aligned}\tag{152}$$

since by the independence of ε_{ij} and ε_{ik} for $j \neq k$,

$$\begin{aligned}\mathbb{E}\left(\langle \varepsilon^{(i)}, w \rangle\right)^2 &= \mathbb{E}\left(\sum_{j=1}^N \varepsilon_{ij} w_j\right)^2 = \mathbb{E}\left(\sum_{j=1}^N \varepsilon_{ij}^2 w_j^2 + \sum_{j \neq k} \varepsilon_{ij} \varepsilon_{ik} w_j w_k\right) \\ &= \sum_{j=1}^N w_j^2 \cdot \mathbb{E}(\varepsilon_{ij}^2) + \sum_{j \neq k} w_j w_k \mathbb{E}(\varepsilon_{ij}) \mathbb{E}(\varepsilon_{ik}) = \sigma^2\|w\|^2.\end{aligned}\tag{153}$$

Combining (149), (150), and (152) we get

$$\mathbb{E}\left(\frac{1}{M}\|\tilde{A}w - y\|^2\right) = \frac{1}{M}\|Aw - y\|^2 + \sigma^2\|w\|^2 = MSE_{\sigma^2}(w),\tag{154}$$

where MSE_{σ^2} is defined in (147). Hence, when the input data A is randomly perturbed, then the mean squared error without a regularization term, but averaged over the perturbations, equals the L^2 -regularized mean-squared error computed on the unperturbed data A .

Let us consider the sequence $\{E_k\}_{k \geq 1}$ of independent and identically distributed perturbation matrices and set $\tilde{A}_k = A + E_k$. Then, by the strong law of large numbers, we get

$$\frac{1}{K} \sum_{k=1}^K \|\tilde{A}_k w - y\|^2 \rightarrow \mathbb{E}\|\tilde{A}w - y\|^2, \quad K \rightarrow +\infty,\tag{155}$$

with probability one. Fix $K \in \mathbb{N}$ and construct the augmented training set

$$\tilde{D}_K = \{(\tilde{a}_k^{(j)}, \tilde{y}_k^{(j)})\}_{j=1, \dots, M, k=1, \dots, K}, \quad \tilde{a}_k^{(j)} = a^{(j)} + \varepsilon_k^{(j)}, \quad \tilde{y}_k^{(j)} = y_j,$$

where $\varepsilon_k^{(j)}$ is the j th row of E_k . Invoking (154) and (155), training a single neuron μ on $\tilde{D}_K$ with the loss

$$\mathcal{L}_K(w) = \frac{1}{MK} \sum_{k=1}^K \sum_{j=1}^M (\mu(w, 0, \tilde{a}_k^{(j)}) - \tilde{y}_k^{(j)})^2 = \frac{1}{MK} \sum_{k=1}^K \|\tilde{A}_k w - y\|^2$$

is asymptotically equivalent to training the same neuron μ on the original dataset (61) with the regularised mean-squared error (147). The perturbed dataset $\tilde{D}_K$ arises through *data augmentation*, which thus acts as an implicit form of regularisation. This is also another example of *regularization by noise*, see also Section 9.3.4 in [30].

5 Architecture of feedforward artificial neural networks

In this section, we introduce feed-forward deep neural networks, a foundational class of models in modern machine and deep learning.

We will use the following notation:

- (1) M - number of hidden layers,
- (2) L - layer index, with $L = 0, 1, \dots, M + 1$. The layer $L = 0$ is called the *input layer*, whereas the layer $L = M + 1$ is called the *output layer*, and the layers $L = 1, 2, \dots, M$ are called *hidden layers*,
- (3) $\varphi^{(L)}$ - scalar activation function applied in layer L ,
- (4) N_L - number of artificial neurons in layer L ,
- (5) $w_{ij}^{(L)}$ - weight of the connection from neuron i in layer $L - 1$ to neuron j in layer L ,
- (6) $b^{(L)} = [b_1^{(L)}, \dots, b_{N_L}^{(L)}]^T \in \mathbb{R}^{N_L}$ - bias vector in layer L .

We assume that the input data consists of vectors in $\mathbb{R}^d$. Hence, we set $N_0 := d$ and identify the input vector $x = [x_1, \dots, x_d]^T \in \mathbb{R}^d$ with the output vector of the zero (input) layer:

$$a^{(0)} = [a_1^{(0)}, \dots, a_{N_0}^{(0)}]^T = [x_1, \dots, x_d]^T = x. \quad (156)$$

We first describe how the data propagate from the input layer ($L = 0$) to the first hidden layer ($L = 1$) and then, by induction, how they flow from layer $L - 1$ to the subsequent layer L .

The linear part of k th neuron in the layer $L = 1$ is

$$\Sigma_k^{(1)} = \sum_{j=1}^d w_{jk}^{(1)} x_j + b_k^{(1)} = \sum_{j=1}^{N_0} w_{jk}^{(1)} a_j^{(0)} + b_k^{(1)}, \quad k = 1, \dots, N_1, \quad (157)$$

what we can write in the matrix form as

$$\Sigma^{(1)} = W^{(1)} a^{(0)} + b^{(1)}, \quad (158)$$

with the weights matrix for the first layer given as

$$W^{(1)} = \begin{bmatrix} w_{11}^{(1)} & w_{21}^{(1)} & \dots & w_{N_0 1}^{(1)} \\ w_{12}^{(1)} & w_{22}^{(1)} & \dots & w_{N_0 2}^{(1)} \\ \vdots & \vdots & \ddots & \vdots \\ w_{1 N_1}^{(1)} & w_{2 N_1}^{(1)} & \dots & w_{N_0 N_1}^{(1)} \end{bmatrix} \in \mathbb{R}^{N_1 \times N_0}. \quad (159)$$

We define the affine function

$$A^{(1)}(x) = W^{(1)} x + b^{(1)}, \quad (160)$$

where we explicitly showed how the input vector x enters the neural network by the zero layer. We have $\Sigma^{(1)} = A^{(1)}(a^{(0)})$. The output from the first layer is the vector

$$a^{(1)} = [a_1^{(1)}, \dots, a_{N_1}^{(1)}]^T = (\Phi^{(1)} \circ A^{(1)})(a^{(0)}) \in \mathbb{R}^{N_1}, \quad (161)$$

where

$$\Phi^{(1)}(z) = [\varphi^{(1)}(z_1), \dots, \varphi^{(1)}(z_{N_1})]^T, \quad (162)$$

for all $z \in \mathbb{R}^{N_1}$. The output vector $a^{(1)}$ produced by the first layer serves as the input to the second layer, and the same pattern repeats for each subsequent pair of layers. Below we describe the transition from layer $L - 1$ to layer L .

Assume that the output of the $L - 1$ th layer is $a^{(L-1)} \in \mathbb{R}^{N_{L-1}}$. The linear part of k th neuron in the layer L is

$$\Sigma_k^{(L)} = \sum_{j=1}^{N_{L-1}} w_{jk}^{(L)} a_j^{(L-1)} + b_k^{(L)}, \quad k = 1, \dots, N_L, \quad (163)$$

and in the vector form

$$\Sigma^{(L)} = W^{(L)} a^{(L-1)} + b^{(L)} \in \mathbb{R}^{N_L}, \quad (164)$$

with the following weights matrix $W^{(L)}$ for L th layer

$$W^{(L)} = \begin{bmatrix} w_{11}^{(L)} & w_{21}^{(L)} & \dots & w_{N_{L-1}1}^{(L)} \\ w_{12}^{(L)} & w_{22}^{(L)} & \dots & w_{N_{L-1}2}^{(L)} \\ \vdots & \vdots & \ddots & \vdots \\ w_{1N_L}^{(L)} & w_{2N_L}^{(L)} & \dots & w_{N_{L-1}N_L}^{(L)} \end{bmatrix} \in \mathbb{R}^{N_L \times N_{L-1}}. \quad (165)$$

Define the affine function

$$A^{(L)}(z) = W^{(L)} z + b^{(L)} \in \mathbb{R}^{N_L}, \quad (166)$$

for all $z \in \mathbb{R}^{N_{L-1}}$. We have $\Sigma^{(L)} = A^{(L)}(a^{(L-1)})$. Then the output from the L th layer is the vector

$$a^{(L)} = [a_1^{(L)}, \dots, a_{N_L}^{(L)}]^T = (\Phi^{(L)} \circ A^{(L)})(a^{(L-1)}) \in \mathbb{R}^{N_L}, \quad (167)$$

where

$$\Phi^{(L)}(s) = [\varphi^{(L)}(s_1), \dots, \varphi^{(L)}(s_{N_L})]^T, \quad (168)$$

for all $s \in \mathbb{R}^{N_L}$. We proceed in the way described above for $L = 1, 2, \dots, M + 1$. We define the *neural complexity* as the number of trainable parameters, i.e.:

$$\mathbf{n}_{\text{comp}} = \sum_{L=1}^{M+1} N_L N_{L-1} + \sum_{L=1}^{M+1} N_L, \quad (169)$$

which is the dimension of the weights vector

$$w = [W^{(1)} | \dots | W^{(M+1)} | b^{(1)} | \dots | b^{(M+1)}]. \quad (170)$$

Defining

$$\Psi^{(L)} = \Phi^{(L)} \circ A^{(L)}, \quad (171)$$

by the feedforward neural network, we mean a function

$$\mathcal{N}^F : \mathbb{R}^{\mathbf{n}_{\text{comp}}} \times \mathbb{R}^{N_0} \rightarrow \mathbb{R}^{N_{M+1}}, \quad (172)$$

where

$$\mathcal{N}^F(w, x) = \left(\Psi^{(M+1)} \circ \Psi^{(M)} \circ \dots \circ \Psi^{(1)} \right)(x). \quad (173)$$

The evaluation of $\mathcal{N}^F(w, x)$ for fixed weights w and an input x is called a *forward pass*. The network $\mathcal{N}^F$ is called a *deep neural network* when it contains at least two hidden layers ($M \geq 2$).

The next proposition highlights that the regularity of the neural network $\mathcal{N}^F(w, x)$ as a function of the pair (w, x) is entirely determined by the smoothness of the activation functions $\varphi^{(L)}$ used in its construction.

Proposition 5.1. (i) *If for all $L = 1, \dots, M + 1$ the activation functions $\varphi^{(L)}$ are Borel measurable, then the neural network $\mathcal{N}^F$ is Borel measurable.*

(ii) If for all $L = 1, \dots, M + 1$ the activation functions $\varphi^{(L)}$ are in C^r for some $r \geq 0$, then the neural network $\mathcal{N}^F$ is in C^r .

(iii) If for all $L = 1, \dots, M + 1$ there exist $K_L \in [0, +\infty)$ such that for all $x_1, x_2 \in \mathbb{R}$

$$|\varphi^{(L)}(x_1) - \varphi^{(L)}(x_2)| \leq K_L |x_1 - x_2|, \quad (174)$$

then for all $w \in \mathbb{R}^{\mathbf{n}_{\text{comp}}}$, $x_1, x_2 \in \mathbb{R}^{N_0}$ it holds

$$\|\mathcal{N}^F(w, x_1) - \mathcal{N}^F(w, x_2)\| \leq \left(\prod_{L=1}^{M+1} K_L \cdot \|W^{(L)}\| \right) \|x_1 - x_2\|. \quad (175)$$

Proof. We give a proof only for the (iii) case, since the proofs of (i) and (ii) are straightforward. For $L = 1, \dots, M + 1$, $z_1, z_2 \in \mathbb{R}^{N_{L-1}}$ we have

$$\|A^{(L)}(z_1) - A^{(L)}(z_2)\| = \|W^{(L)}(z_1 - z_2)\| \leq \|W^{(L)}\| \cdot \|z_1 - z_2\|, \quad (176)$$

and for all $s, t \in \mathbb{R}^{N_L}$

$$\|\Phi^{(L)}(s) - \Phi^{(L)}(t)\|^2 = \sum_{j=1}^{N_L} |\varphi^{(L)}(s_j) - \varphi^{(L)}(t_j)|^2 \leq K_L^2 \|s - t\|^2. \quad (177)$$

Therefore, for $L = 1, \dots, M + 1$, $x_1, x_2 \in \mathbb{R}^{N_{L-1}}$

$$\begin{aligned} \|\Psi^{(L)}(x_1) - \Psi^{(L)}(x_2)\| &= \|\Phi^{(L)}(A^{(L)}(x_1)) - \Phi^{(L)}(A^{(L)}(x_2))\| \\ &\leq K_L \|A^{(L)}(x_1) - A^{(L)}(x_2)\| \leq K_L \cdot \|W^{(L)}\| \cdot \|x_1 - x_2\|, \end{aligned} \quad (178)$$

and this implies that

$$\begin{aligned} &\|\mathcal{N}^F(w, x_1) - \mathcal{N}^F(w, x_2)\| \\ &= \left\| \left(\Psi^{(M+1)} \circ \Psi^{(M)} \circ \dots \circ \Psi^{(1)} \right)(x_1) - \left(\Psi^{(M+1)} \circ \Psi^{(M)} \circ \dots \circ \Psi^{(1)} \right)(x_2) \right\| \\ &\leq K_{M+1} \|W^{(M+1)}\| \left\| \left(\Psi^{(M)} \circ \Psi^{(M-1)} \circ \dots \circ \Psi^{(1)} \right)(x_1) \right. \\ &\quad \left. - \left(\Psi^{(M)} \circ \Psi^{(M-1)} \circ \dots \circ \Psi^{(1)} \right)(x_2) \right\| \\ &\leq \dots \leq \left(\prod_{L=1}^{M+1} K_L \|W^{(L)}\| \right) \|x_1 - x_2\|, \end{aligned} \quad (179)$$

which ends the proof. $\square$

If for some L the activation $\varphi^{(L)}$ is the ReLU function, then $\mathcal{N}^F$ might not be differentiable. This might cause additional difficulties when using gradient descent-like optimization algorithms.

After specifying the architecture of the neural network $\mathcal{N}^F$ —namely, the number of hidden layers M , the neuron counts N_L in each layer, and the activation functions $\varphi^{(L)}$ —the next crucial step is to tune the weight parameters using the training data and a loss function suited to the task to be solved. Consider the training dataset

$$D = \{(x_1, y_1), \dots, (x_K, y_K)\}, \quad (180)$$

where $x_j \in \mathbb{R}^d$ and $y_j \in \mathbb{R}^m$. The network $\mathcal{N}^F$ is configured so that the input layer ($L = 0$) matches the dimension of the feature vectors x_j , while the output layer ($L = M + 1$) matches the dimension of the labels (ground truth) y_j . Thus we set $N_0 := d$ and $N_{M+1} := m$. For a general regression task, we aim to minimise the mean-squared error

$$\text{MSE}(w) = \frac{1}{K} \sum_{j=1}^K \|y_j - \mathcal{N}^F(w, x_j)\|^2, \quad (181)$$

where w collects all trainable weights. Because this objective is typically high-dimensional and strongly nonlinear, an analytic solution is unavailable. Consequently, efficient iterative schemes are generally used: (stochastic) gradient descent and its numerous variants. Moreover, since the Lipschitz constant of a feed-forward neural network grows with the magnitude of its weights (see Proposition 5.1), it is crucial, while minimizing the loss function, to keep those weights as small as possible. Different regularization strategies, such as LASSO, L^2 , or probabilistic dropout methods, can be employed to constrain weight growth. For example, the L^2 -regularization approach applied to (181) results in the following loss function to be minimized

$$\text{MSE}_\lambda(w) = \frac{1}{K} \sum_{j=1}^K \|y_j - \mathcal{N}^F(w, x_j)\|^2 + \lambda \|w\|^2, \quad (182)$$

where $\lambda > 0$.

Feedforward neural networks are the most basic and foundational architecture in the field of deep learning. Many advanced architectures, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and even transformer-based models, are build on the principles of feedforward networks.

5.1 Autoencoders

Having described the architecture of feedforward neural networks, it is quite straightforward to define the *autoencoders*. This is a special case of the feedforward neural network $\mathcal{N}^F$, defined in (173), for which we take the following numbers of neurons in the respective layers

$$d = N_0 \geq N_1 \geq \dots \geq N_D \leq N_{D+1} \leq \dots \leq N_{M+1} = d, \quad (183)$$

where $D \in \{1, \dots, M\}$ is the fixed number that corresponds to the so-called *latent layer* or *bottleneck*. Moreover, the layers up to the latent layer are called the *encoder*, while the remaining layers are referred to as the *decoder*.

The main ingredient that makes autoencoders extremely interesting and useful is the method of their training. Namely, the training data set is of the form

$$D = \{(x_1, x_1), \dots, (x_K, x_K)\}, \quad (184)$$

where $x_j \in \mathbb{R}^d$ and the labels are the same as the input data. The loss function to be minimized is as follows

$$\mathcal{L}(w) = \frac{1}{K} \sum_{j=1}^K \|x_j - \mathcal{N}^F(w, x_j)\|^2, \quad (185)$$

and, due to the form of the training set, it can be interpreted as the reconstruction error. Hence, during training, we compress the input data (of dimension d) into a much smaller latent representation of size N_D , and then attempt to decode it back to the original dimension d .

Autoencoders have found significant applications in anomaly detection, since anomalous inputs generally yield higher reconstruction errors compared to regular data. In practice, an autoencoder is trained on normal data so that it learns to compress and reconstruct only the typical patterns present in the training set. When an anomalous sample is encountered, the model fails to reconstruct it accurately, resulting in a higher reconstruction error, which can then be used as an anomaly score. Such methods are widely applied in *cybersecurity*, for example, in detecting suspicious transactions or unusual user activities that deviate from normal behavior, see [44], [24].

Remark 5.1. In the case when $M = 1$ (we have only one hidden layer) and all activation functions are linear we obtain the so-called *linear autoencoder*. The linear autoencoder is closely related (but not equivalent) to PCA, see [29], [13].

5.2 Backpropagation procedure

Assume that the feedforward neural network $\mathcal{N}^F$ is equipped with a loss function $\mathcal{L}$. While training i.e., minimizing $\mathcal{L}$ with respect to weights and biases, at each step we need to compute the gradients $\nabla \mathcal{L}$. The procedure of computing this gradient efficiently is the *backpropagation algorithm*, see [33]. From the mathematical point of view, the core idea of backpropagation is the careful application of the chain rule (8) to the gradient of the loss function $\mathcal{L}$.

Since $\mathcal{L} = \mathcal{L}(a^{(M+1)}, w)$, where is given by (170), we obtain

$$\frac{\partial \mathcal{L}}{\partial W^{(L)}} = \frac{\partial \mathcal{L}}{\partial a^{(M+1)}} \frac{\partial a^{(M+1)}}{\partial a^{(M)}} \cdots \frac{\partial a^{(L+2)}}{\partial a^{(L+1)}} \frac{\partial a^{(L+1)}}{\partial a^{(L)}} \frac{\partial a^{(L)}}{\partial W^{(L)}}, \quad (186)$$

where all involved partial derivatives are suitable Jacobian matrices. (Note that $\frac{\partial a^{(L)}}{\partial w^{(L)}}$ is a $\mathbb{R}^{N_L \times N_L \times N_{L-1}}$ tensor.) In principle, we might compute all the partial derivatives above for $L = 1, 2, \dots, M+1$. However, this would be computationally inefficient, since many of the same partial derivatives would be computed multiple times. Therefore, we will proceed recursively, computing derivatives starting from the output of the neural network and moving backward toward the input.

Denoting by

$$\delta^{(L+1)} = \frac{\partial \mathcal{L}}{\partial a^{(L+1)}} = \frac{\partial \mathcal{L}}{\partial a^{(M+1)}} \frac{\partial a^{(M+1)}}{\partial a^{(M)}} \cdots \frac{\partial a^{(L+2)}}{\partial a^{(L+1)}}, \quad (187)$$

we obtain the following formula

$$\frac{\partial \mathcal{L}}{\partial W^{(L)}} = \delta^{(L+1)} \cdot \frac{\partial a^{(L+1)}}{\partial W^{(L)}} \quad (188)$$

for $L = 1, 2, \dots, M$, since

$$\frac{\partial a^{(L+1)}}{\partial W^{(L)}} = \frac{\partial a^{(L+1)}}{\partial a^{(L)}} \frac{\partial a^{(L)}}{\partial W^{(L)}}. \quad (189)$$

Again, from the chain rule, we have the following recursive formula

$$\delta^{(L)} = \frac{\partial \mathcal{L}}{\partial a^{(L)}} = \frac{\partial \mathcal{L}}{\partial a^{(L+1)}} \frac{\partial a^{(L+1)}}{\partial a^{(L)}} = \delta^{(L+1)} \frac{\partial a^{(L+1)}}{\partial a^{(L)}} \quad (190)$$

for $L = 1, 2, \dots, M$ and we start by computing $\delta^{(M+1)}$. (This derivative depends on the form of the loss function $\mathcal{L}$.) Since $a^{(L+1)} = \Phi^{(L+1)}(A^{(L+1)}(a^{(L)}))$, we get by the chain rule

$$\frac{\partial a^{(L+1)}}{\partial a^{(L)}} = \frac{\partial \Phi^{(L+1)}}{\partial s}(s) \Big|_{s=A^{(L+1)}(a^{(L)})} \cdot \frac{\partial A^{(L+1)}}{\partial a^{(L)}}, \quad (191)$$

where

$$\frac{\partial A^{(L+1)}}{\partial a^{(L)}} = W^{(L+1)}, \quad (192)$$

$$\frac{\partial \Phi^{(L+1)}}{\partial s}(s_1, \dots, s_{N_{L+1}}) = \text{diag}((\varphi^{(L+1)})'(s_i) \mid i = 1, 2, \dots, N_{L+1}). \quad (193)$$

Moreover,

$$\frac{\partial \mathcal{L}}{\partial W^{(L)}} = \delta^{(L)} \cdot \frac{\partial a^{(L)}}{\partial W^{(L)}}, \quad (194)$$

and

$$\frac{\partial a^{(L)}}{\partial W^{(L)}} = \frac{\partial \Phi^{(L)}}{\partial s}(s) \Big|_{s=A^{(L)}(a^{(L-1)})} \cdot \frac{\partial A^{(L)}}{\partial W^{(L)}}, \quad (195)$$

where

$$\frac{\partial \Phi^{(L)}}{\partial s}(s_1, \dots, s_{N_L}) = \text{diag}((\varphi^{(L)})'(s_i) \mid i = 1, 2, \dots, N_L), \quad (196)$$

$$\frac{\partial A^{(L)}}{\partial W^{(L)}} = \left[\frac{\partial A_i^{(L)}}{\partial W_{kl}^{(L)}} \right]_{k,i=1,\dots,m,l=1,\dots,n} = \left[\delta_{ik} a_l^{(L-1)} \right]_{k,i=1,\dots,m,l=1,\dots,n} \quad (197)$$

is a tensor of degree 3. Analogously,

$$\frac{\partial \mathcal{L}}{\partial b^{(L)}} = \delta^{(L)} \cdot \frac{\partial a^{(L)}}{\partial b^{(L)}}, \quad (198)$$

where

$$\frac{\partial a^{(L)}}{\partial b^{(L)}} = \frac{\partial \Phi^{(L)}}{\partial s}(s) \Big|_{s=A^{(L)}(a^{(L-1)})} \cdot \frac{\partial A^{(L)}}{\partial b^{(L)}}, \quad (199)$$

and $\frac{\partial \Phi^{(L)}}{\partial s}(s) \Big|_{s=A^{(L)}(a^{(L-1)})}$ is given in (196), while

$$\frac{\partial A^{(L)}}{\partial b^{(L)}} = Id \in \mathbb{R}^{N_L \times N_L}. \quad (200)$$

We will summarize the backpropagation procedure as follows.

1) Compute for $L = 1, 2, \dots, M + 1$

$$\begin{aligned} & \frac{\partial \Phi^{(L)}}{\partial s}(s_1, \dots, s_{N_L}) \Big|_{s=A^{(L)}(a^{(L-1)})} \\ &= \text{diag}((\varphi^{(L)})'(s_i) \mid i = 1, 2, \dots, N_L) \Big|_{s=A^{(L)}(a^{(L-1)})}. \end{aligned} \quad (201)$$

2) Using derivatives computed in 1), compute for $L = 1, 2, \dots, M$

$$\frac{\partial a^{(L+1)}}{\partial a^{(L)}} = \frac{\partial \Phi^{(L+1)}}{\partial s}(s) \Big|_{s=A^{(L+1)}(a^{(L)})} \cdot W^{(L+1)}. \quad (202)$$

3) (*the recursion*) Compute

$$\delta^{(M+1)} = \frac{\partial \mathcal{L}}{\partial a^{(M+1)}}, \quad (203)$$

and then for $L = M, M - 1, \dots, 1$ compute recursively

$$\delta^{(L)} = \delta^{(L+1)} \frac{\partial a^{(L+1)}}{\partial a^{(L)}}. \quad (204)$$

Having $\delta^{(L)}$, $L = 1, 2, \dots, M + 1$ we compute $\frac{\partial \mathcal{L}}{\partial W^{(L)}}$, $\frac{\partial \mathcal{L}}{\partial b^{(L)}}$ for $L = 1, 2, \dots, M + 1$ using (194), (198).

5.3 Residual connections

Standard feedforward neural networks $\mathcal{N}^F$ suffer from *vanishing* or *exploding gradients*, which makes the learning procedure unstable or ineffective as depth M increases (see Exercise 16). One of the remedies proposed to solve this problem was the introduction of *residual connections*, see [30]. Residual connections are additional links in a neural network that skip over one or more layers and add the input directly to the output. Mathematical formulation, in the case of skipping one layer, is as follows

$$\begin{aligned} a^{(0)} &= x, \\ a^{(L+1)} &= a^{(L)} + \Psi^{(L+1)}(a^{(L)}) = \mathcal{F}^{(L+1)}(a^{(L)}), \quad L = 0, 1, \dots, M. \end{aligned} \quad (205)$$

where

$$\mathcal{F}^{(L)} = Id + \Psi^{(L)}, \quad (206)$$

and where we assume that $N_L = N_{L+1}$. The feedforward neural network with residual connections is now defined by

$$\mathcal{N}^{F-res} = \mathcal{F}^{(M+1)} \circ \mathcal{F}^{(M)} \circ \dots \circ \mathcal{F}^{(1)}. \quad (207)$$

When we perform the backpropagation algorithm in the case of the residual network $\mathcal{N}^{F-res}$, we compute recursively $\delta^{(L)}$ as in (190), where this time

$$\frac{\partial a^{(L+1)}}{\partial a^{(L)}} = Id + \text{diag}((\varphi^{(L+1)})'(s_i) \mid i = 1, 2, \dots, N_{L+1}) \Big|_{s=A^{(L+1)}(a^{(L)})} \cdot W^{(L+1)},$$

and the additional Id term helps prevent $\delta^{(L)}$ from vanishing.

Residual connections are mainly applied in computer vision, speech recognition, time-series analysis, reinforcement learning, etc.

Remark 5.2. There is a deep connection between residual networks and ODEs. Namely, the forward pass (205) can be viewed as a step of the Euler scheme applied to a suitable ODE, where the step size is equal to 1. Such an ODE is called the Neural ODE, and its derivation and application to training of neural networks are described in [8], [16].

6 Gradient descent algorithm and its convergence

6.1 Definition of the GD algorithm

Let $\mathcal{L}: \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable function. In machine and deep learning, $\mathcal{L}$ is typically the loss function, which must be minimized with respect to the trainable parameters $w \in \mathbb{R}^n$ (the network weights). From a mathematical perspective, this leads to the optimization problem

$$\inf_{w \in \mathbb{R}^n} \mathcal{L}(w). \quad (208)$$

A classical numerical scheme for solving (208) is gradient descent, whose definition we now recall, see [37].

Fix $w^0 \in \mathbb{R}^n$, $\eta \in (0, +\infty)$ and set

$$w^{k+1} = w^k - \eta \nabla \mathcal{L}(w^k) \quad (209)$$

for $k = 0, 1, 2, \dots$

Below we state the first result on the convergence property of the GD algorithm.

Proposition 6.1. *Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$.*

(i) *If the sequence $\{w^k\}_{k \geq 0}$, given by (209), converges then $\nabla \mathcal{L}(w^k) \rightarrow 0$ as $k \rightarrow +\infty$ and $\{w^k\}_{k \geq 0}$ converges to a critical point of $\mathcal{L}$.*

(ii) *If $\sum_{k=0}^{+\infty} \|\nabla \mathcal{L}(w^k)\| < +\infty$ then the sequence $\{w^k\}_{k \geq 0}$, given by (209), converges to the critical point of $\mathcal{L}$.*

Proof. Ad. (i). Let us assume that there exists $g \in \mathbb{R}^n$ such that $\lim_{k \rightarrow +\infty} w^k = g$. Then for all $k \geq 0$

$$0 \leq \eta \cdot \|\nabla \mathcal{L}(w^k)\| = \|w^{k+1} - w^k\| \leq \|w^{k+1} - g\| + \|w^k - g\|, \quad (210)$$

which implies that $\|\nabla \mathcal{L}(w^k)\| \rightarrow 0$ as $k \rightarrow +\infty$. Moreover, by the continuity of the gradient $\nabla \mathcal{L}$, we get

$$\nabla \mathcal{L}(g) = \nabla \mathcal{L}(\lim_{k \rightarrow +\infty} w^k) = \lim_{k \rightarrow +\infty} \nabla \mathcal{L}(w^k) = 0, \quad (211)$$

and therefore, g is the critical point of $\mathcal{L}$.

Ad. (ii). For all $M \geq 1$ we have

$$w^M - w^0 = \sum_{k=0}^{M-1} (w^{k+1} - w^k) = -\eta \cdot \sum_{k=0}^{M-1} \nabla \mathcal{L}(w^k). \quad (212)$$

Due to the assumption $\sum_{k=0}^{+\infty} \|\nabla \mathcal{L}(w^k)\| < +\infty$ and the fact that $(\mathbb{R}^n, \|\cdot\|)$ is Banach space, the sequence

$\sum_{k=0}^{+\infty} \nabla \mathcal{L}(w^k)$ is absolutely convergent and thus convergent in $\mathbb{R}^n$. Hence, we have

$$g := \lim_{k \rightarrow +\infty} w^k = w^0 - \eta \sum_{k=0}^{+\infty} \nabla \mathcal{L}(w^k) \in \mathbb{R}^n, \quad (213)$$

so the sequence $\{w^k\}_{k \geq 0}$ is convergent. Therefore, by (i) we get that it converges to the critical point of $\mathcal{L}$. $\square$

6.2 Convergence theorem for L -smooth functions

In the theorem below, we assume only that the gradient of $\mathcal{L}$ is Lipschitz continuous. We aim to understand what, if anything, can be said about the convergence of the gradient descent algorithm when no convexity assumptions are imposed.

Theorem 6.1. *Let us assume that the function $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ is L -smooth for some $L \in (0, +\infty)$. Then for any $\eta \in (0, 2/L)$, $w^0 \in \mathbb{R}^n$ we have:*

$$(A) \sum_{k=0}^{+\infty} \|\nabla \mathcal{L}(w^k)\|^2 = +\infty \Leftrightarrow \lim_{k \rightarrow +\infty} \mathcal{L}(w^k) = -\infty,$$

$$(B) \sum_{k=0}^{+\infty} \|\nabla \mathcal{L}(w^k)\|^2 < +\infty \Leftrightarrow \exists \mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)] : \forall k \geq 0, \mathcal{L}(w^k) \geq \mathcal{L}^*,$$

(C) either

$$(C1) \lim_{k \rightarrow +\infty} \mathcal{L}(w^k) = -\infty$$

or

$$(C2) \exists \mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)] : \forall k \geq 0, \mathcal{L}(w^k) \geq \mathcal{L}^*.$$

In the (C2)-case we additionally have:

– $\lim_{k \rightarrow +\infty} \nabla \mathcal{L}(w^k) = 0$, the sequence $\{\mathcal{L}(w^k)_{k \geq 0}\}$ is convergent to a real number, and every accumulation point of $\{w_k\}_{k \geq 0}$ is also a critical points of $\mathcal{L}$, and

–

$$\exists C^* \in [0, +\infty) : \forall M \geq 1 \min_{0 \leq k \leq M-1} \|\nabla \mathcal{L}(w^k)\| \leq \frac{C^*}{\sqrt{M}}. \quad (214)$$

Proof. From (611) we have for all $k \geq 0$ that

$$\begin{aligned} & \langle \nabla \mathcal{L}(w^k), w^{k+1} - w^k \rangle - \frac{L}{2} \|w^{k+1} - w^k\|^2 \\ & \leq \mathcal{L}(w^{k+1}) - \mathcal{L}(w^k) \\ & \leq \langle \nabla \mathcal{L}(w^k), w^{k+1} - w^k \rangle + \frac{L}{2} \|w^{k+1} - w^k\|^2. \end{aligned} \quad (215)$$

Since $w^{k+1} - w^k = -\eta \nabla \mathcal{L}(w^k)$, we have from (215) that for all $k \geq 0$

$$(-\beta) \|\nabla \mathcal{L}(w^k)\|^2 \leq \mathcal{L}(w^{k+1}) - \mathcal{L}(w^k) \leq (-\alpha) \|\nabla \mathcal{L}(w^k)\|^2, \quad (216)$$

with

$$\begin{aligned} \alpha &= \eta \left(1 - \frac{L\eta}{2}\right), \\ \beta &= \eta \left(1 + \frac{L\eta}{2}\right). \end{aligned} \quad (217)$$

We assume that $\eta \in (0, 2/L)$, which implies that

$$0 < \alpha < \eta < \beta. \quad (218)$$

From (216) and (266) we have for all $k \geq 0$

$$\mathcal{L}(w^{k+1}) - \mathcal{L}(w^k) \leq (-\alpha) \|\nabla \mathcal{L}(w^k)\|^2 \leq 0, \quad (219)$$

hence, the sequence $\{\mathcal{L}(w^k)\}_{k \geq 0}$ is **non-increasing**.

For all $M \geq 1$ we get, by summing the terms in (216) for $k = 0, 1, \dots, M-1$, that

$$(-\beta) \sum_{k=0}^{M-1} \|\nabla \mathcal{L}(w^k)\|^2 \leq \sum_{k=0}^{M-1} (\mathcal{L}(w^{k+1}) - \mathcal{L}(w^k)) \leq (-\alpha) \sum_{k=0}^{M-1} \|\nabla \mathcal{L}(w^k)\|^2, \quad (220)$$

and, due to the fact that

$$\sum_{k=0}^{M-1} (\mathcal{L}(w^{k+1}) - \mathcal{L}(w^k)) = \mathcal{L}(w^M) - \mathcal{L}(w^0), \quad (221)$$

we get for all $M \geq 1$ that

$$\mathcal{L}(w^0) - \beta \sum_{k=0}^{M-1} \|\nabla \mathcal{L}(w^k)\|^2 \leq \mathcal{L}(w^M) \leq \mathcal{L}(w^0) - \alpha \sum_{k=0}^{M-1} \|\nabla \mathcal{L}(w^k)\|^2, \quad (222)$$

and, equivalently,

$$\frac{\mathcal{L}(w^0) - \mathcal{L}(w^M)}{\beta} \leq \sum_{k=0}^{M-1} \|\nabla \mathcal{L}(w^k)\|^2 \leq \frac{\mathcal{L}(w^0) - \mathcal{L}(w^M)}{\alpha}, \quad (223)$$

since $\beta > \alpha > 0$. Directly from (222) and (223) we get the equivalence (A).

We now justify the equivalence (B).

($\Rightarrow$) Let us assume that $S_\infty := \sum_{k=0}^{+\infty} \|\nabla \mathcal{L}(w^k)\|^2 < +\infty$. Then, for all $M \geq 1$ we have

$$\sum_{k=0}^{M-1} \|\nabla \mathcal{L}(w^k)\|^2 \leq S_\infty < +\infty, \quad (224)$$

and by (266), (222) we get for all $M \geq 1$

$$-\infty < \mathcal{L}(w^0) - \beta S_\infty \leq \mathcal{L}(w^0) - \beta \sum_{k=0}^{M-1} \|\nabla \mathcal{L}(w^k)\|^2 \leq \mathcal{L}(w^M), \quad (225)$$

and $\mathcal{L}(w^M) \leq \mathcal{L}(w^0)$, since $\{\mathcal{L}(w^k)\}_{k \geq 1}$ is non-increasing. This gives for all $M \geq 1$

$$\mathcal{L}(w^0) \geq \mathcal{L}(w^M) \geq \mathcal{L}^*, \quad (226)$$

where

$$\mathcal{L}^* = \mathcal{L}(w^0) - \beta S_\infty \in (-\infty, \mathcal{L}(w^0)]. \quad (227)$$

($\Leftarrow$) Let us assume that $\exists \mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)] : \forall k \geq 0, \mathcal{L}(w^k) \geq \mathcal{L}^*$. Then by (223) we have for all $M \geq 1$

$$\sum_{k=0}^{M-1} \|\nabla \mathcal{L}(w^k)\|^2 \leq \frac{\mathcal{L}(w^0) - \mathcal{L}(w^M)}{\alpha} \leq \frac{\mathcal{L}(w^0) - \mathcal{L}^*}{\alpha} < +\infty, \quad (228)$$

which implies that the sequence $\left\{ \sum_{k=0}^{M-1} \|\nabla \mathcal{L}(w^k)\|^2 \right\}_{M \geq 1}$ is non-decreasing and bounded-above. Therefore, it is convergent and

$$\sum_{k=0}^{+\infty} \|\nabla \mathcal{L}(w^k)\|^2 \leq \frac{\mathcal{L}(w^0) - \mathcal{L}^*}{\alpha} < +\infty. \quad (229)$$

Hence, we have shown the equivalence (B).

Finally, we prove the statement (C). Note that only the following two mutually exclusive cases are possible:

$$\sum_{k=0}^{+\infty} \|\nabla \mathcal{L}(w^k)\|^2 = +\infty, \quad (230)$$

$$\sum_{k=0}^{+\infty} \|\nabla \mathcal{L}(w^k)\|^2 < +\infty. \quad (231)$$

Hence, by (A) and (B) either

$$(C1) \quad \lim_{k \rightarrow +\infty} \mathcal{L}(w^k) = -\infty, \quad (232)$$

or

$$(C2) \quad \exists \mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)] : \forall_{k \geq 0} \mathcal{L}(w^k) \geq \mathcal{L}^*. \quad (233)$$

In the (C2)-case, by the fact that the sequence $\{\mathcal{L}(w^k)\}_{k \geq 0}$ is non-increasing, we get that it is convergent and

$$\mathcal{L}(w^0) \geq \lim_{k \rightarrow +\infty} \mathcal{L}(w^k) = \inf_{k \geq 0} \mathcal{L}(w^k) \geq \mathcal{L}^*. \quad (234)$$

Moreover, if (C2) holds, then by (B) we have that $\sum_{k=0}^{+\infty} \|\nabla \mathcal{L}(w^k)\|^2 < +\infty$, and, by the necessary condition for convergence of the series, we get

$$\lim_{k \rightarrow +\infty} \|\nabla \mathcal{L}(w^k)\|^2 = 0, \quad (235)$$

which implies that $\nabla \mathcal{L}(w^k) \rightarrow 0 \in \mathbb{R}^n$ as $k \rightarrow +\infty$. Now, let $\bar{w} \in \mathbb{R}^n$ be any accumulation points of the sequence $\{w^k\}_{k \geq 0}$. Then there exists a subsequence $\{w^{n_k}\}_{k \geq 0}$ such that

$$\lim_{k \rightarrow +\infty} w^{n_k} = \bar{w}. \quad (236)$$

Since $\{\nabla \mathcal{L}(w^k)\}_{k \geq 0}$ converges to 0, any subsequence of this sequence is also convergent to 0, so, in particular,

$$\lim_{k \rightarrow +\infty} \nabla \mathcal{L}(w^{n_k}) = 0. \quad (237)$$

By (237) and the fact that the gradient $\nabla \mathcal{L}$ is continuous on $\mathbb{R}^n$ we have

$$\nabla \mathcal{L}(\bar{w}) = \nabla \mathcal{L}(\lim_{k \rightarrow +\infty} w^{n_k}) = \lim_{k \rightarrow +\infty} \nabla \mathcal{L}(w^{n_k}) = 0, \quad (238)$$

so $\bar{w}$ is a critical point of $\mathcal{L}$. Finally, if (C2) holds then by (223) we have for all $M \geq 1$

$$\begin{aligned} M \min_{0 \leq k \leq M-1} \|\nabla \mathcal{L}(w^k)\|^2 &\leq \sum_{k=0}^{M-1} \|\nabla \mathcal{L}(w^k)\|^2 \leq \frac{\mathcal{L}(w^0) - \mathcal{L}(w^M)}{\alpha} \\ &\leq \frac{\mathcal{L}(w^0) - \mathcal{L}^*}{\alpha} < +\infty, \end{aligned} \quad (239)$$

which gives the estimate (214). $\square$

6.3 Convergence theorem for L -smooth and convex functions

Assuming that $\mathcal{L}$ is also convex, we can establish that the sequence $\{\mathcal{L}(w^k)\}_{k \geq 0}$ converges to the global minimum value of $\mathcal{L}$.

Theorem 6.2. *Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ be convex and L -smooth for some $L \in (0, +\infty)$. We assume that the set of all global minimums of $\mathcal{L}$ is non-empty, i.e. $\{w^* \in \mathbb{R}^n \mid \mathcal{L}(w^*) = \mathcal{L}^*\} \neq \emptyset$, where $\mathcal{L}^* := \inf_{y \in \mathbb{R}^n} \mathcal{L}(y) > -\infty$.*

Then for all $\eta \in (0, 1/L]$, $w^0 \in \mathbb{R}^n$ it holds:

(1) there exists $c_1 \in (0, +\infty)$ such that for all $k \geq 1$

$$0 \leq \mathcal{L}(w^k) - \mathcal{L}^* \leq \frac{c_1}{k}, \quad (240)$$

and

$$0 \leq \mathcal{L}(\bar{w}_k) - \mathcal{L}^* \leq \frac{c_1}{k}, \quad (241)$$

$$\text{where } \bar{w}_k = \frac{1}{k} \sum_{j=1}^k w^j,$$

(2) there exists $c_2 \in (0, +\infty)$ such that for all $k \geq 1$

$$\|\nabla \mathcal{L}(w^k)\| \leq \frac{c_2}{\sqrt{k}}. \quad (242)$$

Proof. By (216) we get for all $k \geq 0$

$$\mathcal{L}(w^{k+1}) \leq \mathcal{L}(w^k) - \alpha \|\nabla \mathcal{L}(w^k)\|^2, \quad (243)$$

where $\alpha = \eta(1 - \frac{L\eta}{2}) \in [\eta/2, \eta)$, since $\eta \in (0, 1/L]$. Hence, the sequence $\{\mathcal{L}(w^k)\}_{k \geq 0}$ is **non-increasing**. Now, let $w^* \in \mathbb{R}^n$ be a global minimum of $\mathcal{L}$. Then, by Lemma 13.5 (2) we get for all $k \geq 0$ that

$$\mathcal{L}(w^k) \leq \mathcal{L}(w^*) - \langle w^* - w^k, \nabla \mathcal{L}(w^k) \rangle. \quad (244)$$

From (243), (244) we have for all $k \geq 0$

$$\mathcal{L}(w^{k+1}) \leq \mathcal{L}(w^*) - \langle w^* - w^k, \nabla \mathcal{L}(w^k) \rangle - \alpha \|\nabla \mathcal{L}(w^k)\|^2. \quad (245)$$

Since, by Lemma 13.7 and (209)

$$\begin{aligned} -\langle w^* - w^k, \eta \nabla \mathcal{L}(w^k) \rangle &= \frac{1}{2} \|w^* - w^k\|^2 + \frac{\eta^2}{2} \|\nabla \mathcal{L}(w^k)\|^2 \\ &\quad - \frac{1}{2} \|w^* - (w^k - \eta \nabla \mathcal{L}(w^k))\|^2 \\ &= \frac{1}{2} \|w^* - w^k\|^2 + \frac{\eta^2}{2} \|\nabla \mathcal{L}(w^k)\|^2 - \frac{1}{2} \|w^* - w^{k+1}\|^2 \end{aligned} \quad (246)$$

we obtain from (245) for all $k \geq 0$

$$\begin{aligned} \mathcal{L}(w^{k+1}) &\leq \mathcal{L}(w^*) + \frac{1}{\eta} \left[-\langle w^* - w^k, \eta \nabla \mathcal{L}(w^k) \rangle \right] - \alpha \|\nabla \mathcal{L}(w^k)\|^2 \\ &= \mathcal{L}(w^*) + \frac{1}{\eta} \left[\frac{1}{2} \|w^* - w^k\|^2 + \frac{\eta^2}{2} \|\nabla \mathcal{L}(w^k)\|^2 - \frac{1}{2} \|w^* - w^{k+1}\|^2 \right] \\ &\quad - \alpha \|\nabla \mathcal{L}(w^k)\|^2 \\ &= \mathcal{L}(w^*) + \frac{1}{2\eta} \left[\|w^* - w^k\|^2 - \|w^* - w^{k+1}\|^2 \right] + \left(\frac{\eta}{2} - \alpha \right) \|\nabla \mathcal{L}(w^k)\|^2 \\ &\leq \mathcal{L}(w^*) + \frac{1}{2\eta} \left[\|w^* - w^k\|^2 - \|w^* - w^{k+1}\|^2 \right], \end{aligned} \quad (247)$$

since $\frac{\eta}{2} - \alpha = \frac{\eta}{2}(L\eta - 1) \leq 0$ for $\eta \in (0, 1/L]$. Hence, for all $M \geq 1$

$$\begin{aligned} \sum_{k=0}^{M-1} (\mathcal{L}(w^{k+1}) - \mathcal{L}(w^*)) &\leq \frac{1}{2\eta} \sum_{k=0}^{M-1} \left[\|w^* - w^k\|^2 - \|w^* - w^{k+1}\|^2 \right] \\ &= \frac{1}{2\eta} \left[\|w^* - w^0\|^2 - \|w^* - w^M\|^2 \right] \leq \frac{1}{2\eta} \|w^* - w^0\|^2. \end{aligned} \quad (248)$$

The sequence $\{\mathcal{L}(w^k)\}_{k \geq 0}$ is non-increasing hence for all $M \geq 1$

$$\mathcal{L}(w^0) \geq \mathcal{L}(w^1) \geq \dots \geq \mathcal{L}(w^M), \quad (249)$$

and

$$\sum_{k=0}^{M-1} \mathcal{L}(w^{k+1}) \geq M\mathcal{L}(w^M). \quad (250)$$

This implies

$$\sum_{k=0}^{M-1} (\mathcal{L}(w^{k+1}) - \mathcal{L}(w^*)) = \sum_{k=0}^{M-1} \mathcal{L}(w^{k+1}) - M\mathcal{L}(w^*) \geq M(\mathcal{L}(w^M) - \mathcal{L}(w^*)),$$

and by (248)

$$\mathcal{L}(w^M) - \mathcal{L}(w^*) \leq \frac{1}{M} \sum_{k=0}^{M-1} (\mathcal{L}(w^{k+1}) - \mathcal{L}(w^*)) \leq \frac{1}{M} \cdot \frac{1}{2\eta} \|w^* - w^0\|^2 = c_1/M, \quad (251)$$

with

$$c_1 = \frac{1}{2\eta} \|w^* - w^0\|^2. \quad (252)$$

Moreover, by (248) and Jensen inequality we get

$$\mathcal{L}\left(\frac{1}{M} \sum_{k=0}^{M-1} w^{k+1}\right) - \mathcal{L}(w^*) \leq \frac{1}{M} \sum_{k=0}^{M-1} \mathcal{L}(w^{k+1}) - \mathcal{L}(w^*) \leq \frac{c_1}{M}. \quad (253)$$

Since $\mathcal{L}(w^*) = \inf_{y \in \mathbb{R}^n} \mathcal{L}(y) = \mathcal{L}^*$ we get (240) and (241). From Lemma 13.4 and (240) we obtain (242). $\square$

Example. (*Regularity of the loss function in linear smoothing*)

Let $A \in \mathbb{R}^{M \times N}$, $y \in \mathbb{R}^M$, and consider the following function $\mathcal{L} : \mathbb{R}^N \rightarrow [0, +\infty)$

$$\mathcal{L}(w) = \|Aw - y\|^2. \quad (254)$$

By Theorem 3.1, the set of global minimums is non-empty for the function $\mathcal{L}$. Moreover, we have

$$\nabla \mathcal{L}(w) = 2(A^T Aw - A^T y) = 2A^T(Aw - y), \quad (255)$$

$$\|\nabla \mathcal{L}(w_1) - \nabla \mathcal{L}(w_2)\| = \|2A^T A(w_1 - w_2)\| \leq 2\|A^T A\| \cdot \|w_1 - w_2\|, \quad (256)$$

for $w_1, w_2 \in \mathbb{R}^N$, and

$$\nabla^2 \mathcal{L}(w) = 2A^T A, \quad w \in \mathbb{R}^N. \quad (257)$$

Since $A^T A \in \mathbb{R}^{N \times N}$ is symmetric and positive semi-definite, $\mathcal{L}$ is L -smooth (with $L = 2\|A^T A\|$) and convex. Therefore, by Theorem 6.2, the GD sequence converges to a global minimum of $\mathcal{L}$.

Example. (*Regularity of the binary cross-entropy loss function*)

We show that the function (119) is L -smooth and convex. We have that

$$\|\nabla^2 \mathcal{L}(w)\| \leq \frac{\|X\|^2}{M} \|D(w)\| \leq \frac{\|X\|^2}{4\sqrt{M}}, \quad (258)$$

since for the function $g(x) := x(1-x)$, we have that $0 \leq g(x) \leq 1/4$ for all $x \in [0, 1]$. By Lemma 13.1 get

$$\|\nabla \mathcal{L}(w_1) - \nabla \mathcal{L}(w_2)\| \leq \frac{\|X\|^2}{4\sqrt{M}} \|w_1 - w_2\|, \quad (259)$$

for all $w_1, w_2 \in \mathbb{R}^N$. Note that we can write $D(w) = (D(w))^{1/2}(D(w))^{1/2}$, where

$$(D(w))^{1/2} = \text{diag} \left(\left(\varphi(\langle x_j, w \rangle) \cdot (1 - \varphi(\langle x_j, w \rangle)) \right)^{1/2} \right)_{j=1, \dots, M} \quad (260)$$

and $[(D(w))^{1/2}]^T = (D(w))^{1/2}$. Therefore, for all $v \in \mathbb{R}^N$

$$\begin{aligned} v^T X D(w) X^T v &= v^T X (D(w))^{1/2} (D(w))^{1/2} X^T v \\ &= v^T X [(D(w))^{1/2}]^T (D(w))^{1/2} X^T v \\ &= [(D(w))^{1/2} (X^T v)]^T (D(w))^{1/2} X^T v = \|(D(w))^{1/2} X^T v\|^2 \geq 0, \end{aligned} \quad (261)$$

and this implies that $\nabla^2 \mathcal{L}(w)$ is positive semi-definite for all $w \in \mathbb{R}^N$. Hence, $\mathcal{L}$ is also a convex function. Since $\mathcal{L} \geq 0$, we get by Theorem 6.1, for sufficiently small η , that $\nabla \mathcal{L}(w^k) \rightarrow 0$ and $\mathcal{L}(w^k) \rightarrow \inf_{k \geq 0} \mathcal{L}(w^k)$ as $k \rightarrow +\infty$. Moreover, if there exists (at least one) global minimum w^* of $\mathcal{L}$, then by Theorem 6.2 we have that $\mathcal{L}(w^k) \rightarrow \mathcal{L}(w^*)$ as $k \rightarrow +\infty$.

6.4 Convergence theorem for L -smooth and Polyak-Łojasiewicz type functions

The convexity assumption used in the previous section is rather strong—it is rarely satisfied in the case of deep neural networks. We can relax the convexity assumption by considering Polyak-Łojasiewicz (PL) conditions, see, for example, [12], [18], [20].

Definition 6.1. A differentiable function $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$ is of Polyak-Łojasiewicz (PL) type if

$$\inf_{w \in \mathbb{R}^n} \mathcal{L}(w) > -\infty, \quad (262)$$

and there exists $\mu \in (0, +\infty)$ such that for all $w \in \mathbb{R}^n$

$$\mathcal{L}(w) - \inf_{w \in \mathbb{R}^n} \mathcal{L}(w) \leq \frac{1}{2\mu} \|\nabla \mathcal{L}(w)\|^2. \quad (263)$$

Theorem 6.3. Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ be L -smooth for some $L \in (0, +\infty)$ and of Polyak-Łojasiewicz type for some $\mu \in (0, +\infty)$. Then

(i) $\mu \in (0, L]$,

(ii) the set of all global minimums of $\mathcal{L}$ is non-empty, i.e. $\mathcal{M} := \{w^* \in \mathbb{R}^n \mid \mathcal{L}(w^*) = \mathcal{L}^*\} \neq \emptyset$, where $\mathcal{L}^* := \inf_{y \in \mathbb{R}^n} \mathcal{L}(y) > -\infty$,

(iii) for all $\eta \in (0, 1/(3L)]$, $w^0 \in \mathbb{R}^n$ it holds

$$\begin{aligned} 0 &\leq (2/9)^k (\mathcal{L}(w^0) - \mathcal{L}^*) \\ &\leq \mathcal{L}(w^k) - \mathcal{L}^* \\ &\leq (1 - (5/3)\mu\eta)^k (\mathcal{L}(w^0) - \mathcal{L}^*), \end{aligned} \quad (264)$$

there exists a limit in $\mathbb{R}^n$ for the sequence $\{w^k\}_{k \geq 0}$ defined by (209), and the limit belongs to $\mathcal{M}$.

Proof. Since $\mathcal{L}$ is of PL-type and L -smooth, we get by Lemma 13.4 that

$$0 \leq \mu(\mathcal{L}(w) - \mathcal{L}^*) \leq \frac{1}{2} \|\nabla \mathcal{L}(w)\|^2 \leq L(\mathcal{L}(w) - \mathcal{L}^*) \quad (265)$$

for all $w \in \mathbb{R}^n$. Hence, $\mu \in (0, L]$. From (215) we have that for all $k \geq 0$

$$(-\beta) \|\nabla \mathcal{L}(w^k)\|^2 \leq \mathcal{L}(w^{k+1}) - \mathcal{L}(w^k) \leq (-\alpha) \|\nabla \mathcal{L}(w^k)\|^2, \quad (266)$$

with

$$\begin{aligned} \alpha &= \eta \left(1 - \frac{L\eta}{2}\right), \\ \beta &= \eta \left(1 + \frac{L\eta}{2}\right). \end{aligned} \quad (267)$$

Since $\eta \in (0, 1/(3L)]$, we have

$$\frac{1}{3L} \geq \eta \geq \alpha \geq \frac{5}{6}\eta, \quad (268)$$

and

$$0 < \alpha \leq \eta < \beta \leq \frac{7}{6}\eta \leq \frac{7}{18L}. \quad (269)$$

By (265) we obtain

$$(-\alpha)\|\nabla\mathcal{L}(w^k)\|^2 \leq -2\mu\alpha(\mathcal{L}(w^k) - \mathcal{L}^*), \quad (270)$$

$$(-\beta)\|\nabla\mathcal{L}(w^k)\|^2 \geq -2L\beta(\mathcal{L}(w^k) - \mathcal{L}^*). \quad (271)$$

Defining by $z_k = \mathcal{L}(w^k) - \mathcal{L}^* \geq 0$, we thus get for all $k \geq 0$

$$-2L\beta z_k \leq z_{k+1} - z_k \leq -2\mu\alpha z_k, \quad (272)$$

and

$$z_k(1 - 2L\beta) \leq z_{k+1} \leq z_k(1 - 2\mu\alpha). \quad (273)$$

Since $\mu \in (0, L]$, we obtain from (268), (269)

$$\frac{1}{3} \leq 1 - 2\mu\alpha \leq 1 - \frac{5}{3}\mu\eta < 1, \quad (274)$$

$$\frac{2}{9} \leq 1 - 2L\beta < 1. \quad (275)$$

This and (273) imply (264). From (265) and (264) we have for all $k \geq 0$

$$\|\nabla\mathcal{L}(w^k)\| \leq C_0 q^k, \quad (276)$$

where

$$C_0 = \sqrt{2L}(\mathcal{L}(w^0) - \mathcal{L}^*)^{1/2}, \quad (277)$$

$$q = (1 - (5/3)\mu\eta)^{1/2} \in [1/\sqrt{3}, 1). \quad (278)$$

Hence

$$\sum_{k=0}^{\infty} \|\nabla\mathcal{L}(w^k)\| \leq C_0 \sum_{k=0}^{+\infty} q^k = \frac{C_0}{1-q} < +\infty, \quad (279)$$

and by Proposition 6.1 (ii) the sequence $\{w^k\}_{k \geq 0}$ converges to the critical point $w^* = \lim_{k \rightarrow +\infty} w^k$ of $\mathcal{L}$. By (265) we also have that

$$0 \leq \mu(\mathcal{L}(w^*) - \mathcal{L}^*) \leq \frac{1}{2}\|\nabla\mathcal{L}(w^*)\|^2 = 0. \quad (280)$$

Therefore, $\mathcal{L}$ has at least one global minimum and the sequence $\{w^k\}_{k \geq 0}$ converges to the minimizer of $\mathcal{L}$. Finally, we have for all $M \in \mathbb{N}$

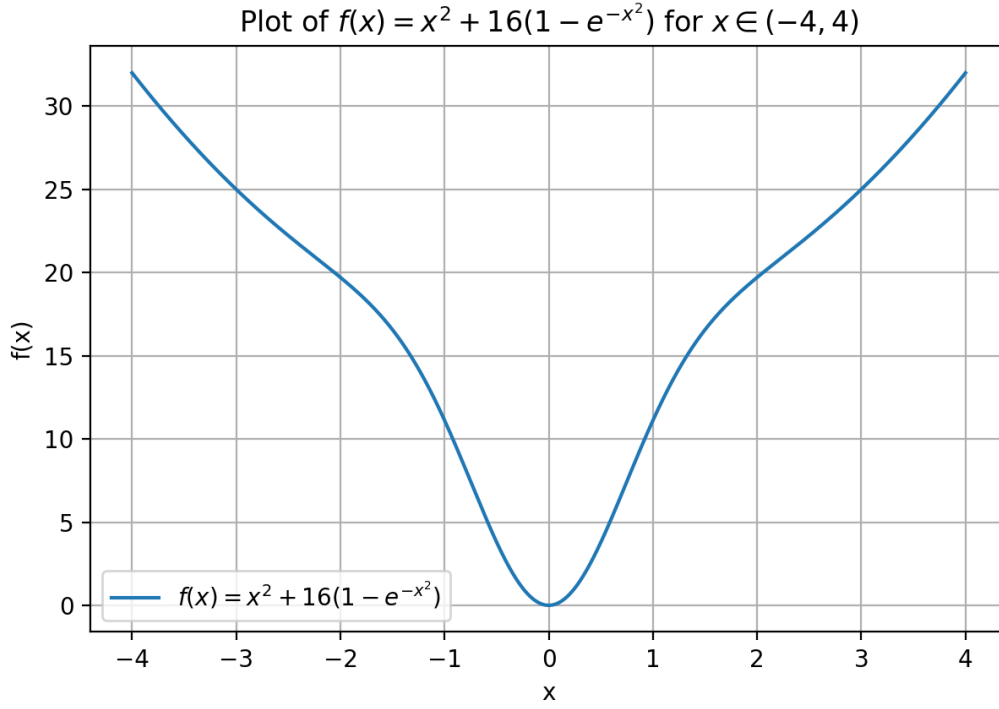
$$\|w^* - w^M\| \leq \eta \sum_{k=M}^{+\infty} \|\nabla\mathcal{L}(w^k)\| \leq \frac{C_0 \eta q^M}{1-q}. \quad (281)$$

□

As a byproduct of Theorem 6.3, we obtained the interesting fact that a function that is both L -smooth and PL-type has at least one global minimizer.

Remark 6.1. The function $\mathcal{L}(w) = w^2 + 3(\sin(x))^2$, $x \in \mathbb{R}$, is L -smooth (with $L = 8$) and of PL-type (with $\mu = 1/40$), but it is not convex function, see [12].

The following fact states another example of a function that is not convex but satisfies the PL condition.



Lemma 6.1. *Let $a > e$, then the function*

$$f(x) = x^2 + a(1 - e^{-x^2}), \quad x \in \mathbb{R} \quad (282)$$

is L -smooth and of PL-type, but it is not a convex function.

Proof. Note that $f(x) = f(-x)$ for all $x \in \mathbb{R}$. Furthermore, $f'(x) = 2x(1 + ae^{-x^2})$, $f''(x) = 2(1 + ae^{-x^2}) - 4ax^2e^{-x^2}$ and $f'(-x) = -f'(x)$, $f''(x) = f''(-x)$ for all $x \in \mathbb{R}$. Hence, f is increasing on $(0, +\infty)$ and decreasing on $(-\infty, 0)$. Moreover, for $x^* = 0$ we have $f'(x^*) = 0$ and $f''(x^*) = 2(1 + a) > 0$. Hence, $x^* = 0$ is the global minimizer of f and

$$\inf_{x \in \mathbb{R}} f(x) = 0 = f(x^*). \quad (283)$$

Since $|f''(x)| \leq 2 + 6a$ for all $x \in \mathbb{R}$, by the mean value theorem we get that f' is globally Lipschitz with the Lipschitz constant $L = 2 + 6a$. Moreover, $e^x \geq 1 + x$ for all $x \in \mathbb{R}$, so

$$e^{-x^2} \geq 1 - x^2, \quad x \in \mathbb{R}. \quad (284)$$

This implies that

$$x^2 \geq \frac{1}{a+1} f(x), \quad x \in \mathbb{R}. \quad (285)$$

Hence, for all $x \in \mathbb{R}$

$$\frac{1}{2} |f'(x)|^2 = 2x^2(1 + ae^{-x^2})^2 \geq 2x^2 \geq \frac{2}{a+1} f(x), \quad (286)$$

and f is of PL-type with $\mu = \frac{2}{1+a}$. Since $a > e$, we have

$$f''(-1) = f''(1) = 2 - 2ae^{-1} < 0, \quad (287)$$

while $f''(0) = 2(1 + a) > 0$, and by the continuity of f'' we get that there exists $x_0 \in (0, 1)$, $x_1 \in (-1, 0)$ for which $f''(x_0) = 0 = f''(x_1)$. Hence, f is not a convex function. $\square$

Remark 6.2. Polyak-Łojasiewicz type assumptions appear in Deep Reinforcement Learning, see, for example, [23].

6.5 Gradient-free descent algorithm

When gradients are unavailable, unreliable, or too costly to obtain, derivative-free optimization becomes a principled and often more efficient alternative, see [21]. Here, we present a method that approximates the gradient $\nabla \mathcal{L}$ by a central divided difference. Take two sequences $h_k, \eta_k \in (0, +\infty)$, $k \geq 0$, and set

$$\begin{aligned} w^0 &\in \mathbb{R}^n, \\ w^{k+1} &= w^k - \eta_k \bar{\nabla}_{h_k} \mathcal{L}(w^k), \quad k = 0, 1, 2, \dots, \end{aligned} \quad (288)$$

where for all $x \in \mathbb{R}^n$

$$\bar{\nabla}_h \mathcal{L}(x) = [\bar{\nabla}_{h,1} \mathcal{L}(x), \dots, \bar{\nabla}_{h,n} \mathcal{L}(x)]^T, \quad (289)$$

$$\bar{\nabla}_{h,j} \mathcal{L}(x) = \frac{\mathcal{L}(x + h e_j) - \mathcal{L}(x - h e_j)}{2h}, \quad (290)$$

and $e_j = [0, \dots, \underbrace{1}_{j\text{th place}}, \dots, 0]^T$, $j = 1, 2, \dots, n$. The common choice is $h_k = \eta_k$, since in that case

$$w^{k+1} = w^k - \frac{1}{2} [\mathcal{L}(w^k + \eta_k e_j) - \mathcal{L}(w^k - \eta_k e_j)]_{j=1,2,\dots,n}, \quad k \geq 0. \quad (291)$$

However, one must be aware that in finite-precision (*fl*) arithmetic, catastrophic cancellation can occur when the quantity $\mathcal{L}(w^k + \eta_k e_j) - \mathcal{L}(w^k - \eta_k e_j)$ is numerically very close to zero. One of the possible remedies is to compute $(\mathcal{L}(w^k \pm \eta_k e_j))$ in long double (128-bit) arithmetic, perform the subtraction in that precision, then cast the result back to 64-bit.

Note that the gradient-free descent algorithm can be seen as an ordinary gradient descent method with additional perturbation, i.e.,

$$w^{k+1} = w^k - \eta_k \nabla \mathcal{L}(w^k) + \eta_k Z_k, \quad k \geq 0, \quad (292)$$

where

$$Z_k = \nabla \mathcal{L}(w^k) - \bar{\nabla}_{h_k} \mathcal{L}(w^k). \quad (293)$$

This representation helps us understand when and why gradient-free methods converge. The key requirement is to maintain sufficient control over the perturbations $\{Z_k\}_{k \geq 0}$. Such control can be achieved by suitably approximating the gradient $\nabla \mathcal{L}$. For instance, the central-difference scheme introduced above yields a smaller truncation error than the first-order forward or backward differences.

6.6 General framework for the study of convergence of the gradient descent algorithm

In this section, we establish a general framework for investigating the convergence of the deterministic gradient descent-like algorithms. In particular, we demonstrate how to handle the case where the gradient is corrupted by deterministic noise. Our approach is a modified version of the theory given in [6]. However, we show some theoretical results under weaker assumptions than in [6]. Namely, in that article, the authors assumed that $\nabla \mathcal{L}$ is L -smooth, when we make use only of the one-sided Lipschitz condition for $\nabla \mathcal{L}$.

Let us assume that for solving the problem (208) we use the following iterative method: fix the sequence $\{\eta_k\}_{k \geq 0}$ of nonnegative real numbers η_k , a sequence of directions $\{d_k\}_{k \geq 0}$, $d_k \in \mathbb{R}^n$ and the starting point $w^0 \in \mathbb{R}^n$, then

$$w^{k+1} = w^k - \eta_k d_k, \quad k = 0, 1, 2, \dots \quad (294)$$

We can then write

$$\begin{aligned} w^{k+1} &= w^k - \eta_k (d_k + \nabla \mathcal{L}(w^k) - \nabla \mathcal{L}(w^k)) \\ &= w^k - \eta_k \nabla \mathcal{L}(w^k) + \eta_k Z_k, \end{aligned} \quad (295)$$

where

$$Z_k = \nabla \mathcal{L}(w^k) - d_k, \quad (296)$$

measures how far the direction d_k differs from the gradient $\nabla \mathcal{L}(w^k)$. Hence, Z_k can be interpreted as the deterministic noise involved when computing the gradient $\nabla \mathcal{L}(w^k)$. On the other hand, any method (295) can be seen as the algorithm (294) with the directions $d_k = \nabla \mathcal{L}(w^k) - Z_k$. Hence, it what follows we will consider the following iterative scheme

$$w^{k+1} = w^k - \eta_k \nabla \mathcal{L}(w^k) + \eta_k Z_k, \quad k = 0, 1, 2, \dots \quad (297)$$

We impose the following assumptions.

(A1) Assume that $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ and $\nabla \mathcal{L}$ satisfies the upper one-sided Lipschitz condition, i.e.: there exists $L \in [0, +\infty)$ such that for all $x, y \in \mathbb{R}^n$

$$\langle \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y), x - y \rangle \leq L \|x - y\|^2. \quad (298)$$

(A2) For all $k \geq 0$

$$\eta_k \in (0, 1/L]. \quad (299)$$

(A3) (*Deterministic noise control*) For the sequence $\{Z_k\}_{k \geq 0}$ we assume that

$$\sum_{k=0}^{+\infty} \eta_k \|Z_k\|^2 < +\infty. \quad (300)$$

From (A1) and Lemma 13.3

$$\begin{aligned} \mathcal{L}(w^{k+1}) - \mathcal{L}(w^k) &\leq -\eta_k \langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}(w^k) - Z_k \rangle + \frac{L}{2} \eta_k^2 \|\nabla \mathcal{L}(w^k) - Z_k\|^2 \\ &= -\eta_k \|\nabla \mathcal{L}(w^k)\|^2 + \eta_k \langle \nabla \mathcal{L}(w^k), Z_k \rangle \\ &\quad + \frac{L}{2} \eta_k^2 \|\nabla \mathcal{L}(w^k)\|^2 - L \eta_k^2 \langle \nabla \mathcal{L}(w^k), Z_k \rangle + \frac{L}{2} \eta_k^2 \|Z_k\|^2 \\ &= -\eta_k \left(1 - \frac{L\eta_k}{2}\right) \|\nabla \mathcal{L}(w^k)\|^2 + \eta_k (1 - L\eta_k) \langle \nabla \mathcal{L}(w^k), Z_k \rangle \\ &\quad + \frac{L}{2} \eta_k^2 \|Z_k\|^2. \end{aligned} \quad (301)$$

From the Cauchy-Schwarz inequality, we get

$$\langle \nabla \mathcal{L}(w^k), Z_k \rangle \leq \|\nabla \mathcal{L}(w^k)\| \cdot \|Z_k\| \leq \frac{1}{2} \left(\|\nabla \mathcal{L}(w^k)\|^2 + \|Z_k\|^2 \right). \quad (302)$$

Hence, by the assumption (A2), we obtain

$$\begin{aligned} \mathcal{L}(w^{k+1}) &\leq \mathcal{L}(w^k) - \eta_k \left(1 - \frac{L\eta_k}{2}\right) \|\nabla \mathcal{L}(w^k)\|^2 \\ &\quad + \frac{1}{2} \eta_k (1 - L\eta_k) \|\nabla \mathcal{L}(w^k)\|^2 + \frac{1}{2} \eta_k (1 - L\eta_k) \|Z_k\|^2 \\ &\quad + \frac{L}{2} \eta_k^2 \|Z_k\|^2 \\ &= \mathcal{L}(w^k) - \frac{1}{2} \eta_k \|\nabla \mathcal{L}(w^k)\|^2 + \frac{1}{2} \eta_k \|Z_k\|^2 \end{aligned} \quad (303)$$

for all $k \geq 0$. In order to establish the convergence of the sequence $\{\mathcal{L}(w^k)\}_{k \geq 0}$, we recall after [6] the following useful lemma.

Lemma 6.2. *Let Y_k, W_k, U_k be three sequences such that W_k is nonnegative for all k . Assume that*

$$Y_{k+1} \leq Y_k - W_k + U_k, \quad k = 0, 1, 2, \dots, \quad (304)$$

and that the series $\sum_{k=0}^K U_k$ converges as $K \rightarrow +\infty$. Then either $Y_k \rightarrow -\infty$ or else Y_k converges to a finite

value and $\sum_{k=0}^{+\infty} W_k < +\infty$.

We apply Lemma 6.2 for $Y_k := \mathcal{L}(w^k)$, $W_k := \frac{1}{2}\eta_k \|\nabla \mathcal{L}(w^k)\|^2$, $U_k := \frac{1}{2}\eta_k \|Z_k\|^2$ and, due to the assumption (A3), we get that either

$$\mathcal{L}(w^k) \rightarrow -\infty$$

as $k \rightarrow +\infty$ or else there exists

$$\lim_{k \rightarrow +\infty} \mathcal{L}(w^k) \in \mathbb{R}$$

and

$$\sum_{k=0}^{+\infty} \eta_k \|\nabla \mathcal{L}(w^k)\|^2 < +\infty. \quad (305)$$

If we additionally assume that

$$(A4) \quad \sum_{k=0}^{\infty} \eta_k = +\infty, \quad (306)$$

then from (A3), (A4), Lemma 13.8 and (305) we have

$$\liminf_{k \rightarrow +\infty} \|\nabla \mathcal{L}(w^k)\| = 0 = \liminf_{k \rightarrow +\infty} \|Z_k\|. \quad (307)$$

We summarize the results obtained in the following theorem.

Theorem 6.4. *Under the assumptions (A1)-(A3) we have that either*

$$\mathcal{L}(w^k) \rightarrow -\infty$$

as $k \rightarrow +\infty$ or else there exists

$$\lim_{k \rightarrow +\infty} \mathcal{L}(w^k) \in \mathbb{R},$$

and

$$\sum_{k=0}^{+\infty} \eta_k \|\nabla \mathcal{L}(w^k)\|^2 < +\infty. \quad (308)$$

If there exists $\mathcal{L}^* \in \mathbb{R}$ such that $\mathcal{L}(w^k) \geq \mathcal{L}^*$ for all $k \geq 0$, then under the assumptions (A1)-(A4) we have for all $M \geq 1$

$$\min_{0 \leq k \leq M-1} \|\nabla \mathcal{L}(w^k)\| \leq \left(\frac{2(\mathcal{L}(w^0) - \mathcal{L}^*) + \sum_{k=0}^{M-1} \eta_k \|Z_k\|^2}{\sum_{k=0}^{M-1} \eta_k} \right)^{1/2}, \quad (309)$$

and

$$\liminf_{k \rightarrow +\infty} \|\nabla \mathcal{L}(w^k)\| = 0. \quad (310)$$

Remark 6.3. The gradient-free descent algorithm (292) is of the form (297). If the approximation error (which can be interpreted as noise)

$$\|Z_k\| = \|\nabla \mathcal{L}(w^k) - \bar{\nabla}_{h_k} \mathcal{L}(w^k)\| \quad (311)$$

satisfies (A3), then from Theorem 6.4 the algorithm (292) has the stationarity properties.

Remark 6.4. In particular, assumption (A3) holds if there exists $n_0 \in \mathbb{N}$ such that $Z_k = 0$ for all $k \geq n_0$.

Remark 6.5. In general, the sequence $\{\mathcal{L}(w^k)\}_{k \geq 0}$, where $\{w^k\}_{k \geq 0}$ is defined by (297), is not necessarily monotone. However, if the noise Z_k is not too large, i.e.: if for all $k \geq 0$

$$\|Z_k\| \leq \|\nabla \mathcal{L}(w^k)\|, \quad (312)$$

then $\{\mathcal{L}(w^k)\}_{k \geq 0}$ is nonincreasing.

7 Stochastic gradient descent algorithm and its convergence

7.1 Definition of SGD algorithm

In the Machine Learning context, the common task is to minimize a function $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$ in the following form

$$\mathcal{L}(x) = \frac{1}{M} \sum_{j=1}^M \mathcal{L}_j(x), \quad x \in \mathbb{R}^n, \quad (313)$$

for some $M \in \mathbb{N}$ and $\mathcal{L}_j : \mathbb{R}^n \rightarrow \mathbb{R}$, $j = 1, 2, \dots, M$. For such a function, the GD method, with varying learning rate, has the following for

$$w^{k+1} = w^k - \eta_k \cdot \frac{1}{M} \sum_{j=1}^M \nabla \mathcal{L}_j(w^k), \quad k = 0, 1, \dots \quad (314)$$

When M is large, computation of the full gradient $\nabla \mathcal{L}$ might be computationally very costly and numerically inefficient. However, in this case, we can write $\nabla \mathcal{L}$ as the expectation

$$\nabla \mathcal{L}(x) = \frac{1}{M} \sum_{j=1}^M \nabla \mathcal{L}_j(x) = \mathbb{E}(\nabla \mathcal{L}_J(x)), \quad (315)$$

where $J : \Omega \rightarrow \{1, 2, \dots, M\}$ is a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ that is uniformly distributed on $\{1, 2, \dots, M\}$, i.e.,

$$\mathbb{P}(J = j) = \frac{1}{M}, \quad \text{for } j = 1, 2, \dots, M. \quad (316)$$

Hence, we can use the following (Monte Carlo) one-sample approximation

$$\nabla \mathcal{L}(x) \approx \nabla \mathcal{L}_J(x), \quad (317)$$

with $J \sim U(\{1, 2, \dots, M\})$. Using this, the stochastic gradient descent (SGD) algorithm, which uses only one random sample per iteration, is defined as follows: take $\{\eta_k\}_{k \geq 0}$, $\{J_k\}_{k \geq 1}$ -an iid sequence from $U(\{1, 2, \dots, M\})$, $w^0 \in \mathbb{R}^n$, then for $k = 0, 1, \dots$

$$w^{k+1} = w^k - \eta_k \cdot \nabla \mathcal{L}_{J_{k+1}}(w^k). \quad (318)$$

For the SGD algorithm we consider from the start learning rates η_k that are converging to zero, since for the non-decaying learning rates, some results show a lack of convergence of the SGD method, see Chapter 7.2.2 in [18].

We now discuss the basic properties of the SGD method. Set

$$\mathcal{F}_0 = \{\emptyset, \Omega\}, \quad (319)$$

$$\mathcal{F}_k = \sigma(J_1, J_2, \dots, J_k), \quad k = 1, 2, \dots \quad (320)$$

Note that $\mathcal{F}_k \subset \mathcal{F}_{k+1}$ for all $k \geq 0$, and

$$\sigma(w^0) \subset \mathcal{F}_0, \quad (321)$$

and by the induction we have for all $k \geq 1$

$$\sigma(w^k) \subset \mathcal{F}_k. \quad (322)$$

Hence the sequence of random vectors $\{w^k\}_{k \geq 0}$ is adapted to the filtration $\{\mathcal{F}_k\}_{k \geq 0}$. Moreover, since w^k is $\mathcal{F}_k$ -measurable and $\sigma(J_{k+1})$ is independent of $\mathcal{F}_k$ we obtain

$$\begin{aligned} \mathbb{E}[\nabla \mathcal{L}_{J_{k+1}}(w^k) \mid \mathcal{F}_k] &= \mathbb{E}[\nabla \mathcal{L}_{J_{k+1}}(x) \mid \mathcal{F}_k] \Big|_{x=w^k} \\ &= \mathbb{E}[\nabla \mathcal{L}_{J_{k+1}}(x)] \Big|_{x=w^k} = \nabla \mathcal{L}(w^k). \end{aligned} \quad (323)$$

Hence, conditionally on $\mathcal{F}_k$, $\nabla \mathcal{L}_{J_{k+1}}(w^k)$ is an unbiased estimator of $\nabla \mathcal{L}(w^k)$, and the conditional expectation

$$\begin{aligned}\mathbb{E}(w^{k+1} \mid \mathcal{F}_k) &= w^k - \eta_k \cdot \mathbb{E}[\nabla \mathcal{L}_{J_{k+1}}(w^k) \mid \mathcal{F}_k] \\ &= w^k - \eta_k \nabla \mathcal{L}(w^k)\end{aligned}\tag{324}$$

is therefore equal to one step in the gradient descent method. Moreover, for the SGD sequence, we have that

$$w^{k+1} = w^k - \eta_k \nabla \mathcal{L}(w^k) + \eta_k Z_k, \quad k = 0, 1, \dots,\tag{325}$$

where

$$Z_k = \nabla \mathcal{L}(w^k) - \nabla \mathcal{L}_{J_{k+1}}(w^k) = \mathbb{E}[\nabla \mathcal{L}_{J_{k+1}}(w^k) \mid \mathcal{F}_k] - \nabla \mathcal{L}_{J_{k+1}}(w^k),\tag{326}$$

and $\sigma(Z_k) \subset \mathcal{F}_{k+1}$, $\mathbb{E}(Z_k \mid \mathcal{F}_k) = 0$, $\mathbb{E}(Z_k) = 0$. So the SGD algorithm can be seen as an ordinary GD algorithm with a *randomly perturbed* (full) gradient $\nabla \mathcal{L}$. In practice, introducing such an unbiased noise can improve scalability and sometimes helps to escape from shallow local minima of the loss function $\mathcal{L}$.

Example. Recall after (64) that for linear smoothing, the loss function is the mean square error

$$MSE(w) = \frac{1}{M} \sum_{j=1}^M \left(\langle a^{(j)}, w \rangle - y_j \right)^2 = \frac{1}{M} \sum_{j=1}^M MSE_j(w),\tag{327}$$

and it is of the form (313). We have that

$$\nabla MSE_j(w) = 2 \left(\langle a^{(j)}, w \rangle - y_j \right) a^{(j)},\tag{328}$$

and the SGD algorithm has the form

$$w^{k+1} = w^k - 2\eta_k \left(\langle a^{(J_{k+1})}, w^k \rangle - y_{J_{k+1}} \right) a^{(J_{k+1})}, \quad k = 0, 1, \dots.\tag{329}$$

7.2 Convergence theorem for L -smooth functions with bounded gradient

We present results on the convergence of the SGD algorithm in the non-convex setting, which widely appears when considering deep neural networks. For more general results in that direction, see, for example, [22], [6].

Theorem 7.1. *Let us assume that:*

- the function $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ is L -smooth for some $L \in (0, +\infty)$,
- there exists $D \in [0, +\infty)$ such that for all $j = 1, 2, \dots, M$, $x \in \mathbb{R}^n$,

$$\|\nabla \mathcal{L}_j(x)\| \leq D,\tag{330}$$

- for the sequence $\{\eta_k\}_{k \geq 0}$ of positive real numbers it holds

$$\sum_{k=0}^{+\infty} \eta_k = +\infty, \quad \sum_{k=0}^{+\infty} \eta_k^2 < +\infty.\tag{331}$$

Then for any $w^0 \in \mathbb{R}^n$ we have:

$$(A) \quad \mathbb{E} \left(\sum_{k=0}^{+\infty} \eta_k \cdot \|\nabla \mathcal{L}(w^k)\|^2 \right) = +\infty \Leftrightarrow \lim_{k \rightarrow +\infty} \mathbb{E}(\mathcal{L}(w^k)) = -\infty,$$

$$(B) \quad \mathbb{E} \left(\sum_{k=0}^{+\infty} \eta_k \cdot \|\nabla \mathcal{L}(w^k)\|^2 \right) < +\infty \Leftrightarrow \exists \mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)] : \forall k \geq 0 \mathbb{E}(\mathcal{L}(w^k)) \geq \mathcal{L}^*,$$

(C) either

$$(C1) \lim_{k \rightarrow +\infty} \mathbb{E}(\mathcal{L}(w^k)) = -\infty$$

or

$$(C2) \exists \mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)] : \forall k \geq 0 \mathbb{E}(\mathcal{L}(w^k)) \geq \mathcal{L}^*.$$

In the (C2)-case we additionally have:

– there exists $C \in [0, +\infty)$ such that

$$\limsup_{k \rightarrow +\infty} \mathbb{E}(\mathcal{L}(w^k)) \leq \mathcal{L}(w^0) - \mathbb{E}\left(\sum_{k=0}^{+\infty} \eta_k \cdot \|\nabla \mathcal{L}(w^k)\|^2\right) + C \sum_{k=0}^{+\infty} \eta_k^2 < +\infty, \quad (332)$$

$$\liminf_{k \rightarrow +\infty} \mathbb{E}(\mathcal{L}(w^k)) \geq \mathcal{L}(w^0) - \mathbb{E}\left(\sum_{k=0}^{+\infty} \eta_k \cdot \|\nabla \mathcal{L}(w^k)\|^2\right) - C \sum_{k=0}^{+\infty} \eta_k^2 > -\infty, \quad (333)$$

– there exists $C \in [0, +\infty)$ such that for all $N \geq 1$

$$\min_{0 \leq k \leq N-1} \mathbb{E}\|\nabla \mathcal{L}(w^k)\|^2 \leq \frac{\mathcal{L}(w^0) - \mathcal{L}^* + C \sum_{k=0}^{N-1} \eta_k^2}{\sum_{k=0}^{N-1} \eta_k}, \quad (334)$$

$$\liminf_{k \rightarrow +\infty} \mathbb{E}\|\nabla \mathcal{L}(w^k)\|^2 = 0, \quad (335)$$

and

$$\mathbb{P}\left(\inf_{k \geq 0} \|\nabla \mathcal{L}(w^k)\| = 0\right) = 1. \quad (336)$$

Proof. Let us denote by

$$A_{j,N} = \sum_{k=0}^{N-1} \eta_k^j, \quad (337)$$

$$\Delta \mathcal{L}_k = \mathcal{L}(w^{k+1}) - \mathcal{L}(w^k), \quad (338)$$

$$\Delta w_k = w^{k+1} - w^k, \quad (339)$$

$$S_N = \sum_{k=0}^{N-1} \eta_k \cdot \|\nabla \mathcal{L}(w^k)\|^2, \quad (340)$$

for $j = 1, 2$, $N \geq 1$, $k \geq 0$. Then, by (611) we get for all $k \geq 0$

$$\begin{aligned} -\eta_k \langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}_{J_{k+1}}(w^k) \rangle - \frac{L}{2} \|\Delta w_k\|^2 &\leq \Delta \mathcal{L}_k \\ &\leq -\eta_k \langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}_{J_{k+1}}(w^k) \rangle + \frac{L}{2} \|\Delta w_k\|^2, \end{aligned} \quad (341)$$

and, since (318) holds, we have

$$L_k \leq \Delta \mathcal{L}_k \leq U_k \quad (342)$$

where, by (330), we get

$$\begin{aligned} U_k &= -\eta_k \langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}_{J_{k+1}}(w^k) \rangle + \frac{L}{2} \eta_k^2 \cdot \|\nabla \mathcal{L}_{J_{k+1}}(w^k)\|^2 \\ &\leq -\eta_k \langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}_{J_{k+1}}(w^k) \rangle + C \eta_k^2, \end{aligned} \quad (343)$$

$$\begin{aligned} L_k &= -\eta_k \langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}_{J_{k+1}}(w^k) \rangle - \frac{L}{2} \eta_k^2 \cdot \|\nabla \mathcal{L}_{J_{k+1}}(w^k)\|^2 \\ &\geq -\eta_k \langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}_{J_{k+1}}(w^k) \rangle - C \eta_k^2, \end{aligned} \quad (344)$$

with $C = LD^2/2$. This implies

$$\begin{aligned} -\eta_k \mathbb{E} \left(\langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}_{J_{k+1}}(w^k) \rangle \right) - C\eta_k^2 &\leq \mathbb{E}(\Delta \mathcal{L}_k) \\ &\leq -\eta_k \mathbb{E} \left(\langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}_{J_{k+1}}(w^k) \rangle \right) + C\eta_k^2, \end{aligned} \quad (345)$$

where by the Cauchy-Schwarz inequality and (330) we have for all $w \in \mathbb{R}^n$

$$\|\nabla \mathcal{L}(w)\| \leq \frac{1}{M} \sum_{j=1}^M \|\nabla \mathcal{L}_j(w)\| \leq D, \quad (346)$$

and

$$\begin{aligned} \mathbb{E} \left(\left| \langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}_{J_{k+1}}(w^k) \rangle \right| \right) &\leq \mathbb{E} \left(\|\nabla \mathcal{L}(w^k)\| \cdot \|\nabla \mathcal{L}_{J_{k+1}}(w^k)\| \right) \\ &\leq D^2 < +\infty. \end{aligned} \quad (347)$$

From the basic properties of conditional expectation

$$\mathbb{E} \left(\langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}_{J_{k+1}}(w^k) \rangle \right) = \mathbb{E} \left[\mathbb{E} \left(\langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}_{J_{k+1}}(w^k) \rangle \mid \mathcal{F}_k \right) \right].$$

By the facts that $\sigma(w^k) \subset \mathcal{F}_k$ and $\{\sigma(J_{k+1}), \mathcal{F}_k\}$ are independent σ -fields we get

$$\mathbb{E} \left(\langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}_{J_{k+1}}(w^k) \rangle \mid \mathcal{F}_k \right) = \mathbb{E} \left(\langle \nabla \mathcal{L}(x), \nabla \mathcal{L}_{J_{k+1}}(x) \rangle \right) \Big|_{x=w^k}, \quad (348)$$

where for all $x \in \mathbb{R}^n$

$$\begin{aligned} \mathbb{E} \left(\langle \nabla \mathcal{L}(x), \nabla \mathcal{L}_{J_{k+1}}(x) \rangle \right) &= \frac{1}{M} \sum_{j=1}^M \langle \nabla \mathcal{L}(x), \nabla \mathcal{L}_j(x) \rangle \\ &= \langle \nabla \mathcal{L}(x), \frac{1}{M} \sum_{j=1}^M \nabla \mathcal{L}_j(x) \rangle = \langle \nabla \mathcal{L}(x), \nabla \mathcal{L}(x) \rangle = \|\nabla \mathcal{L}(x)\|^2. \end{aligned} \quad (349)$$

Therefore

$$\mathbb{E} \left(\langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}_{J_{k+1}}(w^k) \rangle \right) = \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2, \quad (350)$$

which, together with (345), implies for all $k \geq 0$

$$-\eta_k \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 - C\eta_k^2 \leq \mathbb{E}(\Delta \mathcal{L}_k) \leq -\eta_k \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 + C\eta_k^2. \quad (351)$$

Summing up for $k = 0, 1, \dots, N-1$, $N \geq 1$ we obtain

$$\begin{aligned} -\sum_{k=0}^{N-1} \eta_k \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 - C \sum_{k=0}^{N-1} \eta_k^2 &\leq \mathbb{E}(\mathcal{L}(w^N)) - \mathcal{L}(w^0) \\ &\leq -\sum_{k=0}^{N-1} \eta_k \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 + C \sum_{k=0}^{N-1} \eta_k^2. \end{aligned} \quad (352)$$

Hence,

$$\begin{aligned} \mathcal{L}(w^0) - \sum_{k=0}^{N-1} \eta_k \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 - C \sum_{k=0}^{N-1} \eta_k^2 &\leq \mathbb{E}(\mathcal{L}(w^N)) \\ &\leq \mathcal{L}(w^0) - \sum_{k=0}^{N-1} \eta_k \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 + C \sum_{k=0}^{N-1} \eta_k^2, \end{aligned} \quad (353)$$

or, equivalently

$$\begin{aligned}\mathcal{L}(w^0) - \mathbb{E}(\mathcal{L}(w^N)) - C \sum_{k=0}^{N-1} \eta_k^2 &\leq \sum_{k=0}^{N-1} \eta_k \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 \\ &\leq \mathcal{L}(w^0) - \mathbb{E}(\mathcal{L}(w^N)) + C \sum_{k=0}^{N-1} \eta_k^2.\end{aligned}\tag{354}$$

Using the notation (337), we can write the inequalities above as

$$\mathcal{L}(w^0) - \mathbb{E}(S_N) - CA_{2,N} \leq \mathbb{E}(\mathcal{L}(w^N)) \leq \mathcal{L}(w^0) - \mathbb{E}(S_N) + CA_{2,N},\tag{355}$$

and

$$\mathcal{L}(w^0) - \mathbb{E}(\mathcal{L}(w^N)) - CA_{2,N} \leq \mathbb{E}(S_N) \leq \mathcal{L}(w^0) - \mathbb{E}(\mathcal{L}(w^N)) + CA_{2,N},\tag{356}$$

for all $N \geq 1$. Since we assume that $A_{2,\infty} := \lim_{N \rightarrow +\infty} A_{2,N} = \sum_{k=0}^{+\infty} \eta_k^2 < +\infty$, from (355), (356) we directly get the equivalence (A).

We now show the equivalence (B). Since the sequence of random variables $\{S_N\}_{N \geq 1}$ is non-decreasing and positive, we get from the Lebesgue monotone convergence theorem and Tonelli's theorem for sums and integrals that

$$\lim_{N \rightarrow +\infty} \mathbb{E}(S_N) = \mathbb{E}(S_\infty) = \sum_{k=0}^{+\infty} \eta_k \cdot \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2,\tag{357}$$

where

$$S_\infty = \sum_{k=0}^{+\infty} \eta_k \cdot \|\nabla \mathcal{L}(w^k)\|^2.\tag{358}$$

($\Rightarrow$) If $\mathbb{E}(S_\infty) < +\infty$ then for all $N \geq 1$ we have by (357)

$$\mathbb{E}(S_N) = \sum_{k=0}^{N-1} \eta_k \cdot \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 \leq \sum_{k=0}^{+\infty} \eta_k \cdot \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 = \mathbb{E}(S_\infty) < +\infty,\tag{359}$$

and by (355), for all $N \geq 1$

$$\begin{aligned}-\infty < \mathcal{L}(w^0) - \mathbb{E}(S_\infty) - CA_{2,\infty} &\leq \mathcal{L}(w^0) - \mathbb{E}(S_N) - CA_{2,N} \\ &\leq \mathbb{E}(\mathcal{L}(w^N)).\end{aligned}\tag{360}$$

Hence, for

$$-\infty < \mathcal{L}^* := \mathcal{L}(w^0) - \mathbb{E}(S_\infty) - CA_{2,\infty} \leq \mathcal{L}(w^0),\tag{361}$$

we have for all $N \geq 0$ that

$$\mathbb{E}(\mathcal{L}(w^N)) \geq \mathcal{L}^*.\tag{362}$$

($\Leftarrow$) Let us assume that $\exists \mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)] : \forall k \geq 0 \mathbb{E}(\mathcal{L}(w^k)) \geq \mathcal{L}^*$. Using (356) we have for all $N \geq 1$ that

$$\mathbb{E}(S_N) \leq \mathcal{L}(w^0) - \mathbb{E}(\mathcal{L}(w^N)) + CA_{2,N} \leq \mathcal{L}(w^0) - \mathcal{L}^* + CA_{2,\infty} < +\infty.\tag{363}$$

Hence, the sequence $\{\mathbb{E}(S_N)\}_{N \geq 1}$ is bounded from above and non-decreasing. Therefore, it is convergent and by (357)

$$\mathbb{E}(S_\infty) = \lim_{N \rightarrow +\infty} \mathbb{E}(S_N) \leq \mathcal{L}(w^0) - \mathcal{L}^* + CA_{2,\infty} < +\infty.\tag{364}$$

Hence, the proof of the equivalence (B) is thus finished.

Assuming (C2), we have that $\mathbb{E}(S_\infty) < +\infty$, and then, by (355), we get (333), and (332). Moreover, from (356)

$$\begin{aligned} \min_{0 \leq k \leq N-1} \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 \cdot A_{1,N} &\leq \mathbb{E}(S_N) \leq \mathcal{L}(w^0) - \mathbb{E}(\mathcal{L}(w^N)) + CA_{2,N} \\ &\leq \mathcal{L}(w^0) - \mathcal{L}^* + CA_{2,N}, \end{aligned} \quad (365)$$

which implies (334). Since $\mathbb{E}(S_\infty) < +\infty$ and $A_{1,\infty} = +\infty$, by Lemma 13.8 we get (335). By Fatou's lemma and (335)

$$0 \leq \mathbb{E} \left(\liminf_{k \rightarrow +\infty} \|\nabla \mathcal{L}(w^k)\|^2 \right) \leq \liminf_{k \rightarrow +\infty} \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 = 0, \quad (366)$$

and hence

$$\mathbb{P}(\liminf_{k \rightarrow +\infty} \|\nabla \mathcal{L}(w^k)\|^2 = 0) = 1. \quad (367)$$

Finally, with probability one

$$0 = \liminf_{k \rightarrow +\infty} \|\nabla \mathcal{L}(w^k)\|^2 = \sup_{k \geq 0} \inf_{m \geq k} \|\nabla \mathcal{L}(w^m)\|^2 \geq \inf_{m \geq 0} \|\nabla \mathcal{L}(w^m)\|^2 \geq 0, \quad (368)$$

which gives (336). $\square$

The significant issue with the SGD sequence (318) is that we cannot assure (with probability one or on average) that the sequence $\{\mathcal{L}(w^k)\}_{k \geq 0}$ is nonincreasing. In fact, when observing learning curves while a neural network is trained, we rarely observe strictly decreasing values of a loss function. Therefore, the natural question arises if we can strengthen assumptions of Theorem 7.1 to assure, at least, convergence of $\{\mathcal{L}(w^k)\}_{k \geq 0}$? The answer is positive and for the proof of convergence, we will use the following (very powerful) classical result; see [35].

Theorem 7.2 ([35], Robbins-Siegmund theorem of convergence for *almost submartingales*). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \dots$ a sequence of sub- σ -algebras of $\mathcal{F}$. For each $k = 1, 2, \dots$ let z_k, β_k, ξ_k and ζ_k be non-negative $\mathcal{F}_k$ -measurable random variables such that*

$$\mathbb{E}(z_{k+1} \mid \mathcal{F}_k) \leq z_k(1 + \beta_k) + \xi_k - \zeta_k. \quad (369)$$

Then $\lim_{k \rightarrow +\infty} z_k$ exists and is finite and $\sum_{k=1}^{+\infty} \zeta_k < +\infty$ almost surely on

$$\left\{ \sum_{k=1}^{+\infty} \beta_k < +\infty, \sum_{k=1}^{+\infty} \xi_k < +\infty \right\}. \quad (370)$$

Theorem 7.3. *Let $w^0 \in \mathbb{R}^n$ and assume that:*

- *the function $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ is L -smooth for some $L \in (0, +\infty)$,*
- *there exists $D \in [0, +\infty)$ such that for all $j = 1, 2, \dots, M$, $x \in \mathbb{R}^n$,*

$$\|\nabla \mathcal{L}_j(x)\| \leq D, \quad (371)$$

- *for the sequence $\{\eta_k\}_{k \geq 0}$ of positive real numbers it holds*

$$\sum_{k=0}^{+\infty} \eta_k = +\infty, \sum_{k=0}^{+\infty} \eta_k^2 < +\infty, \quad (372)$$

- there exists $\mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)]$ such that for all $k \geq 0$

$$\mathcal{L}(w^k) \geq \mathcal{L}^*, \quad (373)$$

with probability one.

Then the sequence $\{\mathcal{L}(w^k)\}_{k \geq 0}$ is convergent almost surely.

Proof. Since we assume (373), we have that for all $k \geq 0$ that

$$z_k = \mathcal{L}(w^k) - \mathcal{L}^* \geq 0 \quad (374)$$

almost surely, $\sigma(z_k) \subset \mathcal{F}_k$, and $\mathbb{E}(\mathcal{L}(w^k)) \geq \mathcal{L}^*$. Hence, by Theorem 7.1 we get that $\mathbb{E}\left(\sum_{k=0}^{+\infty} \eta_k \cdot \|\nabla \mathcal{L}(w^k)\|^2\right) < +\infty$. Moreover, by (342) we have for all $k \geq 0$ that

$$z_{k+1} \leq z_k + U_k, \quad (375)$$

almost surely. This implies that for all $M \in \mathbb{N}$

$$0 \leq z_M \leq z_0 + \sum_{k=0}^{M-1} U_k. \quad (376)$$

Moreover, for all $k \geq 0$ and almost surely

$$|U_k| \leq \eta_k D^2 (1 + L\eta_k/2), \quad (377)$$

so $\mathbb{E}|U_k| < +\infty$ and, by (376), $\mathbb{E}(z_k) < +\infty$. By (343), (348), and (349) we have for all $k \geq 0$ that

$$\mathbb{E}(U_k \mid \mathcal{F}_k) \leq -\eta_k \|\nabla \mathcal{L}(w^k)\|^2 + C\eta_k^2, \quad (378)$$

and this gives

$$\mathbb{E}(z_{k+1} \mid \mathcal{F}_k) \leq z_k + C\eta_k^2 - \eta_k \|\nabla \mathcal{L}(w^k)\|^2. \quad (379)$$

Hence, by applying Theorem 7.2 we get that there exists finite

$$\lim_{k \rightarrow +\infty} z_k = \lim_{k \rightarrow +\infty} (\mathcal{L}(w^k) - \mathcal{L}^*)$$

almost surely. This implies the thesis. $\square$

Example. (*Boundedness of gradients for the binary cross-entropy loss function*)

By (123) and Lemma 3.2 (i) we get for all $w \in \mathbb{R}^N$, $j = 1, \dots, M$

$$\|\nabla \ell_j(w)\| \leq (|y_j| + |\varphi(\langle x_j, w \rangle)|) \cdot \|x_j\| \leq 2 \max_{1 \leq j \leq M} \|x_j\| < +\infty, \quad (380)$$

since $y_j \in \{0, 1\}$. Therefore, for the fixed training dataset, the gradients of ℓ_j are uniformly bounded. Furthermore, the loss (119) is L -smooth and nonnegative. Hence, Theorem 7.3 implies the a.s. convergence of the stochastic gradient descent procedure.

Example. Consider the mean square loss function

$$\mathcal{L}(w) = \frac{1}{K} \sum_{j=1}^K \|y_j - \mathcal{N}^F(w, x_j)\|^2, \quad (381)$$

where $w \in \mathbb{R}^n$, $x_j \in \mathbb{R}^N$, $y_j \in \mathbb{R}^m$, and $\mathcal{N}^F : \mathbb{R}^n \times \mathbb{R}^N \rightarrow \mathbb{R}^m$ is the feedforward neural network. We have that

$$\nabla \mathcal{L}(w) = \frac{2}{K} \sum_{j=1}^K (\mathcal{N}^F(w, x_j) - y_j)^T \nabla \mathcal{N}^F(w, x_j), \quad (382)$$

and hence, by the Cauchy-Schwarz inequality,

$$\begin{aligned} \|\nabla \mathcal{L}(w)\|^2 &\leq \frac{4}{K^2} \sum_{j=1}^K \|y_j - \mathcal{N}^F(w, x_j)\|^2 \cdot \sum_{j=1}^K \|\nabla \mathcal{N}^F(w, x_j)\|^2 \\ &= \frac{4}{K} \sum_{j=1}^K \|\nabla \mathcal{N}^F(w, x_j)\|^2 \cdot \mathcal{L}(w). \end{aligned} \quad (383)$$

Moreover, for the function (381) we have

$$\mathcal{L}(w) \leq 2 \max_{1 \leq j \leq K} \|y_j\|^2 + \frac{2}{K} \sum_{j=1}^K \|\mathcal{N}^F(w, x_j)\|^2. \quad (384)$$

Assume that the data points $\{(x_j, y_j)\}_{j=1}^K$ are i.i.d. samples drawn from an unknown distribution Q . Then, by the strong law of large numbers, we have almost surely

$$\frac{1}{K} \sum_{j=1}^K \|\nabla \mathcal{N}^F(w, x_j)\|^2 \longrightarrow \mathbb{E} \|\nabla \mathcal{N}^F(w, X)\|^2, \quad K \rightarrow \infty, \quad (385)$$

$$\frac{1}{K} \sum_{j=1}^K \|\mathcal{N}^F(w, x_j)\|^2 \longrightarrow \mathbb{E} \|\mathcal{N}^F(w, X)\|^2, \quad K \rightarrow \infty, \quad (386)$$

assuming that $\mathbb{E} \|\nabla \mathcal{N}^F(w, X)\|^2 < +\infty$, $\mathbb{E} \|\mathcal{N}^F(w, X)\|^2 < +\infty$. The preceding estimates motivate the following heuristic. Assume the training data are normalised, and the network $\mathcal{N}^F(w, \cdot)$ has a gradient that mean-square average $\frac{1}{K} \sum_{j=1}^K \|\nabla \mathcal{N}^F(w, x_j)\|^2$ remains bounded for sufficiently large K . Then there exists a constant $C \geq 0$ such that, for every $w \in \mathbb{R}^n$,

$$\|\nabla \mathcal{L}(w)\|^2 \leq C \mathcal{L}(w). \quad (387)$$

(See Theorem 6.1 on page 180 in [26], where the assumption (387) is used to prove convergence of the stochastic gradient descent.) If, in addition, the neural network $\mathcal{N}^F(w, x)$ is uniformly bounded as a function of (w, x) , then the loss gradient $\nabla \mathcal{L}$ is bounded as well. Consequently, with a sufficiently large training set and a suitably chosen architecture, the loss (381) might possess a bounded gradient.

7.3 Gradient clipping

In supervised learning, the very common case is that the loss function $\mathcal{L}$, which we aim to minimize, is non-negative. Moreover, from the proof of Theorem 7.1 we see that we need the boundedness (371) of the gradients $\nabla \mathcal{L}_j$ only on the SGD sequence $\{w_k\}_{k \geq 0}$. In practice, we assure such a sequential boundedness of the gradients by using the procedure called *gradient clipping*, see [43].

From a mathematical point of view, gradient clipping is a procedure of applying to the computed gradient the following function

$$R_D(x) = \begin{cases} x, & \text{if } \|x\| \leq D, \\ D \frac{x}{\|x\|}, & \text{if } \|x\| \geq D. \end{cases} \quad (388)$$

where $D \in (0, +\infty)$ is a fixed *threshold*. Function R_D is a *retraction*, i.e. a continuous mapping from $\mathbb{R}^n$ to $B(0, D) = \{x \in \mathbb{R}^n \mid \|x\| \leq D\}$, such that $R_D(x) = x$ for all $x \in B(0, D)$. Then the gradient clipping procedure applied to the SGD sequence results in

$$\bar{w}^{k+1} = \bar{w}^k - \eta_k \cdot R_D(\nabla \mathcal{L}_{J_{k+1}}(\bar{w}^k)), \quad k = 0, 1, \dots \quad (389)$$

The main properties of the function R_D are as follows.

Proposition 7.1. ([1], [15])

(i) For all $x, y \in \mathbb{R}^n$

$$\|R_D(x) - R_D(y)\| \leq \|x - y\|, \quad (390)$$

$$\|R_D(x)\| = \min\{\|x\|, D\}, \quad (391)$$

$$\|x - R_D(x)\| = \max\{0, \|x\| - D\}, \quad (392)$$

$$\|R_D(x)\| + \|x - R_D(x)\| = \|x\|, \quad (393)$$

$$(D - \|R_D(x)\|) \cdot \|x - R_D(x)\| = 0 \quad (394)$$

and

$$\langle x, R_D(x) \rangle = \|x\| \cdot \min\{\|x\|, D\} \geq (\min\{\|x\|, D\})^2. \quad (395)$$

(ii) R_D is a metric projection on the ball $B(0, D)$, i.e., for all $x \in \mathbb{R}^n$

$$R_D(x) = \{y \in B(0, D) \mid \|x - y\| = \text{dist}(x, B(0, D))\}, \quad (396)$$

$$\text{and } \text{dist}(x, B(0, D)) = \inf_{z \in B(0, D)} \|x - z\|.$$

There are also other methods of gradient clipping, see [43].

7.4 Minibatch stochastic gradient descent

For a function of the form (313), instead of selecting a single randomly chosen $\mathcal{L}_j$ at each iteration, we can sample a mini-batch of functions $\{\mathcal{L}_{j_1}, \dots, \mathcal{L}_{j_b}\}$, where b denotes the batch size. In this case, having a subset B of indices $\{1, 2, \dots, M\}$ we approximate the full gradient $\nabla \mathcal{L}$ by

$$\nabla \mathcal{L}_B(x) = \frac{1}{|B|} \sum_{j \in B} \nabla \mathcal{L}_j(x), \quad (397)$$

where the batch size $|B|$ is equal to b . Note that, when training neural networks, the batch size b is another hyperparameter that must be chosen before the training procedure begins.

For a fixed batch size b the number $C_{b,M}$ of b -elements subsets of an M -elements set $\{1, 2, \dots, M\}$ is equal to $C_{b,M} = \binom{M}{b}$. Each b -elements subset $B \subset \{1, 2, \dots, M\}$ we call a *mini-batch*. On a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ we consider the random variable $\mathcal{B} : \Omega \rightarrow \{B_1, B_2, \dots, B_{C_{b,M}}\}$, where the set $\{B_1, B_2, \dots, B_{C_{b,M}}\}$ contains all possible minibatches of size b . We assume that $\mathcal{B}$ is uniformly distributed on $\{B_1, B_2, \dots, B_{C_{b,M}}\}$. Hence,

$$\mathbb{P}(\mathcal{B} = B_j) = \frac{1}{C_{b,M}}, \quad (398)$$

for $j = 1, 2, \dots, C_{b,M}$, and we write $\mathcal{B} \sim U(\{B_1, B_2, \dots, B_{C_{b,M}}\})$. Fix $\{\eta_k\}_{k \geq 0}$, $\{\mathcal{B}_k\}_{k \geq 1}$ -an iid sequence from $U(\{B_1, B_2, \dots, B_{C_{b,M}}\})$, $w^0 \in \mathbb{R}^n$, then for $k = 0, 1, \dots$

$$w^{k+1} = w^k - \eta_k \cdot \nabla \mathcal{L}_{\mathcal{B}_{k+1}}(w^k) = w^k - \eta_k \cdot \frac{1}{|\mathcal{B}_{k+1}|} \sum_{j \in \mathcal{B}_{k+1}} \nabla \mathcal{L}_j(w^k). \quad (399)$$

We left as an exercise showing that

$$\mathbb{E}(\nabla \mathcal{L}_{\mathcal{B}}(x)) = \frac{1}{C_{b,M}} \sum_{B \in \{B_1, B_2, \dots, B_{C_{b,M}}\}} \nabla \mathcal{L}_B(x) = \nabla \mathcal{L}(x), \quad x \in \mathbb{R}^n. \quad (400)$$

For the convergence theorem of minibatch SGD, under the assumptions of L -smoothness and convexity, we refer to [12]. In the next section, we provide a general setting that allows us to investigate some convergence properties of minibatch SGD under weaker assumptions than those imposed in [12].

Note that if we take $b = M$, then we get the full batch GD as in (209). If $b = 1$, then we are using the SGD algorithm (318), which is also known as *online SGD*. In practice, we assume $1 \leq b \ll M$.

8 On convergence of gradient descent algorithm with random perturbations

As mentioned in (325), the SGD algorithm can be viewed as a gradient descent algorithm corrupted by random noise. In this section, we explore this idea in more detail. Namely, we provide a suitable setting that allows us to investigate the convergence of the random iterative schemes that solve the optimization problem (208). As in the case of the deterministic gradient descent algorithm, we provide analysis under weaker assumptions than those commonly assumed in the literature, see, for example Theorem 6.1 in [26] or Chapter 9 in [14].

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with a filtration $\{\mathcal{F}_k\}_{k \geq 0}$. We consider the following iterative scheme: take $w^0 \in \mathbb{R}^n$ and set

$$w^{k+1} = w^k - \eta_k \nabla \mathcal{L}(w^k) + \eta_k Z_k, \quad k = 0, 1, \dots \quad (401)$$

We impose the following assumptions:

(B1) $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ and $\nabla \mathcal{L}$ satisfies the upper one-sided Lipschitz condition, i.e.: there exists $L \in [0, +\infty)$ such that for all $x, y \in \mathbb{R}^n$

$$\langle \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y), x - y \rangle \leq L \|x - y\|^2. \quad (402)$$

(B2) For all $k \geq 0$

$$\eta_k \in (0, 1/L]. \quad (403)$$

(B3) For all $k \geq 0$

$$\mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 < +\infty. \quad (404)$$

(B4) For all $k \geq 0$

$$\begin{aligned} \sigma(Z_k) &\subset \mathcal{F}_{k+1}, \\ \mathbb{E} \|Z_k\|^2 &< +\infty, \\ \mathbb{E}(Z_k \mid \mathcal{F}_k) &= 0. \end{aligned} \quad (405)$$

(B5) (*Stochastic noise control*) With probability one

$$\sum_{k=0}^{+\infty} \eta_k^2 \cdot \mathbb{E}(\|Z_k\|^2 \mid \mathcal{F}_k) < +\infty. \quad (406)$$

(B6) There exists $\mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)]$ such that for all $k \geq 0$

$$\mathcal{L}(w^k) \geq \mathcal{L}^*, \quad (407)$$

with probability one.

By (B4) and the induction, we get for the sequence (401) that

$$\sigma(w^k) \subset \mathcal{F}_k, \quad (408)$$

and from (B3)

$$\mathbb{E} |\langle \nabla \mathcal{L}(w^k), Z_k \rangle| \leq (\mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2)^{1/2} \cdot (\mathbb{E} \|Z_k\|^2)^{1/2} < +\infty. \quad (409)$$

Hence, by the properties of conditional expectation

$$\mathbb{E} \left(\langle \nabla \mathcal{L}(w^k), Z_k \rangle \mid \mathcal{F}_k \right) = \langle \nabla \mathcal{L}(w^k), \mathbb{E}(Z_k \mid \mathcal{F}_k) \rangle = 0, \quad (410)$$

and

$$\mathbb{E} \left(\langle \nabla \mathcal{L}(w^k), Z_k \rangle \right) = 0, \quad (411)$$

for all $k \geq 0$. From (B1) and Lemma 13.3

$$\begin{aligned} & \mathcal{L}(w^{k+1}) - \mathcal{L}(w^k) \\ & \leq -\alpha(\eta_k) \|\nabla \mathcal{L}(w^k)\|^2 + \gamma(\eta_k) \langle \nabla \mathcal{L}(w^k), Z_k \rangle + \frac{L}{2} \eta_k^2 \|Z_k\|^2 \\ & \leq -\frac{1}{2} \eta_k \|\nabla \mathcal{L}(w^k)\|^2 + \gamma(\eta_k) \langle \nabla \mathcal{L}(w^k), Z_k \rangle + \frac{L}{2} \eta_k^2 \|Z_k\|^2, \end{aligned} \quad (412)$$

where

$$\begin{aligned} \alpha(\eta) &= \eta \left(1 - \frac{L\eta}{2}\right) \\ \gamma(\eta) &= \eta(1 - L\eta), \end{aligned} \quad (413)$$

and $0 \leq \eta/2 \leq \alpha(\eta)$, $\gamma(\eta) \geq 0$ for $\eta \in (0, 1/L]$. By (B6) we have that

$$z_k = \mathcal{L}(w^k) - \mathcal{L}^* \geq 0 \quad (414)$$

for all $k \geq 0$ with probability one. Therefore, from (412) we have for all $k \geq 0$ that

$$z_{k+1} - z_k \leq -\frac{1}{2} \eta_k \|\nabla \mathcal{L}(w^k)\|^2 + \gamma(\eta_k) \langle \nabla \mathcal{L}(w^k), Z_k \rangle + \frac{L}{2} \eta_k^2 \|Z_k\|^2, \quad (415)$$

with probability one. This implies for all $M \in \mathbb{N}$ that

$$\begin{aligned} 0 \leq z_M \leq z_0 - \frac{1}{2} \sum_{k=0}^{M-1} \eta_k \|\nabla \mathcal{L}(w^k)\|^2 + \sum_{k=0}^{M-1} \gamma(\eta_k) \langle \nabla \mathcal{L}(w^k), Z_k \rangle \\ + \frac{L}{2} \sum_{k=0}^{M-1} \eta_k^2 \|Z_k\|^2 \end{aligned} \quad (416)$$

with probability one. Hence, by (B3), (B4), (411) we get for all $M \geq 1$

$$0 \leq \mathbb{E}(z_M) \leq z_0 - \frac{1}{2} \sum_{k=0}^{M-1} \eta_k \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 + \frac{L}{2} \sum_{k=0}^{M-1} \eta_k^2 \mathbb{E} \|Z_k\|^2 < +\infty, \quad (417)$$

and

$$\mathbb{E}|\mathcal{L}(w^M)| \leq \mathbb{E}(z_M) + |\mathcal{L}^*| < +\infty. \quad (418)$$

Hence, from (410) we have for all $k \geq 0$

$$\mathbb{E}(z_{k+1} \mid \mathcal{F}_k) \leq z_k - \frac{1}{2} \eta_k \|\nabla \mathcal{L}(w^k)\|^2 + \frac{L}{2} \eta_k^2 \mathbb{E}(\|Z_k\|^2 \mid \mathcal{F}_k). \quad (419)$$

Using (B5) and Theorem 7.2 we have that $\lim_{k \rightarrow +\infty} z_k = \lim_{k \rightarrow +\infty} (\mathcal{L}(w^k) - \mathcal{L}^*)$ exists and is finite and $\sum_{k=0}^{+\infty} \eta_k \|\nabla \mathcal{L}(w^k)\|^2 < +\infty$ almost surely.

Let us assume now that

$$(B5') \quad \sum_{k=0}^{+\infty} \eta_k^2 \cdot \mathbb{E} \|Z_k\|^2 < +\infty, \quad (420)$$

which implies (B5), since

$$\mathbb{E} \left(\sum_{k=0}^{+\infty} \eta_k^2 \cdot \mathbb{E}(\|Z_k\|^2 \mid \mathcal{F}_k) \right) = \sum_{k=0}^{+\infty} \eta_k^2 \cdot \mathbb{E} \|Z_k\|^2 < +\infty. \quad (421)$$

From (412), (411), (418), (B3), and (B4) we get for all $k \geq 0$

$$\mathbb{E}(\mathcal{L}(w^{k+1})) \leq \mathbb{E}(\mathcal{L}(w^k)) - \frac{1}{2} \eta_k \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 + \frac{L}{2} \eta_k^2 \mathbb{E} \|Z_k\|^2. \quad (422)$$

Since by (B6) we get that $\mathbb{E}(\mathcal{L}(w^k)) \geq \mathcal{L}^*$ for all $k \geq 0$, from (B5') and Lemma 6.2 we obtain that there exist finite $\lim_{k \rightarrow +\infty} \mathbb{E}(\mathcal{L}(w^k))$

and $\sum_{k=0}^{+\infty} \eta_k \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 < +\infty$. Furthermore, from Tonelli's theorem

$$\begin{aligned} \mathbb{E} \left(\sum_{k=0}^{+\infty} \|w^{k+1} - w^k\|^2 \right) &\leq 2 \sum_{k=0}^{+\infty} \eta_k^2 \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 + 2 \sum_{k=0}^{+\infty} \eta_k^2 \mathbb{E} \|Z_k\|^2 \\ &\leq \frac{2}{L} \sum_{k=0}^{+\infty} \eta_k \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 + 2 \sum_{k=0}^{+\infty} \eta_k^2 \mathbb{E} \|Z_k\|^2 < +\infty. \end{aligned} \quad (423)$$

We summarize the obtained results in the following theorem.

Theorem 8.1. *Under the assumptions (B1)-(B6) we have for the sequence (401) that $\lim_{k \rightarrow +\infty} \mathcal{L}(w^k)$ exists and is finite and*

$$\sum_{k=0}^{+\infty} \eta_k \|\nabla \mathcal{L}(w^k)\|^2 < +\infty$$

almost surely. Moreover, if in addition the assumption (B5') holds then there exist finite $\lim_{k \rightarrow +\infty} \mathbb{E}(\mathcal{L}(w^k))$ and

$$\begin{aligned} \sum_{k=0}^{+\infty} \eta_k \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 &< +\infty, \\ \mathbb{E} \left(\sum_{k=0}^{+\infty} \|w^{k+1} - w^k\|^2 \right) &< +\infty. \end{aligned} \quad (424)$$

From (424) we have that the increments $\{w^{k+1} - w^k\}_{k \geq 0}$ converge to 0 in $L^2(\Omega)$ and $\sum_{k=0}^{+\infty} \|w^{k+1} - w^k\|^2 < +\infty$ almost surely, which in turns implies that $\{w^{k+1} - w^k\}_{k \geq 0}$ also almost surely converges to 0.

The following Corollary also states some stationarity of the sequence $\{w^k\}_{k \geq 0}$.

Corollary 8.1. *If the assumptions (B1)-(B6) and (B5') holds, and if*

$$\sum_{k=0}^{+\infty} \eta_k = +\infty, \quad (425)$$

then for all $M \geq 1$

$$\min_{0 \leq k \leq M-1} \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 \leq \frac{2(\mathcal{L}(w^0) - \mathcal{L}^*) + L \sum_{k=0}^{M-1} \eta_k^2 \mathbb{E} \|Z_k\|^2}{\sum_{k=0}^{M-1} \eta_k}, \quad (426)$$

$$\liminf_{k \rightarrow +\infty} \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 = 0, \quad (427)$$

and

$$\liminf_{k \rightarrow +\infty} \|\nabla \mathcal{L}(w^k)\| = 0 \quad (428)$$

almost surely.

From (401) we have that for all $k \geq 1$

$$w^k = w^0 - \sum_{j=0}^{k-1} \eta_j \nabla \mathcal{L}(w^j) + \sum_{j=0}^{k-1} \eta_j Z_j. \quad (429)$$

Using the martingale convergence theorem, we show that the noise term $\sum_{j=0}^{k-1} \eta_j Z_j$ does not explode when $k \rightarrow +\infty$. Let us denote

$$M_k = \sum_{j=0}^{k-1} \eta_j Z_j, \quad k \geq 1, \quad (430)$$

where we set $M_0 := 0$. Note that for all $k \geq 0$

$$\sigma(M_k) \subset \mathcal{F}_k, \quad (431)$$

$$\mathbb{E}(M_{k+1} - M_k \mid \mathcal{F}_k) = \eta_k \cdot \mathbb{E}(Z_k \mid \mathcal{F}_k) = 0, \quad (432)$$

and

$$\|M_k\|^2 \leq \left(\sum_{j=0}^{k-1} \eta_j \|Z_j\| \right)^2 \leq \sum_{j=0}^{k-1} \eta_j^2 \cdot \sum_{j=0}^{k-1} \|Z_j\|^2, \quad (433)$$

$$\mathbb{E}\|M_k\|^2 \leq \sum_{j=0}^{k-1} \eta_j^2 \cdot \sum_{j=0}^{k-1} \mathbb{E}\|Z_j\|^2 < +\infty. \quad (434)$$

Hence, $(M_k, \mathcal{F}_k)_{k \geq 0}$ is a square integrable martingale. We can therefore define the quadratic variation process

$$\langle M \rangle_k = \sum_{j=0}^{k-1} \mathbb{E}(\|M_{j+1} - M_j\|^2 \mid \mathcal{F}_j) = \sum_{j=0}^{k-1} \eta_j^2 \cdot \mathbb{E}(\|Z_j\|^2 \mid \mathcal{F}_j). \quad (435)$$

Since, by (B5), with probability one,

$$\langle M \rangle_\infty := \lim_{k \rightarrow +\infty} \langle M \rangle_k = \sum_{j=0}^{+\infty} \eta_j^2 \cdot \mathbb{E}(\|Z_j\|^2 \mid \mathcal{F}_j) < +\infty, \quad (436)$$

we get by Theorem 12.7 in [26] that there exists a finite random vector M_∞ such that

$$\lim_{k \rightarrow +\infty} M_k = M_\infty \quad \text{almost surely.} \quad (437)$$

Moreover, under the assumption (B5'), we have that

$$\mathbb{E}(\langle M \rangle_\infty) = \sum_{j=0}^{+\infty} \eta_j^2 \cdot \mathbb{E}\|Z_j\|^2 < +\infty, \quad (438)$$

and $\mathbb{E}(\|M_\infty\|^2) < +\infty$. Hence, asymptotically, the noise term M_k stabilizes almost surely and in the mean square sense.

8.1 On the convergence of the minibatch SGD algorithm

Here we use the framework introduced to investigate the convergence of the minibatch SGD method. Firstly, note that we can write (399) as (401), where

$$Z_k = \nabla \mathcal{L}(w^k) - \nabla \mathcal{L}_{\mathcal{B}_{k+1}}(w^k), \quad k \geq 0. \quad (439)$$

Then $\sigma(Z_k) \subset \mathcal{F}_{k+1}$, where $\mathcal{F}_0 = \{\emptyset, \Omega\}$, $\mathcal{F}_k = \sigma(\mathcal{B}_1, \dots, \mathcal{B}_k)$, $k \geq 1$. Moreover, for $k \geq 0$ we get $\sigma(w^k) \subset \mathcal{F}_k$, and

$$\mathbb{E}(Z_k \mid \mathcal{F}_k) = \nabla \mathcal{L}(w^k) - \mathbb{E}(\nabla \mathcal{L}_{\mathcal{B}_{k+1}}(w)) \Big|_{w=w^k} = 0, \quad (440)$$

$$\begin{aligned} \mathbb{E}(\|Z_k\|^2 \mid \mathcal{F}_k) &= \mathbb{E}(\langle Z_k, Z_k \rangle \mid \mathcal{F}_k) \\ &= \mathbb{E} \left[\|\nabla \mathcal{L}(w^k)\|^2 - 2\langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}_{\mathcal{B}_{k+1}}(w^k) \rangle \right. \\ &\quad \left. + \|\nabla \mathcal{L}_{\mathcal{B}_{k+1}}(w^k)\|^2 \mid \mathcal{F}_k \right] \\ &= \mathbb{E} \left[\|\nabla \mathcal{L}_{\mathcal{B}_{k+1}}(w)\|^2 \Big|_{w=w^k} - \|\nabla \mathcal{L}(w^k)\|^2 \right] \leq \mathbb{E} \left[\|\nabla \mathcal{L}_{\mathcal{B}_{k+1}}(w)\|^2 \Big|_{w=w^k} \right]. \end{aligned} \quad (441)$$

Hence, under the assumptions (B1), (B2), (B6), (330), and (372) the thesis of Theorem 8.1, and Corollary 8.1 follow.

9 Stochastic approximation algorithm as gradient descent method with random perturbations

We turn to the more general case, when we want to minimize the loss function $\mathcal{L}$ that is given by expectation. Namely, we assume that

$$\mathcal{L}(w) = \mathbb{E}(H(w, J)), \quad w \in \mathbb{R}^n, \quad (442)$$

for a Borel measurable function $H : \mathbb{R}^n \times \mathbb{R}^p \rightarrow \mathbb{R}$ and some random vector $J : \Omega \rightarrow \mathbb{R}^p$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. A stochastic optimization algorithm has the form: $w^0 \in \mathbb{R}^n$ and

$$w^{k+1} = w^k - \eta_k \nabla H(w^k, J_{k+1}), \quad k \geq 0, \quad (443)$$

where $\{J_k\}_{k \geq 1}$ is an iid sequence of random vectors with the same law as J , and $\nabla H(w^k, J_{k+1}) = \nabla H(w, J_{k+1})|_{w=w^k}$ is the gradient of H with respect to w . A classical result on the convergence of such a scheme is due to Robbins and Monro, see, for example, [14], [26]. In this chapter, we demonstrate that the stochastic optimization algorithm can be rewritten as a gradient descent method corrupted by some random noise (401). It will allow us to apply the results from Section 8. To the best of our knowledge, this approach and the assumptions we impose differ slightly from those commonly used in the literature, see, for example, [26]. We believe that our approach provides additional insights into the topic, as we assume only an upper one-sided Lipschitz condition for the loss function $\mathcal{L}$.

We start with some heuristics. Assume that H is sufficiently regular so that

$$\nabla \mathcal{L}(w) = \nabla \mathbb{E}(H(w, J)) = \mathbb{E}(\nabla H(w, J)), \quad (444)$$

then we can write (443) as (401) with

$$Z_k = \nabla \mathcal{L}(w^k) - \nabla H(w^k, J_{k+1}). \quad (445)$$

If we take $\mathcal{F}_0 = \{\emptyset, \Omega\}$ and $\mathcal{F}_k = \sigma(J_1, \dots, J_k)$, $k \geq 1$, then $\sigma(w^k) \subset \mathcal{F}_k$, $\sigma(Z_k) \subset \mathcal{F}_{k+1}$, and by (444)

$$\mathbb{E}(Z_k \mid \mathcal{F}_k) = \nabla \mathcal{L}(w^k) - \mathbb{E}[\nabla H(w, J_{k+1})]_{w=w^k} = 0, \quad (446)$$

since $\sigma(J_{k+1})$ and $\mathcal{F}_k$ are independent. We now investigate the stochastic optimization algorithm rigorously.

We impose the following conditions.

(H1) For all $w \in \mathbb{R}^n$

$$\mathbb{E}|H(w, J)| < +\infty. \quad (447)$$

(H2) For almost all $\omega \in \Omega$ there exist

$$\frac{\partial H}{\partial w_j}(w, J(\omega)) \quad (448)$$

for all $j = 1, 2, \dots, n$, $w \in \mathbb{R}^n$.

(H3) There exist positive random variables Y_j such that $\mathbb{E}Y_j < +\infty$ for $j = 1, \dots, n$ and for almost all $\omega \in \Omega$ and all $w \in \mathbb{R}^n$

$$\left| \frac{\partial H}{\partial w_j}(w, J(\omega)) \right| \leq Y_j(\omega). \quad (449)$$

(H4) For all $k \geq 0$

$$\mathbb{E} \left[\mathbb{E} \|\nabla H(w, J_{k+1})\|^2 \Big|_{w=w^k} \right] < +\infty. \quad (450)$$

(H5) The function $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ and $\nabla \mathcal{L}$ satisfies the upper one-sided Lipschitz condition, i.e.: there exists $L \in [0, +\infty)$ such that for all $x, y \in \mathbb{R}^n$

$$\langle \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y), x - y \rangle \leq L \|x - y\|^2. \quad (451)$$

(H6) For all $k \geq 0$

$$\eta_k \in (0, 1/L]. \quad (452)$$

(H7) (*Stochastic noise control*) With probability one

$$\sum_{k=0}^{+\infty} \eta_k^2 \cdot \mathbb{E} \|\nabla H(w, J_{k+1})\|^2 \Big|_{w=w^k} < +\infty. \quad (453)$$

(H8) There exists $\mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)]$ such that for all $k \geq 0$

$$\mathcal{L}(w^k) \geq \mathcal{L}^*, \quad (454)$$

with probability one.

From (H1)-(H3) and according to Theorem 2.2 in [26] we get that for all $w \in \mathbb{R}^n$

$$|\mathcal{L}(w)| \leq \mathbb{E}|H(w, J)| < +\infty, \quad (455)$$

and (444) holds. Moreover, by the Schwarz inequality and (H4)

$$\mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 \leq \mathbb{E} \left[\mathbb{E} \|\nabla H(w, J_{k+1})\|^2 \Big|_{w=w^k} \right] < +\infty. \quad (456)$$

We also have

$$\mathbb{E} \left[\|\nabla H(w, J_{k+1})\|^2 \Big|_{w=w^k} \right] = \mathbb{E} \left[\mathbb{E} \|\nabla H(w, J_{k+1})\|^2 \Big|_{w=w^k} \right] < +\infty, \quad (457)$$

$$\begin{aligned} & \mathbb{E} |\langle \nabla \mathcal{L}(w^k), \nabla H(w^k, J_{k+1}) \rangle| \\ & \leq (\mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2)^{1/2} \cdot (\mathbb{E} \|\nabla H(w^k, J_{k+1})\|^2)^{1/2} < +\infty, \end{aligned} \quad (458)$$

and

$$\begin{aligned} \mathbb{E} \|Z_k\|^2 & \leq 2\mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 + 2\mathbb{E} \left[\|\nabla H(w, J_{k+1})\|^2 \Big|_{w=w^k} \right] \\ & \leq 4\mathbb{E} \left[\mathbb{E} \|\nabla H(w, J_{k+1})\|^2 \Big|_{w=w^k} \right] < +\infty, \end{aligned} \quad (459)$$

for all $k \geq 0$. Hence, almost surely

$$\begin{aligned} & \mathbb{E}(\|Z_k\|^2 \mid \mathcal{F}_k) = \mathbb{E}(\langle Z_k, Z_k \rangle \mid \mathcal{F}_k) \\ & = \mathbb{E} \left[\|\nabla \mathcal{L}(w^k)\|^2 - 2\langle \nabla \mathcal{L}(w^k), \nabla H(w^k, J_{k+1}) \rangle \right. \\ & \quad \left. + \|\nabla H(w^k, J_{k+1})\|^2 \mid \mathcal{F}_k \right] \\ & = \|\nabla \mathcal{L}(w^k)\|^2 - 2\langle \nabla \mathcal{L}(w^k), \mathbb{E}(\nabla H(w^k, J_{k+1}) \mid \mathcal{F}_k) \rangle \\ & \quad + \mathbb{E} \left[\|\nabla H(w^k, J_{k+1})\|^2 \mid \mathcal{F}_k \right] \\ & = \mathbb{E} \left[\|\nabla H(w^k, J_{k+1})\|^2 \mid \mathcal{F}_k \right] - \|\nabla \mathcal{L}(w^k)\|^2 \\ & = \mathbb{E} \left[\|\nabla H(w, J_{k+1})\|^2 \Big|_{w=w^k} - \|\nabla \mathcal{L}(w^k)\|^2 \right] \leq \mathbb{E} \|\nabla H(w, J_{k+1})\|^2 \Big|_{w=w^k}, \end{aligned}$$

and

$$\begin{aligned} \mathbb{E} \|Z_k\|^2 & = \mathbb{E} \left[\mathbb{E}(\|Z_k\|^2 \mid \mathcal{F}_k) \right] \\ & = \mathbb{E} \left[\mathbb{E} \|\nabla H(w, J_{k+1})\|^2 \Big|_{w=w^k} - \|\nabla \mathcal{L}(w^k)\|^2 \right] \leq \mathbb{E} \left[\mathbb{E} \|\nabla H(w, J_{k+1})\|^2 \Big|_{w=w^k} \right]. \end{aligned}$$

Under assumption (H7), we have that

$$\sum_{k=0}^{+\infty} \eta_k^2 \mathbb{E}(\|Z_k\|^2 \mid \mathcal{F}_k) \leq \sum_{k=0}^{+\infty} \eta_k^2 \mathbb{E} \left[\|\nabla H(w, J_{k+1})\|^2 \Big|_{w=w^k} \right] < +\infty \quad (460)$$

almost surely. If we additionally assume that

$$(H7') \quad \sum_{k=0}^{+\infty} \eta_k^2 \mathbb{E} \left[\mathbb{E} \|\nabla H(w, J_{k+1})\|^2 \Big|_{w=w^k} \right] < +\infty, \quad (461)$$

then also

$$\sum_{k=0}^{+\infty} \eta_k^2 \mathbb{E} \|Z_k\|^2 < +\infty. \quad (462)$$

Hence, we can apply Theorem 8.1 to get the following result.

Theorem 9.1. *Under the assumptions (H1)-(H8) we have for the sequence (443) that $\lim_{k \rightarrow +\infty} \mathcal{L}(w^k)$ exists and is finite and*

$$\sum_{k=0}^{+\infty} \eta_k \|\nabla \mathcal{L}(w^k)\|^2 < +\infty$$

almost surely. Moreover, if in addition the assumption (H7') holds then there exist finite $\lim_{k \rightarrow +\infty} \mathbb{E}(\mathcal{L}(w^k))$ and

$$\begin{aligned} \sum_{k=0}^{+\infty} \eta_k \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 &< +\infty, \\ \mathbb{E} \left(\sum_{k=0}^{+\infty} \|w^{k+1} - w^k\|^2 \right) &< +\infty. \end{aligned} \quad (463)$$

Corollary 9.1. *If the assumptions (H1)-(H8) and (H7') holds, and if*

$$\sum_{k=0}^{+\infty} \eta_k = +\infty, \quad (464)$$

then for all $M \geq 1$

$$\begin{aligned} \min_{0 \leq k \leq M-1} \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 &\leq \frac{2(\mathcal{L}(w^0) - \mathcal{L}^*) + L \sum_{k=0}^{M-1} \eta_k^2 \mathbb{E} \left[\mathbb{E} \|\nabla H(w, J_{k+1})\|^2 \Big|_{w=w^k} \right]}{\sum_{k=0}^{M-1} \eta_k}, \\ \liminf_{k \rightarrow +\infty} \mathbb{E} \|\nabla \mathcal{L}(w^k)\|^2 &= 0, \end{aligned} \quad (465)$$

and

$$\liminf_{k \rightarrow +\infty} \|\nabla \mathcal{L}(w^k)\| = 0 \quad (466)$$

almost surely.

Remark 9.1. Taking a discrete random variable $J \sim U(\{1, 2, \dots, M\})$, the loss function (313) can be written in the form (442).

Remark 9.2. See Section 11.3 where we apply the stochastic approximation algorithm (443) to approximate the solution of the ODE system.

10 Continuous learning and gradient systems

The following system of ordinary differential equations

$$\begin{cases} w'(t) = -\nabla \mathcal{L}(w(t)), & t \geq 0, \\ w(0) = w^0 \end{cases} \quad (467)$$

is called the *gradient system*. We have two complementary viewpoints:

- (V1) Gradient descent (209) can be regarded as the explicit-Euler discretization (at the time points $t_k = k\eta$) of the continuous-time gradient flow (467), when the discretization stepsize is equal to η .
- (V2) Conversely, the gradient flow (467) arises as the limit of gradient descent (209) when the learning rate $\eta \rightarrow 0+$.

To show (V2) define the following continuous piecewise linear interpolation of the (GD) steps (209). For fixed $T, \eta > 0$ take $k_{\max}^\eta = \lceil T/\eta \rceil - 1 < T/\eta \leq \lceil T/\eta \rceil$, $t_k = k\eta$, and let

$$w_\eta(t) = w_{k,\eta}(t), \quad t \in [t_k, t_{k+1}], \quad k = 0, 1, \dots, k_{\max}^\eta, \quad (468)$$

with

$$w_{k,\eta}(t) = w_{k,\eta} - (t - t_k)\nabla \mathcal{L}(w_{k,\eta}), \quad (469)$$

where the (GD) steps are given by

$$w_{0,\eta} = w^0, \quad (470)$$

$$w_{k+1,\eta} = w_{k,\eta} - \eta \nabla \mathcal{L}(w_{k,\eta}). \quad (471)$$

Proposition 10.1. *Let $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$ be L -smooth and $w^0 \in \mathbb{R}^n$. Then for any $T \in (0, +\infty)$*

$$\lim_{\eta \rightarrow 0+} \sup_{0 \leq t \leq T} \|w(t) - w_\eta(t)\| = 0. \quad (472)$$

Proof. Take any $T, \eta > 0$. Set $F(x) := -\nabla \mathcal{L}(x)$ and define

$$\bar{w}_\eta(t) = w_{k,\eta}, \quad t \in [t_k, t_{k+1}), \quad k = 0, 1, \dots, k_{\max}^\eta. \quad (473)$$

Then we can write for all $t \in [0, k_{\max}^\eta + 1]$ that

$$w_\eta(t) = w^0 + \int_0^t F(\bar{w}_\eta(s)) ds. \quad (474)$$

Since for all $t \in [0, T] \subset [0, k_{\max}^\eta + 1]$

$$w(t) = w^0 + \int_0^t F(w(s)) ds, \quad (475)$$

we get for all $t \in [0, T]$ that

$$\begin{aligned} \|w(t) - w_\eta(t)\| &\leq L \int_0^t \|w(s) - \bar{w}_\eta(s)\| ds \leq L \int_0^t \|w(s) - w_\eta(s)\| ds + L \int_0^t \|w_\eta(s) - \bar{w}_\eta(s)\| ds \\ &\leq L \int_0^t \|w(s) - w_\eta(s)\| ds + LB_\eta, \end{aligned} \quad (476)$$

where

$$\begin{aligned}
B_\eta &= \int_0^{t_{k_{\max}^\eta}^\eta} \|w_\eta(s) - \bar{w}_\eta(s)\| ds = \sum_{j=0}^{k_{\max}^\eta} \int_{t_j}^{t_{j+1}} \|w_{j,\eta}(s) - w_{j,\eta}\| ds \\
&= \sum_{j=0}^{k_{\max}^\eta} \int_{t_j}^{t_{j+1}} (s - t_j) \|F(w_{j,\eta})\| ds.
\end{aligned} \tag{477}$$

Since $\mathcal{L}$ is L -smooth, we get for all $x \in \mathbb{R}^n$ that

$$\|F(x)\| = \|\nabla \mathcal{L}(x)\| \leq M_1(1 + \|x\|), \tag{478}$$

where $M_1 = \max\{L, \|\nabla \mathcal{L}(0)\|\}$. Hence, for all $k = 0, 1, \dots, k_{\max}^\eta - 1$

$$\|w_{k+1,\eta}\| \leq \|w_{k,\eta}\| + \eta \|F(w_{k,\eta})\| \leq (1 + \eta M_1) \|w_{k,\eta}\| + \eta M_1, \tag{479}$$

and by the discrete version of the Gronwall's lemma, we get for $k = 0, 1, \dots, k_{\max}^\eta$

$$\begin{aligned}
\|w_{k,\eta}\| &\leq (1 + \eta M_1)^k \|w^0\| + \frac{(1 + \eta M_1)^k - 1}{\eta M_1} \cdot \eta M_1 \\
&\leq e^{k\eta M_1} \|w^0\| + e^{k\eta M_1} - 1 \leq e^{TM_1} \|w^0\| + e^{TM_1} - 1 =: M_2.
\end{aligned} \tag{480}$$

From (477), (478), (480) we obtain that

$$\begin{aligned}
B_\eta &\leq M_1 \sum_{j=0}^{k_{\max}^\eta} \int_{t_j}^{t_{j+1}} (s - t_j) (1 + \|w_{j,\eta}\|) ds \leq \frac{1}{2} M_1 (1 + M_2) \sum_{j=0}^{k_{\max}^\eta} (t_{j+1} - t_j)^2 \\
&= \frac{1}{2} M_1 (1 + M_2) (k_{\max}^\eta + 1) \eta^2 \leq \frac{1}{2} M_1 (1 + M_2) \left(\frac{T}{\eta} + 1\right) \eta^2 = \frac{1}{2} M_1 (1 + M_2) (T + \eta) \eta.
\end{aligned} \tag{481}$$

By (476) and (481) we have for all $t \in [0, T]$ that

$$\|w(t) - w_\eta(t)\| \leq L \int_0^t \|w(s) - w_\eta(s)\| ds + \frac{L}{2} M_1 (1 + M_2) (T + \eta) \eta, \tag{482}$$

and by the continuous version of the Gronwall's lemma we get

$$\sup_{0 \leq t \leq T} \|w(t) - w_\eta(t)\| \leq \frac{L}{2} e^{LT} M_1 (1 + M_2) (T + \eta) \eta. \tag{483}$$

This gives the thesis. $\square$

We refer to the iterative computation of the sequence $\{w^k\}_{k \geq 0}$ generated by gradient descent as *discrete learning* (or simply *learning*), whereas solving the continuous-time gradient flow (467) is called *continuous learning*.

10.1 Stability of the gradient systems

One of the features of (467) is that every critical point w^* of $\mathcal{L}$ defines a stationary solution of the gradient system: for the trajectory $w(t) \equiv w^*$ with $t \geq 0$ we have

$$w'(t) = 0 = -\nabla \mathcal{L}(w(t)) = -\nabla \mathcal{L}(w^*), \tag{484}$$

for all $t \geq 0$. Hence, *analyzing the properties of (467) offers an additional mathematical lens to study the landscape of the loss function $\mathcal{L}$.*

We start by recalling some theorems on the existence and uniqueness of solutions to (467). For the proof of the following theorem see [4], page 47.

Theorem 10.1. *If the function $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ satisfies the following conditions:*

(C1)

$$\exists_{\gamma_1, \gamma_2 \in \mathbb{R}} : \forall_{x \in \mathbb{R}^n} -\langle x, \nabla \mathcal{L}(x) \rangle \leq \gamma_1 \|x\|^2 + \gamma_2, \quad (485)$$

(C2) $\mathbb{R}^n \ni x \rightarrow -\nabla \mathcal{L}(x) \in \mathbb{R}^n$ is locally Lipschitz,

there exists an unique solution $w \in C^1([0, +\infty), \mathbb{R}^n)$ to (467).

The following theorem follows from Lemma 3.19, page 165 in [27].

Theorem 10.2. *If for the function $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ there exists $K \in [0, +\infty)$ such that for all $x, y \in \mathbb{R}^n$*

$$\langle x - y, \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y) \rangle \geq -K \|x - y\|^2, \quad (486)$$

there exists an unique solution $w \in C^1([0, +\infty), \mathbb{R}^n)$ to (467).

Since every L -smooth function is weakly convex, from the theorem above we get the following corollary.

Theorem 10.3. *If $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth then there exists an unique solution $w \in C^1([0, +\infty), \mathbb{R}^n)$ to (467).*

Gradient systems of the form (467) with a convex $\mathcal{L}$ are often referred to in the literature as *dissipative systems*, see [4].

We now prove an analogous theorem to the convergence of the GD algorithm in the case of L -smooth loss functions $\mathcal{L}$.

Theorem 10.4. *Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$, $w^0 \in \mathbb{R}^n$ and let $w \in C^1([0, +\infty); \mathbb{R}^n)$ be (any) solution to (467). Then*

$$(A) \quad \int_0^{+\infty} \|\nabla \mathcal{L}(w(t))\|^2 dt = +\infty \Leftrightarrow \lim_{t \rightarrow +\infty} \mathcal{L}(w(t)) = -\infty,$$

$$(B) \quad \int_0^{+\infty} \|\nabla \mathcal{L}(w(t))\|^2 dt < +\infty \Leftrightarrow \exists_{\mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)]} : \forall_{t \geq 0} \mathcal{L}(w(t)) \geq \mathcal{L}^*,$$

(C) either

$$(C1) \quad \lim_{t \rightarrow +\infty} \mathcal{L}(w(t)) = -\infty,$$

or

$$(C2) \quad \exists_{\mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)]} : \forall_{t \geq 0} \mathcal{L}(w(t)) \geq \mathcal{L}^*.$$

In the (C2)-case we additionally have:

- there exist $\lim_{t \rightarrow +\infty} \mathcal{L}(w(t)) \in \mathbb{R}$, and $\inf_{t \geq 0} \|\nabla \mathcal{L}(w(t))\| = 0$,
- the following estimate holds

$$\exists_{C^* \in [0, +\infty)} : \forall_{t \geq 0} \inf_{0 \leq s \leq t} \|\nabla \mathcal{L}(w(s))\| \leq C^* / \sqrt{t}. \quad (487)$$

Proof. Let us define the function $\varphi : [0, +\infty) \rightarrow \mathbb{R}$ as

$$\varphi(t) = \mathcal{L}(w(t)). \quad (488)$$

Then

$$\varphi'(t) = \langle \nabla \mathcal{L}(w(t)), w'(t) \rangle = -\|\nabla \mathcal{L}(w(t))\|^2 \leq 0, \quad (489)$$

for all $t \geq 0$ and the function $[0, +\infty) \ni t \rightarrow \mathcal{L}(w(t)) \in \mathbb{R}$ is non-increasing. Since $\varphi' \in C([0, +\infty)); \mathbb{R}$, we get for all $t \geq 0$ that

$$\varphi(t) = \varphi(0) + \int_0^t \varphi'(s) ds, \quad (490)$$

so

$$\mathcal{L}(w(t)) = \mathcal{L}(w^0) - \int_0^t \|\nabla \mathcal{L}(w(s))\|^2 ds. \quad (491)$$

The equality (491) gives the first equivalence (A). We now pass to the proof of (B).

($\Rightarrow$) If $\int_0^{+\infty} \|\nabla \mathcal{L}(w(t))\|^2 dt < +\infty$ then for all $t \geq 0$

$$\int_0^t \|\nabla \mathcal{L}(w(s))\|^2 ds \leq \int_0^{+\infty} \|\nabla \mathcal{L}(w(s))\|^2 ds < +\infty, \quad (492)$$

and by (491)

$$\mathcal{L}(w(t)) \geq \mathcal{L}^* := \mathcal{L}(w^0) - \int_0^{+\infty} \|\nabla \mathcal{L}(w(s))\|^2 ds > -\infty. \quad (493)$$

($\Leftarrow$) Assume that $\exists \mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)] : \forall t \geq 0, \mathcal{L}(w(t)) \geq \mathcal{L}^*$. Then from (491) we have for all $t \geq 0$

$$0 \leq \int_0^t \|\nabla \mathcal{L}(w(s))\|^2 ds = \mathcal{L}(w^0) - \mathcal{L}(w(t)) \leq \mathcal{L}(w^0) - \mathcal{L}^* < +\infty, \quad (494)$$

so the function $[0, +\infty) \ni t \rightarrow \int_0^t \|\nabla \mathcal{L}(w(s))\|^2 ds \in [0, +\infty)$ is non-decreasing, and bounded. Therefore

$$\int_0^{+\infty} \|\nabla \mathcal{L}(w(s))\|^2 ds = \sup_{t \geq 0} \int_0^t \|\nabla \mathcal{L}(w(s))\|^2 ds \leq \mathcal{L}(w^0) - \mathcal{L}^* < +\infty. \quad (495)$$

In the (C2)-case we have that the function $[0, +\infty) \ni t \rightarrow \mathcal{L}(w(t)) \in \mathbb{R}$ is non-increasing and bounded from below. Therefore, the exists finite limit $\lim_{t \rightarrow +\infty} \mathcal{L}(w(t))$. Moreover, by Lemma 13.9 we have that $\inf_{t \geq 0} \|\nabla \mathcal{L}(w(t))\| = 0$. In addition, from (494) for all $t \geq 0$

$$t \cdot \inf_{0 \leq s \leq t} \|\nabla \mathcal{L}(w(s))\|^2 \leq \int_0^t \|\nabla \mathcal{L}(w(s))\|^2 ds \leq \mathcal{L}(w^0) - \mathcal{L}^*, \quad (496)$$

from which (487) follows. $\square$

The following proposition follows directly from Theorem 10.4 and Barbălat's Lemma (page 269 in [3]).

Proposition 10.2. *Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$, $w^0 \in \mathbb{R}^n$ and let $w \in C^1([0, +\infty); \mathbb{R}^n)$ be (any) solution to (467). If $\exists \mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)] : \forall t \geq 0, \mathcal{L}(w(t)) \geq \mathcal{L}^*$ and the function $[0, +\infty) \ni t \rightarrow \|\nabla \mathcal{L}(w(t))\|^2 \in [0, +\infty)$ is uniformly continuous then*

$$\lim_{t \rightarrow +\infty} \nabla \mathcal{L}(w(t)) = 0. \quad (497)$$

In the next theorem, we study the stability of the stationary solution $w(t) \equiv w^*$, where w^* is a critical point of $\mathcal{L}$. Our argument assumes *at least* weak convexity of $\mathcal{L}$ - treating more general situations typically calls for a Lyapunov-function approach.

Theorem 10.5. *Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$, $w^0 \in \mathbb{R}^n$ and, $\nabla \mathcal{L}(w^*) = 0$. Then:*

(i) *if $\mathcal{L}$ is K -weakly convex then for all $t \geq 0$*

$$\|w(t) - w^*\| \leq e^{Kt} \|w^0 - w^*\|. \quad (498)$$

(ii) *if $\mathcal{L}$ is convex then for all $t \geq 0$*

$$\|w(t) - w^*\| \leq \|w^0 - w^*\|, \quad (499)$$

and there exists $\lim_{t \rightarrow +\infty} \|w(t) - w^\|$ (so the stationary solution $y(t) = w^*$ is stable); Moreover, if $\mathcal{L} \in C^2(\mathbb{R}^n; \mathbb{R})$ then $\lim_{t \rightarrow +\infty} \nabla \mathcal{L}(w(t)) = 0$,*

(iii) *if $\mathcal{L}$ is strongly convex then*

$$\lim_{t \rightarrow +\infty} \|w(t) - w^*\| = 0, \quad (500)$$

and $\lim_{t \rightarrow +\infty} \nabla \mathcal{L}(w(t)) = 0$ (so the stationary solution $y(t) = w^$ is asymptotically stable).*

The proof relies on the following version of Gronwall's lemma.

Lemma 10.1. *Assume that a differentiable function $\varphi : [0, +\infty) \rightarrow \mathbb{R}$ satisfies the following condition:*

$$\exists H \in \mathbb{R} : \forall t \geq 0 \varphi'(t) \leq H \varphi(t). \quad (501)$$

Then, for all $t \geq 0$

$$\varphi(t) \leq e^{Ht} \varphi(0). \quad (502)$$

Proof. Since $e^{-Ht} > 0$, from (501) we have that

$$\begin{aligned} \frac{d}{dt}(e^{-Ht} \varphi(t)) &= -H e^{-Ht} \varphi(t) + e^{-Ht} \varphi'(t) \\ &= e^{-Ht} (\varphi'(t) - H \varphi(t)) \leq 0. \end{aligned} \quad (503)$$

Hence, the function $[0, +\infty) \ni t \rightarrow e^{-Ht} \varphi(t)$ is non-increasing. This implies for all $t \geq 0$ that

$$e^{-Ht} \varphi(t) \leq \varphi(0), \quad (504)$$

and the proof is complete. $\square$

Proof of Theorem 10.5. By Theorem 10.2 if $\mathcal{L}$ is weakly convex, then there exists a unique solution w of (467).

For the function $\varphi(t) = \|w(t) - w^*\|^2$, $t \geq 0$, we have

$$\begin{aligned} \varphi'(t) &= \frac{d}{dt} \langle w(t) - w^*, w(t) - w^* \rangle = 2 \langle w'(t), w(t) - w^* \rangle \\ &= 2 \langle (-\nabla \mathcal{L}(w(t))) - (-\nabla \mathcal{L}(w^*)), w(t) - w^* \rangle. \end{aligned} \quad (505)$$

If $\mathcal{L}$ is K -weakly convex for some $K \geq 0$, then for all $t \geq 0$

$$\varphi'(t) \leq 2K \varphi(t), \quad (506)$$

and Lemma 10.1 gives (498).

If $\mathcal{L}$ is convex, then w^* is a global minimum of $\mathcal{L}$, and for all $t \geq 0$

$$\varphi'(t) \leq 0, \quad (507)$$

which gives (499). Moreover, φ is non-increasing and non-negative. Hence, there exists $\lim_{t \rightarrow +\infty} \varphi(t) \in [0, +\infty)$. If $\mathcal{L} \in C^2$, then due to the convexity of $\mathcal{L}$ we have that for all $x, z \in \mathbb{R}^n$

$$z^T \nabla^2 \mathcal{L}(x) z \geq 0. \quad (508)$$

Hence, for all $t \geq 0$

$$w''(t) = -\frac{d}{dt} \nabla \mathcal{L}(w(t)) = -\nabla^2 \mathcal{L}(w(t)) w'(t), \quad (509)$$

and

$$\langle w'(t), w''(t) \rangle = -(w'(t))^T \nabla^2 \mathcal{L}(w(t)) w'(t) \leq 0. \quad (510)$$

Therefore,

$$\frac{d}{dt} \|w'(t)\|^2 = 2 \langle w'(t), w''(t) \rangle \leq 0, \quad (511)$$

and the function $[0, +\infty) \ni t \rightarrow \|w'(t)\|^2$ is non-increasing. Therefore, $[0, +\infty) \ni t \rightarrow \|\nabla \mathcal{L}(w(t))\|$ is non-increasing and bounded from below. Hence, there exists finite $\lim_{t \rightarrow +\infty} \|\nabla \mathcal{L}(w(t))\| = \inf_{t \geq 0} \|\nabla \mathcal{L}(w(t))\|$. Furthermore, since for all $t \geq 0$ we have $\mathcal{L}(w^0) \geq \mathcal{L}(w(t)) \geq \mathcal{L}(w^*) > -\infty$, we have from Theorem 10.4 (C2) that

$$\lim_{t \rightarrow +\infty} \|\nabla \mathcal{L}(w(t))\| = \inf_{t \geq 0} \|\nabla \mathcal{L}(w(t))\| = 0. \quad (512)$$

Finally, if $\mathcal{L}$ is K -strongly convex for some $K > 0$, then w^* is the unique global minimum of $\mathcal{L}$, and for all $t \geq 0$

$$\varphi'(t) \leq -2K\varphi(t). \quad (513)$$

By Lemma 10.1 we have for all $t \geq 0$

$$\|w(t) - w^*\| \leq e^{-Kt} \|w^0 - w^*\|, \quad (514)$$

and this implies that $\lim_{t \rightarrow +\infty} \|w(t) - w^*\| = 0$. Hence, by the continuity of $\nabla \mathcal{L}$ we obtain

$$\lim_{t \rightarrow +\infty} \nabla \mathcal{L}(w(t)) = \nabla \mathcal{L}(\lim_{t \rightarrow +\infty} w(t)) = \nabla \mathcal{L}(w^*) = 0. \quad (515)$$

□

In the next theorem, we will explore the stability of the gradient systems in the case when the gradient $\nabla \mathcal{L}$ is L -smooth and of Polyak-Łojasiewicz type. Again, under such assumptions, direct analysis is possible without referring to Lyapunov stability theory.

Theorem 10.6. *Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ be L -smooth for some $L \in (0, +\infty)$ and of Polyak-Łojasiewicz type for some $\mu \in (0, +\infty)$. Then*

(i) $\mu \in (0, L]$,

(ii) the set of all global minimums of $\mathcal{L}$ is non-empty, i.e. $\mathcal{M} := \{w^* \in \mathbb{R}^n \mid \mathcal{L}(w^*) = \mathcal{L}^*\} \neq \emptyset$, where $\mathcal{L}^* := \inf_{y \in \mathbb{R}^n} \mathcal{L}(y) > -\infty$,

(iii) for all $t \geq 0$, $w^0 \in \mathbb{R}^n$ it holds

$$(\mathcal{L}(w^0) - \mathcal{L}^*) \cdot e^{-2Lt} \leq \mathcal{L}(w(t)) - \mathcal{L}^* \leq (\mathcal{L}(w^0) - \mathcal{L}^*) \cdot e^{-2\mu t}, \quad (516)$$

there exists $\lim_{t \rightarrow +\infty} w(t)$ and it belongs to $\mathcal{M}$.

Proof. Since $\mathcal{L}$ is of PL-type and L -smooth, we get by Lemma 13.4 that for all $t \geq 0$

$$\mu(\mathcal{L}(w(t)) - \mathcal{L}^*) \leq \frac{1}{2} \|\nabla \mathcal{L}(w(t))\|^2 \leq L(\mathcal{L}(w(t)) - \mathcal{L}^*), \quad (517)$$

where $w = w(t)$ is the unique solution of the gradient flow (467) under the initial condition $w(0) = w^0$. Hence, $\mu \in (0, L]$. Defining by $\psi(t) = \mathcal{L}(w(t)) - \mathcal{L}^*$, we obtain $\psi'(t) = -\|\nabla \mathcal{L}(w(t))\|^2$ and by (517)

$$-2L\psi(t) \leq \psi'(t) \leq -2\mu\psi(t), \quad (518)$$

for all $t \geq 0$. Applying Lemma 10.1 to the functions ψ and $-\psi$ we get for all $t \geq 0$

$$e^{-2Lt}\psi(0) \leq \psi(t) \leq e^{-2\mu t}\psi(0), \quad (519)$$

from which (516) follows.

By (517) and (516) we get for all $t \geq 0$

$$\int_0^t \|\nabla \mathcal{L}(w(s))\| ds \leq c_0 \int_0^t e^{-\mu s} ds = \frac{c_0}{\mu} (1 - e^{-\mu t}). \quad (520)$$

Hence,

$$\int_0^{+\infty} \|\nabla \mathcal{L}(w(s))\| ds = \lim_{t \rightarrow +\infty} \int_0^t \|\nabla \mathcal{L}(w(s))\| ds \leq c_0/\mu < +\infty, \quad (521)$$

where $c_0 = (2L(\mathcal{L}(w^0) - \mathcal{L}^*))^{1/2}$, and therefore

$$w^* := \lim_{t \rightarrow +\infty} w(t) = w^0 - \int_0^{+\infty} \nabla \mathcal{L}(w(s)) ds. \quad (522)$$

From (516), (522) and by the continuity of $\mathcal{L}$

$$\mathcal{L}^* = \lim_{t \rightarrow +\infty} \mathcal{L}(w(t)) = \mathcal{L}(w^*), \quad (523)$$

hence, $w^* \in \mathcal{M}$. Finally, we have for all $t \geq 0$

$$\|w^* - w(t)\| \leq \int_t^{+\infty} \|\nabla \mathcal{L}(w(s))\| ds \leq \frac{c_0}{\mu} e^{-\mu t}. \quad (524)$$

□

By using the Lyapunov function approach, one can show the following general result (We refer to page 142 in [4] for the proof.).

Theorem 10.7. *Let $\mathcal{L} \in C^2(\mathbb{R}^n; \mathbb{R})$. If w^* is an isolated critical point of $\mathcal{L}$ which is also a local minimum, then the stationary solution $y(t) = w^*$ is an asymptotically stable solution of the gradient system (467).*

Theorem 10.7 implies that if the initial value w^0 is sufficiently close to the (isolated) local minimum w^* of $\mathcal{L}$, then $\lim_{t \rightarrow +\infty} w(t) = w^*$.

The next theorem states that closed orbits are impossible in gradient systems because the dynamics follow the downhill direction of a scalar potential, so trajectories cannot cycle back to previously visited states (see also Theorem 7.2.3, page 223 in [39]). For machine-learning optimizers such as (stochastic) gradient descent, this property guarantees that, absent stochastic noise, training will not get stuck in endless oscillations. That

assurance underpins convergence proofs and lets practitioners reason about learning rates, momentum terms, and step sizes without worrying about limit-cycle attractors. In deep networks, where the parameter space is high-dimensional and non-linear, knowing that pure gradient flow (467) cannot orbit means any observed cycling must come from factors like momentum, adaptive scaling, or stochasticity. This gives us insights that guide the design of more stable/efficient optimizers.

Theorem 10.8. *Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ and $w^0 \in \mathbb{R}^n$. If $w \in C^1([0, +\infty); \mathbb{R}^n)$ is periodic (with period equal to some $T > 0$) and if it satisfies (467), then $\forall t \geq 0, w(t) = w^0$ and w^0 is the critical point of $\mathcal{L}$.*

Proof. Since there exists $T > 0$ such that $w(t + T) = w(t)$ for all $t \geq 0$, we have $w(T) = w(0) = w^0$ and

$$0 = \mathcal{L}(w(T)) - \mathcal{L}(w(0)) = - \int_0^T \|\nabla \mathcal{L}(w(s))\|^2 ds = - \int_0^T \|w'(s)\|^2 ds. \quad (525)$$

Because $[0, +\infty) \ni t \rightarrow \|w'(t)\|^2 \in [0, +\infty)$ is continuous, we get that $w'(t) = 0$ for all $t \in [0, T]$. Therefore, for all $t \in [0, T]$

$$w(t) = w^0 + \int_0^t w'(s) ds = w^0. \quad (526)$$

By the periodicity of w we have for all $t \in [0, T]$

$$w(t + T) = w(t) = w^0, \quad (527)$$

then for all $t \in [T, 2T]$

$$w(t + T) = w(t) = w^0, \quad (528)$$

and by the induction we get that $w(t) = w^0$ for all $t \in [0, +\infty) = \bigcup_{n \geq 0} [nT, (n+1)T]$. This also implies that

$$0 = w'(t) = -\nabla \mathcal{L}(w(t)) = -\nabla \mathcal{L}(w^0), \quad (529)$$

so w^0 is the critical point of $\mathcal{L}$. $\square$

10.2 Discontinuous gradient systems

In classical gradient-flow theory, one assumes that the loss landscape is smooth enough for the gradient $\nabla \mathcal{L}$ to be continuous, so the associated ordinary differential equation (467) defines a well-posed dynamical system whose trajectories inherit desirable properties such as uniqueness and continuous dependence on initial data. The moment we build a neural network with non-differentiable activation functions, such as ReLU, this framework breaks down: $\nabla \mathcal{L}$ may be undefined on entire manifolds or exhibit jump discontinuities. Two complementary strategies are commonly adopted to restore a rigorous analytical footing:

- (1) **Filippov differential inclusions** - One treats the gradient flow as an ODE with a discontinuous right-hand side and replaces it by the so-called set-valued *Filippov extension*, see [11]. Existence (and sometimes uniqueness) of solutions is then recovered by invoking general results for differential inclusions. This viewpoint aligns naturally with notions of Fréchet or Clarke subgradients used in modern deep-learning theory.
- (2) **Regularization by noise** - Alternatively, one perturbs the system with small stochastic forcing, yielding the stochastic differential equation (SDE)

$$dw(t) = -\nabla \mathcal{L}(w(t)) dt + \sigma(w(t)) dW_t,$$

where W_t is Brownian motion and σ is chosen so that the drift remains meaningful even where $\nabla \mathcal{L}$ is discontinuous. The noise “fills in” the gaps, ensuring well-posedness and often imparting favorable regularizing effects (e.g. smoothing of invariant measures, avoidance of spurious minima), see, for example, [40]. In practice, this mirrors the behaviour of stochastic gradient descent with mini-batch noise, see, for example, [17] or Theorem 10.3, page 334 in [5].

Both perspectives highlight that, once ReLU or similar activations enter the picture, it is no longer sufficient to analyse smooth gradient flows - one must apply the non-smooth analysis, whether in the deterministic Filippov sense or in the stochastic-regularised setting.

10.3 Approximation of gradient systems

Another key feature of gradient systems (467) is that applying numerical discretisation schemes to them naturally yields new algorithms for minimising the loss function $\mathcal{L}$. As we have already mentioned, by utilizing Euler discretization of (467) on the mesh $t_k = k\eta$, $k = 0, 1, 2, \dots$ we obtain the gradient descent algorithm (209). If we use two-stage Runge-Kutta discretization of (467) on the non-equidistant mesh $t_k = k\eta_k$, then we get the following optimization scheme

$$\begin{aligned} w^0 &\in \mathbb{R}^n, \\ w^{k+\frac{1}{2}} &= w^k - \frac{1}{2}\eta_k \nabla \mathcal{L}(w^k), \\ w^{k+1} &= w^k - \eta_k \nabla \mathcal{L}(w^{k+\frac{1}{2}}), \quad k = 0, 1, 2, \dots \end{aligned} \quad (530)$$

In the literature, this method is also referred to as explicit midpoint GD optimization, see [18]. We can use the most famous 4th-order Runge-Kutta scheme to obtain

$$\begin{aligned} w^0 &\in \mathbb{R}^n, \\ w^{k+1} &= w^k + \frac{\eta_k}{6}(K_1 + 2K_2 + 2K_3 + K_4), \quad k = 0, 1, \dots, \end{aligned} \quad (531)$$

where in each step

$$\begin{aligned} K_1 &= -\nabla \mathcal{L}(w^k), \\ K_2 &= -\nabla \mathcal{L}(w^k + \frac{\eta_k}{2}K_1), \\ K_3 &= -\nabla \mathcal{L}(w^k + \frac{\eta_k}{2}K_2), \\ K_4 &= -\nabla \mathcal{L}(w^k + \eta_k K_3). \end{aligned}$$

Furthermore, numerical analysis already offers a rich toolbox of *randomised algorithms* for ordinary differential equations. By applying these solvers to the gradient flow (467), we automatically obtain new stochastic optimisation algorithms whose update rules inherit the accuracy and stability guarantees of the underlying randomised ODE schemes. For example, by using randomized two-stage Runge-Kutta method to (467) we arrive at

$$\begin{aligned} w^0 &\in \mathbb{R}^n, \\ w_\tau^k &= w^k - \eta_k \tau_{k+1} \nabla \mathcal{L}(w^k), \\ w^{k+1} &= w^k - \eta_k \nabla \mathcal{L}(w_\tau^k), \quad k = 0, 1, 2, \dots \end{aligned} \quad (532)$$

where $\{\tau_k\}_{k \geq 1}$ is an iid sequence from the uniform distribution $U(0, 1)$. We call this method *randomized two-stage gradient descent*. Analogously, we can define the following *randomized two-stage stochastic gradient descent method* for the functions (313):

$$\begin{aligned} w^0 &\in \mathbb{R}^n, \\ w_\tau^k &= w^k - \eta_k \tau_{k+1} \nabla \mathcal{L}_{\xi_j}(w^k), \\ w^{k+1} &= w^k - \eta_k \nabla \mathcal{L}_{\zeta_j}(w_\tau^k), \quad j = 1, 2, \dots, \end{aligned} \quad (533)$$

where $(\tau_k)_{k \in \mathbb{N}}$ is an iid sequence sampled from $U(0, 1)$, $(\xi_j)_{j \geq 1}$ and $(\zeta_j)_{j \geq 1}$ are iid sequences sampled from $U(\{1, 2, \dots, M\})$, and finally $(\tau_j)_{j \geq 1}$, $(\xi_j)_{j \geq 1}$, $(\zeta_j)_{j \geq 1}$ are independent. See also [32] for other examples of approximation schemes that turn out to be efficient optimization algorithms.

10.3.1 Convergence of explicit midpoint gradient descent algorithm

In what follows, we utilize the framework established in Section 6.6 to demonstrate the convergence of (530). Assume that $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ is L -smooth for some $L \in (0, +\infty]$, bounded from below and that there exists $D \geq 0$ such that for all $w \in \mathbb{R}^n$

$$\|\nabla \mathcal{L}(w)\| \leq D. \quad (534)$$

For the sequences of learning rates $\{\eta_k\}_{k \geq 0}$ we assume (299), (306) and

$$\sum_{k=0}^{+\infty} \eta_k^2 < +\infty. \quad (535)$$

We rewrite (530) as

$$w^{k+1} = w^k - \eta_k \nabla \mathcal{L}(w^k) + \eta_k Z_k, \quad (536)$$

with

$$Z_k = \nabla \mathcal{L}(w^k) - \nabla \mathcal{L}(w^{k+\frac{1}{2}}). \quad (537)$$

We have that

$$\|Z_k\| \leq L \|w^k - w^{k+\frac{1}{2}}\| \leq \frac{L}{2} \eta_k \|\nabla \mathcal{L}(w^k)\|, \quad (538)$$

and by (299)

$$\|Z_k\| \leq \frac{1}{2} \|\nabla \mathcal{L}(w^k)\|, \quad (539)$$

which implies that the sequence $\{\mathcal{L}(w^k)\}_{k \geq 0}$ is non-increasing, see Remark 6.5. Since $\mathcal{L}$ is bounded from below we get that there exist finite $\lim_{k \rightarrow +\infty} \mathcal{L}(w^k)$. We also have that

$$\|Z_k\| \leq \frac{LD}{2} \eta_k, \quad (540)$$

and hence

$$\sum_{k=0}^{+\infty} \eta_k \|Z_k\|^2 \leq \left(\frac{LD}{2}\right)^2 \frac{1}{L} \sum_{k=0}^{+\infty} \eta_k^2 < +\infty. \quad (541)$$

By Theorem 6.4 we get that $\liminf_{k \rightarrow +\infty} \|\nabla \mathcal{L}(w^k)\| = 0$.

11 Ordinary differential equations and artificial neural networks

11.1 Identification of ODE-based model via artificial neural networks

In this section, we show how to calibrate a model given by a system of differential equations to the observed data. Assume that the real data-generating process is deterministic and satisfies the following ODE

$$\begin{aligned} x'(t) &= f(t, x(t)), \quad t \in [a, b], \\ x(a) &= \eta. \end{aligned} \quad (542)$$

Hence, we assume that there exists (in general) an unknown function f such that we can interpret the observed data

$$x_0, x_1, \dots, x_M \quad (543)$$

to be the values of $x(t_k)$ at (observation) time points t_k , i.e.: $x_k = x(t_k)$ for all $k = 0, 1, \dots, M$, where $a = t_0 < t_1 < \dots < t_M = b$ and $\eta = x_0$. Our goal is to obtain an approximation to f based on the data set (543).

We assume that f belongs to the class of Lipschitz functions

$$\mathcal{F}_L = \{f \in C([a, b] \times \mathbb{R}^d; \mathbb{R}^d) \mid \forall_{x, y \in \mathbb{R}^d, t \in [a, b]} \|f(t, x) - f(t, y)\| \leq L\|x - y\|\} \quad (544)$$

where $L \in [0, +\infty)$ is the parameter of the class. (Note that if $f \in \mathcal{F}_L$, then (542) has a unique solution $x \in C^1([a, b]; \mathbb{R}^d)$.) We seek a suitable approximation to f that also belongs to the class $\mathcal{F}_L$. Firstly, we need to define a suitable loss function to solve the problem. Let

$$\bar{\mathcal{L}}(g) = \sum_{k=0}^{M-1} (\Delta t_k)^2 \cdot \|x'(t_k) - g(t_k, x(t_k))\|^2 \quad (545)$$

for any $g \in \mathcal{F}_L$, where $\Delta t_k = t_{k+1} - t_k$. If x satisfies (542) with the unknown right-hand side function f then

$$\inf_{g \in \mathcal{F}_L} \bar{\mathcal{L}}(g) = 0 = \bar{\mathcal{L}}(f). \quad (546)$$

However, there are two main issues with the loss function $\bar{\mathcal{L}}$:

- 1) it uses the values of the derivative $x'(t_k)$, which are not available in our setting (we have access only to the values of $x(t_k)$),
- 2) minimization of $\bar{\mathcal{L}}$ over the whole class $\mathcal{F}_L$ is an infinite dimensional problem.

To overcome the above issues, we use the following loss function

$$\mathcal{L}(w) = \frac{1}{M} \bar{\mathcal{L}}(\mathcal{N}^F(w, \cdot)), \quad w \in \mathbb{R}^N, \quad (547)$$

where $\mathcal{N}^F = \mathcal{N}^F(w, [t, x]) : \mathbb{R}^N \times \mathbb{R}^{d+1} \rightarrow \mathbb{R}^d$ is the deep feedforward artificial neural network, while

$$\tilde{\mathcal{L}}(g) = \sum_{k=0}^{M-1} (\Delta t_k)^2 \cdot \left\| \frac{\Delta x_k}{\Delta t_k} - g(t_k, x_k) \right\|^2, \quad (548)$$

with $\Delta x_k = x_{k+1} - x_k = x(t_{k+1}) - x(t_k)$. Then

$$\mathcal{L}(w) = \frac{1}{M} \sum_{k=0}^{M-1} (\Delta t_k)^2 \cdot \left\| \frac{\Delta x_k}{\Delta t_k} - \mathcal{N}^F(w, [t_k, x_k]) \right\|^2, \quad (549)$$

and we seek the optimal weights

$$w^* \in \operatorname{argmin}_{w \in \mathbb{R}^N} \mathcal{L}(w). \quad (550)$$

From the point of view of supervised learning, $\mathcal{L} = \mathcal{L}(w)$ is a weighted mean square loss function for the feedforward neural network $\mathcal{N}^F$ and the following training set

$$D = \left\{ \left([t_0, x_0], \frac{\Delta x_0}{\Delta t_0} \right), \dots, \left([t_{M-1}, x_{M-1}], \frac{\Delta x_{M-1}}{\Delta t_{M-1}} \right) \right\}, \quad (551)$$

where each $[t_j, x_j]$ is the input for $\mathcal{N}^F$, while $\frac{\Delta x_j}{\Delta t_j}$ is the label for the input $[t_j, x_j]$. After minimizing (549) we obtain $\mathcal{N}^F(w^*, \cdot)$ that approximates f .

Note that due to Proposition 5.1 the neural network $\mathcal{N}^F = \mathcal{N}^F(w, [t, x])$ is, for all w , Lipschitz function of $[t, x]$. However, the Lipschitz constant of $\mathcal{N}^F$ depends on the weights w . Hence, we need to use some regularization techniques to ensure that the Lipschitz constant does not grow during the minimization process. Hence, we use the following loss function

$$\mathcal{L}_\lambda(w) = \frac{1}{M} \sum_{k=0}^{M-1} (\Delta t_k)^2 \cdot \left\| \frac{\Delta x_k}{\Delta t_k} - \mathcal{N}^F(w, [t_k, x_k]) \right\|^2 + \lambda \|w\|^2, \quad (552)$$

with $\lambda > 0$. Minimization of the loss function above results in $\mathcal{N}^F(w^*, \cdot) \in \mathcal{F}_{L^*}$ for some $L^* \in [0, +\infty)$. However, we cannot ensure that $L^* = L$, since the Lipschitz constant of f is generally unknown.

After obtaining the optimal weights w^* , the goodness of fit might be tested in the following way. Compute the values $\{x_k^*\}_{k=0, \dots, M}$ given by the neural network-adjusted Euler scheme given by

$$\begin{aligned} x_0^* &= x_0 = \eta, \\ x_{k+1}^* &= x_k^* + \Delta t_k \cdot \mathcal{N}^F(w^*, [t_k, x_k^*]), \quad k = 0, 1, \dots, M-1, \end{aligned} \quad (553)$$

and compute

$$\frac{1}{M} \sum_{k=1}^M \|x_k - x_k^*\|^2. \quad (554)$$

Remark 11.1. We have

$$\mathcal{L}(w) = \frac{1}{M} \sum_{k=0}^{M-1} \|r_k(w)\|^2, \quad (555)$$

where

$$r_k(w) = x_{k+1} - x_k - \Delta t_k \cdot \mathcal{N}^F(w, [t_k, x_k]), \quad (556)$$

is the one-step Euler error obtained when the feedforward neural network $\mathcal{N}^F(w, [t_k, x_k])$ is used to approximate the right-hand-side function f . Moreover, we can use the (minibatch) SGD algorithm to minimize the loss (555).

11.2 Time-dependent parameter estimation for ODE-based models via ANNs

Consider the following system of ODEs

$$\begin{aligned} x'(t) &= f(t, x(t), \Theta(t)), \quad t \in [a, b], \\ x(a) &= \eta, \end{aligned} \quad (557)$$

where the right-hand side function $f : [a, b] \times \mathbb{R}^d \times \mathbb{R}^s \rightarrow \mathbb{R}^d$ is chosen in advance. The aim is to approximate the time-dependent parameter function $\Theta : [a, b] \rightarrow \mathbb{R}^s$ based on discrete observations $x_0, x_1, \dots, x_M$ of the function $x = x(t)$ at $t \in \{t_0, \dots, t_M\}$, where $a = t_0 < t_1 < \dots < t_M = b$. Such a situation may arise, for

example, when we wish to estimate time-dependent parameters of the epidemiological SIRD model, assuming we have access to the actual values of S, I, R , and D .

Let

$$\bar{\mathcal{L}}(\psi) = \sum_{k=0}^{M-1} (\Delta t_k)^2 \cdot \|x'(t_k) - f(t_k, x(t_k), \psi(t_k))\|^2 \quad (558)$$

for any $\psi \in B([a, b])$, where $B([a, b])$ is a Banach space of all Borel measurable and bounded, in the supremum, functions $\psi : [a, b] \rightarrow \mathbb{R}^s$. We have that

$$\inf_{\psi \in B([a, b])} \bar{\mathcal{L}}(\psi) = \bar{\mathcal{L}}(\Theta), \quad (559)$$

where Θ is the unknown parameter function that we want to estimate. Since we cannot evaluate the above loss function, we use the following one

$$\mathcal{L}(w) = \frac{1}{M} \tilde{\mathcal{L}}(\mathcal{N}^F(w, \cdot)), \quad w \in \mathbb{R}^N, \quad (560)$$

where $\mathcal{N}^F = \mathcal{N}^F(w, t) : \mathbb{R}^N \times \mathbb{R} \rightarrow \mathbb{R}^s$ is the deep feedforward artificial neural network, while

$$\tilde{\mathcal{L}}(\psi) = \sum_{k=0}^{M-1} (\Delta t_k)^2 \cdot \left\| \frac{\Delta x_k}{\Delta t_k} - f(t_k, x_k, \psi(t_k)) \right\|^2. \quad (561)$$

Hence,

$$\mathcal{L}(w) = \frac{1}{M} \sum_{k=0}^{M-1} (\Delta t_k)^2 \cdot \left\| \frac{\Delta x_k}{\Delta t_k} - f(t_k, x_k, \mathcal{N}^F(w, t_k)) \right\|^2, \quad w \in \mathbb{R}^N, \quad (562)$$

and we seek the optimal weights

$$w^* \in \operatorname{argmin}_{w \in \mathbb{R}^N} \mathcal{L}(w). \quad (563)$$

Then $\mathcal{N}^F(w^*, \cdot)$ is the neural network that approximates the parameter function $\Theta(\cdot)$. We can rewrite (562) as

$$\mathcal{L}(w) = \frac{1}{M} \sum_{k=0}^{M-1} \|R_k(w)\|^2, \quad (564)$$

with

$$R_k(w) = \Delta x_k - \Delta t_k \cdot f(t_k, x_k, \mathcal{N}^F(w, t_k)). \quad (565)$$

Hence, a (minibatch) SGD algorithm might be used to minimize (564).

Remark 11.2. The loss identifies Θ only up to observational equivalence. If two parameter functions Θ_1, Θ_2 satisfy

$$f(t, x(t), \Theta_1(t)) = f(t, x(t), \Theta_2(t))$$

along the observed trajectory, then they produce the same value of the loss. Therefore, minimizing the loss identifies Θ only up to observational equivalence unless some identifiability assumption is imposed.

Remark 11.3. See [7] where the authors calibrated the SIRD model, with one time-dependent parameter, to the actual data by minimizing a suitable loss function, however, without using neural networks. The approach described in this section provides a method that allows for considering and calibrating the SIRD model (and beyond) with all parameters being time-dependent.

11.3 PINNs - solving ODEs with artificial neural networks

Assume that $-\infty < a < b < +\infty$, $\eta \in \mathbb{R}^d$ and $f \in \mathcal{F}_L$ are fixed and we aim to approximate the solution $x \in C^1([a, b]; \mathbb{R}^d)$ of the initial value problem (542). Again, we wish to approximate x by a feedforward neural network.

Let us define the following nonlinear functional

$$\bar{\mathcal{L}}(z) = \|z(a) - \eta\|^2 + \int_a^b \|z'(t) - f(t, z(t))\|^2 dt, \quad (566)$$

for all $z \in C^1([a, b]; \mathbb{R}^d)$. Note that

$$\inf_{z \in C^1([a, b]; \mathbb{R}^d)} \bar{\mathcal{L}}(z) = 0 = \bar{\mathcal{L}}(x), \quad (567)$$

where the infimum is attained for the unique solution $x = x(\eta, f)$ of (542). Again, the minimization of (566) is an infinite-dimensional problem. Hence, we need to discretize both its form (we need to approximate the integral in (566)) and its domain. Hence, take $a = t_0 < \dots < t_M = b$ and consider the following custom loss function

$$\begin{aligned} \mathcal{L}(w) = & \|\mathcal{N}^F(w, a) - \eta\|^2 \\ & + \sum_{k=0}^{M-1} \Delta t_k \cdot \left\| \frac{\partial}{\partial t} \mathcal{N}^F(w, t)|_{t=t_k} - f(t_k, \mathcal{N}^F(w, t_k)) \right\|^2 \end{aligned} \quad (568)$$

that is used to adjust the weight of the neural network $\mathcal{N}^F : \mathbb{R}^N \times \mathbb{R} \rightarrow \mathbb{R}^d$ during the minimization process. We seek $w^* \in \operatorname{argmin}_{w \in \mathbb{R}^N} \mathcal{L}(w)$ and then $\mathcal{N}^F(w^*, \cdot) \approx x(\cdot)$. Note that for the equidistant discretization of the interval $[a, b]$, i.e., $t_k = a + kh$, $k = 0, 1, \dots, M$, $h = (b - a)/M$, we have

$$\mathcal{L}(w) = \|\mathcal{N}^F(w, a) - \eta\|^2 + \frac{1}{M} \sum_{k=0}^{M-1} \mathcal{L}_k(w), \quad (569)$$

where

$$\mathcal{L}_k(w) = (b - a) \cdot \left\| \frac{\partial}{\partial t} \mathcal{N}^F(w, t)|_{t=t_k} - f(t_k, \mathcal{N}^F(w, t_k)) \right\|^2. \quad (570)$$

Hence, the (minibatch) SGD algorithm is suitable to minimize loss (569).

Finally, we show how to define a second loss function that is an alternative to (569). Note that we can write

$$\bar{\mathcal{L}}(z) = \mathbb{E} \left[\|z(a) - \eta\|^2 + (b - a) \|z'(\tau) - f(\tau, z(\tau))\|^2 \right], \quad (571)$$

where $\tau \sim U(a, b)$. Therefore, we can consider the following loss function, which approximates $\bar{\mathcal{L}}$,

$$\mathcal{L}(w) = \mathbb{E}(H(w, \tau)), \quad (572)$$

where

$$H(w, \tau) = \|\mathcal{N}^F(w, a) - \eta\|^2 + (b - a) \left\| \frac{\partial}{\partial t} \mathcal{N}^F(w, \tau) - f(\tau, \mathcal{N}^F(w, \tau)) \right\|^2. \quad (573)$$

From Section 9 we know that to minimize (572) we can use the following version of the SGD algorithm

$$w^{k+1} = w^k - \eta_k \nabla H(w, \tau_{k+1}) \Big|_{w=w^k}, \quad k \geq 0, \quad (574)$$

where $\{\tau_k\}_{k \geq 1}$ is an iid sequence from $U(a, b)$. Note that this iterative scheme is close to the SGD algorithm applied to (569).

Remark 11.4. The combination of differential equations with artificial neural networks, briefly outlined in this chapter, is commonly referred to as *PINNs* (*Physics-Informed Neural Networks*), see [34], [18]. PINNs embed the governing physical laws (expressed as differential equations) directly into the loss function of the artificial neural network, which constrains the solution space and reduces the need for large training data sets. This framework has been applied to a wide range of forward and inverse problems, for example, fluid dynamics and heat transfer, material science, and biomedical modeling, etc. PINNs demonstrate both accuracy and data efficiency.

Remark 11.5. We refer to [2], where the authors introduced the stochastic PINNs (SPINNs) - a mathematical framework to approximate the solutions of stochastic differential equations (SDEs) driven by Lévy noise via artificial neural networks.

12 Time-dependent parameter estimation in the Black-Scholes model

In financial mathematics, one often considers the Black-Scholes model with time-dependent coefficients, i.e.:

$$dS(t) = \mu(t)S(t)dt + \sigma(t)S(t)dW(t), \quad t \in [0, T], \quad (575)$$

$$S(0) = S_0, \quad (576)$$

where $T, S_0 \in (0, +\infty)$, $\mu : [0, T] \rightarrow \mathbb{R}$, $\sigma : [0, T] \rightarrow (0, +\infty)$ are Borel measurable bounded functions, and $W = (W(t))_{t \in [0, T]}$ is a one-dimensional Wiener process. Assuming that we have observed the real prices of a risky asset

$$S(t_0), S(t_1), \dots, S(t_N), \quad (577)$$

at points $0 = t_0 < t_1 < \dots < t_N = T$ and assuming that the real prices follows the SDE (575), the aim is to estimate the functions μ, σ basing on the data (577). When μ and σ are constant on $[0, T]$, then the estimation of such coefficients can be made by well-known techniques, see, for example, [31]. The more challenging task is when we allow μ, σ to depend on t in a nonlinear way. In this case, we can use deep feedforward neural networks to model such time-dependent coefficients. In this section, we show a simplified approach adopted from [19], since we reformulate estimation of μ, σ as a *heteroscedastic regression* problem. In addition, we show two ways of discretizing the equation (575) that lead to two methods of estimating of the functions μ and σ^2 .

12.1 Arithmetic returns approach

We discretize the equation (575) assuming that the real data (577) follows the Euler-Maruyama scheme

$$S(t_{i+1}) = S(t_i) + \mu(t_i)S(t_i)\Delta t_i + \sigma(t_i)S(t_i)\Delta W_i, \quad (578)$$

$i = 0, 1, \dots, N-1$, where $\Delta W_i = W(t_{i+1}) - W(t_i)$, $\Delta t_i = t_{i+1} - t_i$. (This assumption is justified when we have access to a large number N of observations in $[0, T]$ and when $\max_{0 \leq i \leq N-1} \Delta t_i$ is small.) Since ΔW_i has the same distribution as $\sqrt{\Delta t_i}\xi_i$, where ξ_i is normally distributed with zero mean and unit variance, we can write (578) as

$$Y_i = \bar{\mu}(t_i) + \sigma(t_i)\xi_i, \quad i = 0, 1, \dots, N-1, \quad (579)$$

where

$$Y_i = \frac{\Delta S_i}{\sqrt{\Delta t_i}S(t_i)}, \quad (580)$$

$$\bar{\mu}(t_i) = \sqrt{\Delta t_i} \cdot \mu(t_i), \quad (581)$$

and $\{\xi_i\}_{i=0,1,\dots,N-1}$ is the iid sequence of random variable from $N(0, 1)$. (Note that $\sqrt{\Delta t_i}Y_i = \Delta S_i/S(t_i)$ are the arithmetic returns.) Hence, the random variables $Y_0, Y_1, \dots, Y_{N-1}$ are independent and

$$Y_i \sim N(\bar{\mu}(t_i), \sigma^2(t_i)). \quad (582)$$

This corresponds to the heteroscedastic regression, since the variance depends on time, see, for example, Section 4.3.3 in [10], or Section 5.3.4 in [30]. The joint density distribution of $[Y_0, Y_1, \dots, Y_{N-1}]$ is

$$f_{(Y_0, Y_1, \dots, Y_{N-1})}(y_0, y_1, \dots, y_{N-1}) = \prod_{i=0}^{N-1} f_{Y_i}(y_i), \quad (583)$$

where

$$f_{Y_i}(y_i) = \frac{1}{\sqrt{2\pi\sigma^2(t_i)}} \exp\left(-\frac{(y_i - \bar{\mu}(t_i))^2}{2\sigma^2(t_i)}\right). \quad (584)$$

We approximate the functions $[\bar{\mu}, \sigma^2]$ via the feedforward neural network

$$\mathcal{N}^F : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^2, \quad (585)$$

where $\mathcal{N}^F(t, w) = [\mathcal{N}_1^F(t, w), \mathcal{N}_2^F(t, w)]$ and $\mathcal{N}_1^F$ approximates $\bar{\mu}$, while $(\mathcal{N}_2^F(t, w))^2$ approximates σ^2 . We define the loss function as a negative log-likelihood function that follows from the joint density (583). Hence, the loss function for the neural network $\mathcal{N}^F$ is as follows

$$\mathcal{L}(w) = \frac{1}{2} \sum_{j=0}^{N-1} \left[\ln[(\mathcal{N}_2^F(t_j, w))^2] + \left(\frac{Y_j - \mathcal{N}_1^F(t_j, w)}{\mathcal{N}_2^F(t_j, w)} \right)^2 \right], \quad (586)$$

where we omitted the constant term $\frac{N}{2} \ln(2\pi)$. The optimal weights w^* are obtained by minimizing the loss above. The estimated parameter functions are

$$\hat{\mu}(t) = \frac{1}{\sqrt{\Delta t_i}} \mathcal{N}_1^F(t, w^*), \quad t \in [t_i, t_{i+1}), i = 0, 1, \dots, N-1, \quad (587)$$

$$\widehat{\sigma^2}(t) = (\mathcal{N}_2^F(t, w^*))^2, \quad t \in [0, T]. \quad (588)$$

12.2 Log returns approach

By applying the Itô formula to (575), we get that

$$d(\ln(S(t))) = m(t)dt + \sigma(t)dW(t), \quad (589)$$

where

$$m(t) = \mu(t) - \frac{1}{2}\sigma^2(t). \quad (590)$$

Therefore, for $i = 0, 1, \dots, N-1$

$$\begin{aligned} \ln\left(\frac{S(t_{i+1})}{S(t_i)}\right) &= \int_{t_i}^{t_{i+1}} m(s)ds + \int_{t_i}^{t_{i+1}} \sigma(s)dW(s) \\ &\approx m(t_i)\Delta t_i + \sigma(t_i)\Delta W_i = m(t_i)\Delta t_i + \sigma(t_i)\sqrt{\Delta t_i}\xi_i \end{aligned} \quad (591)$$

where we replaced the Wiener increment ΔW_i by $\sqrt{\Delta t_i}\xi_i$, where $\xi_i \sim N(0, 1)$. Hence, we assume that the real data follows the following discrete scheme for $i = 0, 1, \dots, N-1$

$$L_i = \bar{m}(t_i) + \sigma(t_i)\xi_i, \quad (592)$$

where

$$L_i = \frac{1}{\sqrt{\Delta t_i}} \ln\left(\frac{S(t_{i+1})}{S(t_i)}\right), \quad (593)$$

$$\bar{m}(t_i) = \sqrt{\Delta t_i} \cdot m(t_i), \quad (594)$$

and $\{\xi_i\}_{i=0,1,\dots,N-1}$ is the iid sequence of random variable from $N(0, 1)$. (Note that $\sqrt{\Delta t_i}L_i = \ln\left(\frac{S(t_{i+1})}{S(t_i)}\right)$ are the log returns.) The random variables $L_0, L_1, \dots, L_{N-1}$ are independent and

$$L_i \sim N(\bar{m}(t_i), \sigma^2(t_i)). \quad (595)$$

Therefore, we can now proceed analogously as in the Subsection 12.1 and we approximate the functions $[\bar{m}, \sigma^2]$ via the feedforward neural network

$$\mathcal{N}^F : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^2, \quad (596)$$

where $\mathcal{N}^F(t, w) = [\mathcal{N}_1^F(t, w), \mathcal{N}_2^F(t, w)]$ and $\mathcal{N}_1^F$ approximates $\bar{m}$, while $(\mathcal{N}_2^F(t, w))^2$ approximates σ^2 . The loss function for the neural network $\mathcal{N}^F$ is defined as follows

$$\mathcal{L}(w) = \frac{1}{2} \sum_{j=0}^{N-1} \left[\ln[(\mathcal{N}_2^F(t_j, w))^2] + \left(\frac{L_j - \mathcal{N}_1^F(t_j, w)}{\mathcal{N}_2^F(t_j, w)} \right)^2 \right], \quad (597)$$

where we omitted the constant term $\frac{N}{2} \ln(2\pi)$. The optimal weights w^* are obtained by minimizing the loss above. The estimated parameter functions are

$$\hat{m}(t) = \frac{1}{\sqrt{\Delta t_i}} \mathcal{N}_1^F(t, w^*), \quad t \in [t_i, t_{i+1}), i = 0, 1, \dots, N-1, \quad (598)$$

$$\widehat{\sigma^2(t)} = (\mathcal{N}_2^F(t, w^*))^2, \quad t \in [0, T], \quad (599)$$

$$\hat{\mu}(t) = \hat{m}(t) + \frac{1}{2} \widehat{\sigma^2(t)}, \quad t \in [t_i, t_{i+1}), i = 0, 1, \dots, N-1. \quad (600)$$

Remark 12.1. For both approaches we underline that estimating the drift function $\mu(t)$ from a single observed path is statistically much harder than estimating the volatility $\sigma(t)$. The volatility is informed by quadratic variation, whereas the drift is dominated by stochastic fluctuations at high frequency. Therefore, the drift estimate may be unstable.

13 Appendix - technical tools from mathematical analysis

Definition 13.1. We call a function $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$ L -smooth for some $L \in [0, +\infty)$ if $\mathcal{L}$ is differentiable and the gradient $\nabla \mathcal{L}$ is L -Lipschitz continuous, i.e.: for all $x, y \in \mathbb{R}^n$

$$\|\nabla \mathcal{L}(x) - \nabla \mathcal{L}(y)\| \leq L\|x - y\|. \quad (601)$$

Note that if $\mathcal{L}$ is L -smooth then $\nabla \mathcal{L}$ is continuous and $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$. For the proof of the following lemma, we refer to page 211 in [36].

Lemma 13.1. Let $f : U \rightarrow \mathbb{R}^m$ be a differentiable function defined on an open and convex set U . If there exists a constant $C \in [0, +\infty)$ such that $\|f'(x)\| \leq C$ for all $x \in U$, then

$$\|f(x) - f(y)\| \leq C\|x - y\|, \quad (602)$$

for all $x, y \in U$.

Theorem 13.1. Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$. Then for all $x, y \in \mathbb{R}^n$

$$\mathcal{L}(y) - \mathcal{L}(x) = \int_0^1 \langle \nabla \mathcal{L}(x + t(y - x)), y - x \rangle dt. \quad (603)$$

Proof. Fix arbitrary $x, y \in \mathbb{R}^n$ and define the function $g : [0, 1] \rightarrow \mathbb{R}$ as follows

$$g(t) = \mathcal{L}(x + t(y - x)) = \mathcal{L}(L_{x,y}(t)), \quad (604)$$

where $L_{x,y} : [0, 1] \rightarrow \mathbb{R}^n$, $L_{x,y}(t) = x + t(y - x)$ is a line segment between x and y . Then $g(0) = \mathcal{L}(x)$, $g(1) = \mathcal{L}(y)$, $g \in C^1([0, 1]; \mathbb{R})$, and

$$g'(t) = \langle \nabla \mathcal{L}(z), L'_{x,y}(t) \rangle|_{z=L_{x,y}(t)} = \langle \nabla \mathcal{L}(x + t(y - x)), y - x \rangle, \quad (605)$$

since $L'_{x,y}(t) = y - x$. By the fundamental theorem of calculus, we thus get

$$\mathcal{L}(y) - \mathcal{L}(x) = g(1) - g(0) = \int_0^1 g'(t) dt = \int_0^1 \langle \nabla \mathcal{L}(x + t(y - x)), y - x \rangle dt, \quad (606)$$

which ends the proof. $\square$

Lemma 13.2 (Descent Lemma). Let $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$ be L -smooth for some $L \in [0, +\infty)$. Then for all $x, y \in \mathbb{R}^n$

$$\left| \mathcal{L}(y) - \mathcal{L}(x) - \langle \nabla \mathcal{L}(x), y - x \rangle \right| \leq \frac{L}{2} \|x - y\|^2. \quad (607)$$

Proof. By Theorem 13.1 we have for all $x, y \in \mathbb{R}^n$ that

$$\begin{aligned} \mathcal{L}(y) - \mathcal{L}(x) &= \int_0^1 \langle \nabla \mathcal{L}(x + t(y - x)) - \nabla \mathcal{L}(x), y - x \rangle dt \\ &\quad + \int_0^1 \langle \nabla \mathcal{L}(x), y - x \rangle dt \\ &= \int_0^1 \langle \nabla \mathcal{L}(x + t(y - x)) - \nabla \mathcal{L}(x), y - x \rangle dt + \langle \nabla \mathcal{L}(x), y - x \rangle, \end{aligned} \quad (608)$$

and hence

$$\left| \mathcal{L}(y) - \mathcal{L}(x) - \langle \nabla \mathcal{L}(x), y - x \rangle \right| = \left| \int_0^1 \langle \nabla \mathcal{L}(x + t(y - x)) - \nabla \mathcal{L}(x), y - x \rangle dt \right|. \quad (609)$$

Moreover, by the Schwarz inequality and Lipschitz continuity of $\nabla \mathcal{L}$,

$$\begin{aligned} & \left| \int_0^1 \langle \nabla \mathcal{L}(x + t(y - x)) - \nabla \mathcal{L}(x), y - x \rangle dt \right| \\ & \leq \int_0^1 |\langle \nabla \mathcal{L}(x + t(y - x)) - \nabla \mathcal{L}(x), y - x \rangle| dt \\ & \leq \int_0^1 \|\nabla \mathcal{L}(x + t(y - x)) - \nabla \mathcal{L}(x)\| \cdot \|y - x\| dt \\ & \leq L \|y - x\|^2 \cdot \int_0^1 t dt = \frac{L}{2} \|y - x\|^2. \end{aligned} \quad (610)$$

Combining (609) and (610), we get the thesis. $\square$

Note that the inequality (607) is equivalent to the following:

$$\langle \nabla \mathcal{L}(x), y - x \rangle - \frac{L}{2} \|y - x\|^2 \leq \mathcal{L}(y) - \mathcal{L}(x) \leq \langle \nabla \mathcal{L}(x), y - x \rangle + \frac{L}{2} \|y - x\|^2, \quad (611)$$

for all $x, y \in \mathbb{R}^n$.

In a more detailed way, we now investigate the relation between the inequality (611) and the regularity of the gradient of $\mathcal{L}$.

Lemma 13.3. *Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$. Then the following conditions are equivalent:*

(i) $\nabla \mathcal{L}$ satisfies the upper one-sided Lipschitz condition, i.e., there exists $L \in [0, +\infty)$ such that for all $x, y \in \mathbb{R}^n$

$$\langle \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y), x - y \rangle \leq L \|x - y\|^2. \quad (612)$$

(ii) There exists $L \in [0, +\infty)$ such that for all $x, y \in \mathbb{R}^n$

$$\mathcal{L}(y) - \mathcal{L}(x) \leq \langle \nabla \mathcal{L}(x), y - x \rangle + \frac{L}{2} \|y - x\|^2. \quad (613)$$

Proof. ($\Rightarrow$) For arbitrary fixed $x, y \in \mathbb{R}^n$ define the functions

$$F(t) = \langle \nabla \mathcal{L}(x + t(y - x)) - \nabla \mathcal{L}(x), y - x \rangle, \quad (614)$$

$$G(t) = Lt \|x - y\|^2, \quad (615)$$

for all $t \in [0, 1]$. Since $\nabla \mathcal{L} \in C(\mathbb{R}^n; \mathbb{R}^n)$, we have that $F \in C([0, 1]; \mathbb{R})$, $G \in C([0, 1]; [0, +\infty))$. Moreover, $F(0) = 0 = G(0)$ and, by (612), for all $t \in (0, 1]$

$$\begin{aligned} F(t) &= \frac{1}{t} \langle \nabla \mathcal{L}(x + t(y - x)) - \nabla \mathcal{L}(x), x + t(y - x) - x \rangle \\ &\leq \frac{1}{t} L \|t(y - x)\|^2 = G(t). \end{aligned} \quad (616)$$

Hence, $F(t) \leq G(t)$ for all $t \in [0, 1]$, and

$$\int_0^1 F(t) dt \leq \int_0^1 G(t) dt = \frac{L}{2} \|x - y\|^2. \quad (617)$$

By Theorem 13.1 we have

$$\begin{aligned}\mathcal{L}(y) - \mathcal{L}(x) &= \int_0^1 F(t)dt + \int_0^1 \langle \nabla \mathcal{L}(x), y - x \rangle dt \\ &\leq \int_0^1 G(t)dt + \langle \nabla \mathcal{L}(x), y - x \rangle = \frac{L}{2} \|x - y\|^2 + \langle \nabla \mathcal{L}(x), y - x \rangle,\end{aligned}\tag{618}$$

and we obtain (ii).

($\Leftarrow$) Assuming (ii) we have that there exists $L \in [0, +\infty)$ such that for all $x, y \in \mathbb{R}^n$

$$\begin{aligned}\mathcal{L}(y) - \mathcal{L}(x) &\leq \langle \nabla \mathcal{L}(x), y - x \rangle + \frac{L}{2} \|y - x\|^2, \\ \mathcal{L}(x) - \mathcal{L}(y) &\leq \langle \nabla \mathcal{L}(y), x - y \rangle + \frac{L}{2} \|y - x\|^2.\end{aligned}\tag{619}$$

Adding the above inequalities side by side, we obtain

$$\begin{aligned}0 &\leq L \|x - y\|^2 + \langle \nabla \mathcal{L}(x), y - x \rangle + \langle \nabla \mathcal{L}(y), x - y \rangle \\ &= L \|x - y\|^2 - \langle \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y), x - y \rangle,\end{aligned}\tag{620}$$

and the proof of (i) is complete. $\square$

Corollary 13.1. *Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$. Then the following conditions are equivalent:*

(i) $\nabla \mathcal{L}$ satisfies the lower one-sided Lipschitz condition, i.e., there exists $K \in [0, +\infty)$ such that for all $x, y \in \mathbb{R}^n$

$$\langle \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y), x - y \rangle \geq -K \|x - y\|^2.\tag{621}$$

(ii) There exists $K \in [0, +\infty)$ such that for all $x, y \in \mathbb{R}^n$

$$\mathcal{L}(y) - \mathcal{L}(x) \geq \langle \nabla \mathcal{L}(x), y - x \rangle - \frac{K}{2} \|y - x\|^2.\tag{622}$$

Proof. The thesis follows from Lemma 13.3 applied to the function $x \rightarrow -\nabla \mathcal{L}$. $\square$

Directly from Lemma 13.3 and Corollary 13.1 we have the following theorem.

Theorem 13.2. *Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$. Then the following conditions are equivalent:*

(i) There exist $L_1, L_2 \in [0, +\infty)$ such that for all $x, y \in \mathbb{R}^n$

$$-L_1 \|x - y\|^2 \leq \langle \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y), x - y \rangle \leq L_2 \|x - y\|^2.\tag{623}$$

(ii) There exist $L_1, L_2 \in [0, +\infty)$ such that for all $x, y \in \mathbb{R}^n$

$$\begin{aligned}\langle \nabla \mathcal{L}(x), y - x \rangle - \frac{L_1}{2} \|y - x\|^2 &\leq \mathcal{L}(y) - \mathcal{L}(x) \\ &\leq \langle \nabla \mathcal{L}(x), y - x \rangle + \frac{L_2}{2} \|y - x\|^2.\end{aligned}\tag{624}$$

From Theorem 13.2 we obtain what follows.

Corollary 13.2. *Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$. Then the following conditions are equivalent:*

(i) There exist $L \in [0, +\infty)$ such that for all $x, y \in \mathbb{R}^n$

$$\left| \langle \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y), x - y \rangle \right| \leq L \|x - y\|^2.\tag{625}$$

(ii) There exist $L \in [0, +\infty)$ such that for all $x, y \in \mathbb{R}^n$

$$\left| \mathcal{L}(y) - \mathcal{L}(x) - \langle \nabla \mathcal{L}(x), y - x \rangle \right| \leq \frac{L}{2} \|y - x\|^2. \quad (626)$$

If $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ and $\nabla \mathcal{L}$ is Lipschitz continuous, then by the Schwarz inequality we get for all $x, y \in \mathbb{R}^n$

$$\begin{aligned} \left| \langle \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y), x - y \rangle \right| &\leq \|\nabla \mathcal{L}(x) - \nabla \mathcal{L}(y)\| \cdot \|x - y\| \\ &\leq L \|x - y\|^2. \end{aligned} \quad (627)$$

Hence, we can ask if the condition (625) implies the Lipschitz continuity of $\nabla \mathcal{L}$. The answer is positive, see, for example, Lemma 2.1 in [42].

Lemma 13.4. *Let $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$ be L -smooth for some $L \in (0, +\infty)$. If $\inf_{y \in \mathbb{R}^n} \mathcal{L}(y) > -\infty$ then for all $x \in \mathbb{R}^n$*

$$\frac{1}{2L} \|\nabla \mathcal{L}(x)\|^2 \leq \mathcal{L}(x) - \inf_{y \in \mathbb{R}^n} \mathcal{L}(y). \quad (628)$$

If, additionally, $w^* \in \mathbb{R}^n$ is a critical point of $\mathcal{L}$, then for all $x \in \mathbb{R}^n$

$$\frac{1}{L} \|\nabla \mathcal{L}(x)\| \leq \|x - w^*\|. \quad (629)$$

Proof. From (611) we have for all $x \in \mathbb{R}^n$ and $\eta \in (0, +\infty)$ that

$$\begin{aligned} \mathcal{L}(x - \eta \nabla \mathcal{L}(x)) - \mathcal{L}(x) &\leq \langle \nabla \mathcal{L}(x), -\eta \nabla \mathcal{L}(x) \rangle + \frac{L\eta^2}{2} \|\nabla \mathcal{L}(x)\|^2 \\ &= -\eta \left(1 - \frac{L\eta}{2}\right) \|\nabla \mathcal{L}(x)\|^2, \end{aligned} \quad (630)$$

and by taking $\eta := 1/L$ we get for all $x \in \mathbb{R}^n$ that

$$\inf_{y \in \mathbb{R}^n} \mathcal{L}(y) - \mathcal{L}(x) \leq \mathcal{L}(x - (1/L) \nabla \mathcal{L}(x)) - \mathcal{L}(x) \leq -\frac{1}{2L} \|\nabla \mathcal{L}(x)\|^2, \quad (631)$$

which implies the thesis.

Since $\nabla \mathcal{L}(w^*) = 0$, we get for all $x \in \mathbb{R}^n$ that

$$\|\nabla \mathcal{L}(x)\| = \|\nabla \mathcal{L}(x) - \nabla \mathcal{L}(w^*)\| \leq L \|x - w^*\|. \quad (632)$$

□

We recall some properties of convex functions, see, for example, [25].

Lemma 13.5. (1) *Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ be a convex function. If $\nabla \mathcal{L}(w^*) = 0$ then w^* is the global minimum of $\mathcal{L}$.*

(2) *If $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ then $\mathcal{L}$ is convex if and only if for all $x, y \in \mathbb{R}^n$*

$$\mathcal{L}(y) \geq \mathcal{L}(x) + \langle \nabla \mathcal{L}(x), y - x \rangle. \quad (633)$$

(3) *A function $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ is convex if and only if for all $x, y \in \mathbb{R}^n$*

$$\langle x - y, \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y) \rangle \geq 0. \quad (634)$$

In particular, from the Lemma above we have that if the C^1 -function $\mathcal{L}$ is convex, then $\mathbb{R}^n \ni x \rightarrow -\nabla \mathcal{L}(x) \in \mathbb{R}^n$ satisfies the one-sided Lipschitz (OSL) condition with the OSL constant equal to zero, i.e.:

$$\langle x - y, -\nabla \mathcal{L}(x) - (-\nabla \mathcal{L}(y)) \rangle = -\langle x - y, \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y) \rangle \leq 0. \quad (635)$$

Definition 13.2. Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$. The function $\mathcal{L}$ is:

- (i) *strongly convex* if there exists $K > 0$ such that $\mathcal{L}(x) - \frac{K}{2}\|x\|^2$ is convex,
- (ii) *weakly convex* (or *semi-convex*) if there exists $K \geq 0$ such that $\mathcal{L}(x) + \frac{K}{2}\|x\|^2$ is convex.

Theorem 13.3. Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$. Then

(i) $\mathcal{L}$ is strongly convex iff

$$\exists_{K>0} : \forall_{x,y \in \mathbb{R}^n} \langle x - y, \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y) \rangle \geq K\|x - y\|^2, \quad (636)$$

(ii) $\mathcal{L}$ is convex iff

$$\forall_{x,y \in \mathbb{R}^n} \langle x - y, \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y) \rangle \geq 0, \quad (637)$$

(iii) $\mathcal{L}$ is weakly convex iff

$$\exists_{K>0} : \forall_{x,y \in \mathbb{R}^n} \langle x - y, \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y) \rangle \geq -K\|x - y\|^2, \quad (638)$$

In summary:

strong convexity $\Rightarrow$ convexity $\Rightarrow$ weak convexity

Corollary 13.3. If $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth then it is weakly convex.

We recall after Corollary 2.20 in [28], one of the main properties of strongly convex functions.

Lemma 13.6. If $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous and strongly convex, then there exists a unique minimizer of $\mathcal{L}$.

Lemma 13.7. For all $a, b \in \mathbb{R}^n$ we have

$$-\langle a, b \rangle = \frac{1}{2}\|a\|^2 + \frac{1}{2}\|b\|^2 - \frac{1}{2}\|a + b\|^2. \quad (639)$$

Proof. From the bilinearity of the inner product, we have that

$$\begin{aligned} \|a + b\|^2 &= \langle a + b, a + b \rangle = \langle a, a \rangle + \langle a, b \rangle + \langle b, a \rangle + \langle b, b \rangle \\ &= \|a\|^2 + 2\langle a, b \rangle + \|b\|^2, \end{aligned} \quad (640)$$

from which the thesis follows. $\square$

Lemma 13.8. Assume that for two non-negative sequences $\{a_n\}_{n \geq 1}$, $\{b_n\}_{n \geq 1}$ we have

$$\sum_{n=1}^{+\infty} a_n b_n < +\infty, \quad \sum_{n=1}^{+\infty} a_n = +\infty. \quad (641)$$

Then

$$\liminf_{n \rightarrow +\infty} b_n = 0, \quad (642)$$

and

$$\inf_{n \geq 1} b_n = 0. \quad (643)$$

Proof. Assume that $\liminf_{n \rightarrow +\infty} b_n = L > 0$. Then there exists $n_0 \geq 1$ such that for all $n \geq n_0$

$$b_n \geq L/2. \quad (644)$$

Since $a_n \geq 0$, this implies

$$\begin{aligned} +\infty &> \sum_{n=1}^{+\infty} a_n b_n \geq \sum_{n=n_0}^{+\infty} a_n b_n \geq \frac{L}{2} \sum_{n=n_0}^{+\infty} a_n \\ &= \frac{L}{2} \left(\sum_{n=1}^{+\infty} a_n - \sum_{n=1}^{n_0-1} a_n \right) = +\infty, \end{aligned} \quad (645)$$

which is a contradiction. Moreover,

$$0 = \liminf_{n \rightarrow +\infty} b_n = \sup_{n \geq 1} \inf_{m \geq n} b_m \geq \inf_{m \geq 1} b_m \geq 0, \quad (646)$$

and we get (643). $\square$

Lemma 13.9. Assume that for a Borel measurable function $f : [0, +\infty) \rightarrow [0, +\infty)$ we have

$$\int_0^{+\infty} f(t) dt < +\infty. \quad (647)$$

Then $\inf_{t \geq 0} f(t) = 0$. If additionally, f is non-increasing then $\lim_{t \rightarrow +\infty} f(t) = 0$.

Proof. Let us denote by $I = \int_0^{+\infty} f(t) dt < +\infty$. Then, for all $t \geq 0$

$$t \cdot \inf_{0 \leq u \leq t} f(u) \leq \int_0^t f(u) du \leq I, \quad (648)$$

which implies that

$$\lim_{t \rightarrow +\infty} \inf_{0 \leq u \leq t} f(u) = 0. \quad (649)$$

The function $g(t) = \inf_{0 \leq u \leq t} f(u)$ is non-increasing, non-negative, and $\inf_{t \geq 0} g(t) = \inf_{t \geq 0} f(t)$. Therefore

$$0 = \lim_{t \rightarrow +\infty} g(t) = \inf_{t \geq 0} g(t) = \inf_{t \geq 0} f(t). \quad (650)$$

If f is non-increasing then we get

$$\lim_{t \rightarrow +\infty} f(t) = \inf_{t \geq 0} f(t) = 0. \quad (651)$$

$\square$

Lemma 13.10. Let $\{X_n\}_{n \geq 0}$ be a sequence of nonnegative random variables such that

$$\liminf_{n \rightarrow +\infty} \mathbb{E}(X_n) = 0. \quad (652)$$

Then

$$\liminf_{n \rightarrow +\infty} X_n = 0 \quad (653)$$

almost surely.

Proof. From Fatou's lemma

$$0 \leq \mathbb{E}(\liminf_{n \rightarrow +\infty} X_n) \leq \liminf_{n \rightarrow +\infty} \mathbb{E}(X_n) = 0, \quad (654)$$

and hence $\mathbb{E}(\liminf_{n \rightarrow +\infty} X_n) = 0$ which implies the thesis. $\square$

14 Exercises

Exercise 1. Under the assumptions of Theorem 6.1 show that in the (C2)-case the following estimate holds

$$\mathcal{L}(w^0) - \beta \sum_{k=0}^{+\infty} \|\nabla \mathcal{L}(w^k)\|^2 \leq \lim_{k \rightarrow +\infty} \mathcal{L}(w^k) \leq \mathcal{L}(w^0) - \alpha \sum_{k=0}^{+\infty} \|\nabla \mathcal{L}(w^k)\|^2 \quad (655)$$

Exercise 2. Under the assumptions of Theorem 6.1 show that in the (C2)-case the following estimate holds for all $M \geq 1$

$$0 \leq \left(\frac{\mathcal{L}(w^0) - \mathcal{L}(w^M)}{\beta M} \right)^{1/2} \leq \max_{0 \leq k \leq M-1} \|\nabla \mathcal{L}(w^k)\|. \quad (656)$$

Exercise 3. Let us assume that for the function $\mathcal{L} \in C^2(\mathbb{R}; \mathbb{R})$ there exist $d_0, D_0 \in (0, +\infty)$ such that for all $x \in \mathbb{R}$

$$d_0 \leq \mathcal{L}''(x) \leq D_0. \quad (657)$$

- (1) Show that $\mathcal{L}$ has a unique global minimum w^* ,
- (2) Take any $q \in \left(\frac{D_0 - d_0}{D_0 + d_0}, 1 \right)$. Give a proof that for any $\eta \in \left[\frac{1-q}{d_0}, \frac{1+q}{D_0} \right]$, $w^0 \in \mathbb{R}$ the (GD) algorithm (209) converges to the global minimum w^* .
- (3) Fix arbitrary $w^0 \in \mathbb{R}$ and consider the following initial value problem (also called the *gradient ODE*)

$$w(0) = w^0, \quad w'(t) = -\mathcal{L}'(w(t)) \text{ for } t \in [0, +\infty). \quad (658)$$

Show that $\lim_{t \rightarrow +\infty} w(t) = w^*$.

- (4) Show that the function $\mathcal{L}(x) = x^2 - \cos(x)$ satisfies (682) with suitable d_0, D_0 . Perform numerical experiments in Python for this function and the gradient descent method.

Hint: for (2) use the Banach contraction mapping principle.

Exercise 4. Let us assume that $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth, and let w^* be a critical point of $\mathcal{L}$. Show that for all $y \in \mathbb{R}^n$

$$|\mathcal{L}(y) - \mathcal{L}(w^*)| \leq \frac{L}{2} \|y - w^*\|^2. \quad (659)$$

Exercise 5. Investigate the behavior of the GD method for the function $\mathcal{L}(x) = x^2$, $x \in \mathbb{R}$.

Exercise 6. Consider the function $\mathcal{L}(x) = \frac{1}{3}x^3$, $x \in \mathbb{R}$.

- (1) For the sequence $\{w^k\}_{k \geq 0}$ given by (209) show that

$$\lim_{k \rightarrow +\infty} w^k = -\infty, \text{ if } w^0 \in (-\infty, 0) \cup (1/\eta, +\infty), \quad (660)$$

$$\lim_{k \rightarrow +\infty} w^k = 0, \text{ if } w^0 \in [0, \frac{1}{\eta}]. \quad (661)$$

- (2) Fix arbitrary $w^0 \in \mathbb{R}$ and consider the following initial value problem (also called the *gradient ODE*)

$$w(0) = w^0, \quad w'(t) = -\mathcal{L}'(w(t)) = -(w(t))^2 \text{ for } t \in [0, +\infty). \quad (662)$$

Solve the above ODE. For $w^0 \in [0, +\infty)$ show that $\lim_{t \rightarrow +\infty} w(t) = 0$. What happens for $w_0 < 0$?

Exercise 7. Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$. Show that the following conditions are equivalent:

(i)

$$\exists_{K \in [0, +\infty)} : \forall_{x, y \in \mathbb{R}^n} |\mathcal{L}(x) - \mathcal{L}(y)| \leq K \|x - y\|, \quad (663)$$

(ii)

$$\exists_{K \in [0, +\infty)} : \forall_{x \in \mathbb{R}^n} \|\nabla \mathcal{L}(x)\| \leq K. \quad (664)$$

Moreover, show that if $\mathcal{L}$ is L -smooth then there exists $K \in [0, +\infty)$ such that for all $x \in \mathbb{R}^n$

$$\|\nabla \mathcal{L}(x)\| \leq K(1 + \|x\|). \quad (665)$$

Exercise 8. For the SGD algorithm (318) show that for all $k \geq 0$, $i = 1, \dots, n$

$$\begin{aligned} \text{Var}(w_i^{k+1} \mid \mathcal{F}_k) &= \eta_k^2 \cdot \text{Var}\left(\frac{\partial \mathcal{L}_{J_{k+1}}}{\partial w_i}(w^k) \mid \mathcal{F}_k\right) \\ &= \eta_k^2 \cdot \left[\frac{1}{M} \sum_{j=1}^M \left| \frac{\partial \mathcal{L}_j}{\partial w_i}(w^k) \right|^2 - \left| \frac{\partial \mathcal{L}}{\partial w_i}(w^k) \right|^2 \right]. \end{aligned} \quad (666)$$

Exercise 9. Find a solution of a gradient system for the function $\mathcal{L}(x) = x^4/4$, $x \in \mathbb{R}$, and discuss the behavior of the solution as $t \rightarrow +\infty$.

Exercise 10. For the SGD sequence (318) show, under the assumptions of Theorem 7.1, that $\sum_{k=0}^{+\infty} \|w^{k+1} - w^k\|^2 < +\infty$ almost surely.

Exercise 11. Show that

$$\begin{aligned} \mathbb{E}(U_k \mid \mathcal{F}_k) &= -\eta_k \left(1 - \frac{L\eta_k}{2}\right) \|\nabla \mathcal{L}(w^k)\|^2 \\ &+ \frac{L}{2} \eta_k^2 \cdot \mathbb{E} \left[\|\nabla \mathcal{L}_{J_{k+1}}(w^k) - \nabla \mathcal{L}(w^k)\|^2 \mid \mathcal{F}_k \right], \end{aligned} \quad (667)$$

and

$$\begin{aligned} &\mathbb{E} \left[\|\nabla \mathcal{L}_{J_{k+1}}(w^k) - \nabla \mathcal{L}(w^k)\|^2 \mid \mathcal{F}_k \right] \\ &= \frac{1}{M} \sum_{j=1}^M \|\nabla \mathcal{L}_j(w^k)\|^2 - \|\nabla \mathcal{L}(w^k)\|^2, \end{aligned} \quad (668)$$

where U_k is defined in (343).

Exercise 12. Let $A \in \mathbb{R}^{M \times N}$. Show that:

- (i) the matrix $A^T A$ is symmetric and positive semi-definite,
- (ii) for all $\lambda > 0$ the matrix $A^T A + \lambda I$ is symmetric and positive definite.

Exercise 13. Consider the function $U(x) = x^4 - x^2$, $x \in \mathbb{R}$. Show that U' is locally Lipschitz and for all $x, y \in \mathbb{R}$

$$(x - y)(U'(x) - U'(y)) \geq -2|x - y|^2. \quad (669)$$

Hence, U is weakly convex. However, it is neither convex nor L -smooth.

Exercise 14. Consider the functions $\varphi \in C^2(\mathbb{R}^M; \mathbb{R}^N)$ and $F(w) = \|\varphi(w)\|^2$. Show that for all $w \in \mathbb{R}^M$

$$\nabla F(w) = 2(\varphi(w))^T \nabla \varphi(w), \quad (670)$$

$$\nabla^2 F(w) = 2(\nabla \varphi(w))^T \nabla \varphi(w) + 2 \sum_{i=1}^N \varphi_i(w) \nabla^2 \varphi_i(w) \quad (671)$$

where $\nabla \varphi(w)$ is the Jacobian of φ , and $\nabla^2 \varphi_i(w)$ is the Hessian of φ_i .

Exercise 15. For the SGD algorithm (318) show that the conditional covariance matrix of w^{k+1} satisfies

$$\text{Cov}(w^{k+1} \mid \mathcal{F}_k) = \eta_k^2 \cdot \text{Cov}(Z_k \mid \mathcal{F}_k), \quad k = 0, 1, \dots, \quad (672)$$

where Z_k is defined in (326). Moreover, show that

$$\text{Cov}(Z_k \mid \mathcal{F}_k) = \mathbb{E}(Z_k Z_k^T \mid \mathcal{F}_k) = D(w^k), \quad (673)$$

where for all $w \in \mathbb{R}^n$

$$D(w) = \frac{1}{M} \sum_{j=1}^M \nabla \mathcal{L}_j(w) \nabla^T \mathcal{L}_j(w) - \nabla \mathcal{L}(w) \nabla^T \mathcal{L}(w) \quad (674)$$

is the so-called *diffusion matrix*.

Exercise 16. By using the backpropagation procedure, described in Section 5.2, discuss the problems of *vanishing* and *exploding* gradients in feedforward neural networks. Consider the case of sigmoid and ReLU activation functions.

Exercise 17. Minimize the loss (106) and compute the estimators $\hat{\kappa}_N, \hat{\mu}_N$.

Exercise 18. For $a > 25/3$ consider the function

$$f(x) = x^2 + a \ln(1 + x^2), \quad x \in \mathbb{R}. \quad (675)$$

Show that it is L -smooth and of PL-type, but it is not a convex function.

Exercise 19. Show that in Theorem 10.5 (ii) the assumption $\mathcal{L} \in C^2(\mathbb{R}^n; \mathbb{R})$ can be weakened to $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ and the limit $\lim_{t \rightarrow +\infty} \nabla \mathcal{L}(w(t)) = 0$ still holds.

Hint. Prove that the function $\phi_h(t) = \|w(t+h) - w(t)\|^2$ is nonincreasing for all $h \in \mathbb{R}$ with respect to the variable t .

Exercise 20. Show that the loss function (119) can be written as in (132).

Hint. Use the fact that for the sigmoid activation φ , the following holds

$$-\ln(\varphi(z)) = \ln(1 + e^{-z}), \quad (676)$$

$$-\ln(1 - \varphi(z)) = \ln(1 + e^z), \quad (677)$$

for all $z \in \mathbb{R}$.

Exercise 21. Assume that $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ is L -smooth and consider the gradient descent algorithm with clipping, i.e., $\eta > 0$, $w^0 \in \mathbb{R}^n$ and

$$w^{k+1} = w^k - \eta \cdot R_D(\nabla \mathcal{L}(w^k)), \quad k = 0, 1, \dots, \quad (678)$$

where R_D is defined in (388) for a fixed $D \in (0, +\infty)$. Moreover, assume that $\eta \in (0, 1/L)$ and that there exists $\mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)]$ such that $\mathcal{L}(w) \geq \mathcal{L}^*$ for all $w \in \mathbb{R}^n$. Show that the sequence $\{\mathcal{L}(w^k)\}_{k \geq 0}$ converges and

$$\lim_{k \rightarrow +\infty} \nabla \mathcal{L}(w^k) = 0. \quad (679)$$

15 Solutions to the exercises

Exercise 1

Under the assumptions of Theorem 6.1 show that in the (C2)-case the following estimate holds

$$\mathcal{L}(w^0) - \beta \sum_{k=0}^{+\infty} \|\nabla \mathcal{L}(w^k)\|^2 \leq \lim_{k \rightarrow +\infty} \mathcal{L}(w^k) \leq \mathcal{L}(w^0) - \alpha \sum_{k=0}^{+\infty} \|\nabla \mathcal{L}(w^k)\|^2 \quad (680)$$

Solution. From the proof of Theorem 6.1, for every $k \geq 0$,

$$-\beta \|\nabla \mathcal{L}(w_k)\|^2 \leq \mathcal{L}(w_{k+1}) - \mathcal{L}(w_k) \leq -\alpha \|\nabla \mathcal{L}(w_k)\|^2,$$

where

$$\alpha = \eta \left(1 - \frac{\eta}{2}\right) > 0, \quad \beta = \eta \left(1 + \frac{L\eta}{2}\right) > 0.$$

Summing from $k = 0$ to $M - 1$ gives

$$\mathcal{L}(w_0) - \beta \sum_{k=0}^{M-1} \|\nabla \mathcal{L}(w_k)\|^2 \leq \mathcal{L}(w_M) \leq \mathcal{L}(w_0) - \alpha \sum_{k=0}^{M-1} \|\nabla \mathcal{L}(w_k)\|^2.$$

In the case (C2), Theorem 6.1 gives

$$\sum_{k=0}^{\infty} \|\nabla \mathcal{L}(w_k)\|^2 < \infty$$

and the sequence $\mathcal{L}(w_k)$ has a finite limit. Passing to the limit $M \rightarrow \infty$ in the preceding finite-sum inequalities gives the asserted estimate.

Exercise 2

Under the assumptions of Theorem 6.1 show that in the (C2)-case the following estimate holds for all $M \geq 1$

$$0 \leq \left(\frac{\mathcal{L}(w^0) - \mathcal{L}(w^M)}{\beta M} \right)^{1/2} \leq \max_{0 \leq k \leq M-1} \|\nabla \mathcal{L}(w^k)\|. \quad (681)$$

Solution. The upper descent estimate from Theorem 6.1 gives

$$\mathcal{L}(w_{k+1}) - \mathcal{L}(w_k) \leq -\alpha \|\nabla \mathcal{L}(w_k)\|^2 \leq 0,$$

so $\mathcal{L}(w_M) \leq \mathcal{L}(w_0)$. The lower descent estimate gives, after summation,

$$0 \leq \mathcal{L}(w_0) - \mathcal{L}(w_M) \leq \beta \sum_{k=0}^{M-1} \|\nabla \mathcal{L}(w_k)\|^2.$$

Therefore

$$\frac{\mathcal{L}(w_0) - \mathcal{L}(w_M)}{\beta M} \leq \frac{1}{M} \sum_{k=0}^{M-1} \|\nabla \mathcal{L}(w_k)\|^2 \leq \max_{0 \leq k \leq M-1} \|\nabla \mathcal{L}(w_k)\|^2.$$

Exercise 3

Let us assume that for the function $\mathcal{L} \in C^2(\mathbb{R}; \mathbb{R})$ there exist $d_0, D_0 \in (0, +\infty)$ such that for all $x \in \mathbb{R}$

$$d_0 \leq \mathcal{L}''(x) \leq D_0. \quad (682)$$

- (1) Show that $\mathcal{L}$ has a unique global minimum w^* ,
- (2) Take any $q \in \left(\frac{D_0 - d_0}{D_0 + d_0}, 1\right)$. Give a proof that for any $\eta \in \left[\frac{1-q}{d_0}, \frac{1+q}{D_0}\right]$, $w^0 \in \mathbb{R}$ the (GD) algorithm (209) converges to the global minimum w^* .
- (3) Fix arbitrary $w^0 \in \mathbb{R}$ and consider the following initial value problem (also called the *gradient ODE*)

$$w(0) = w^0, \quad w'(t) = -\mathcal{L}'(w(t)) \text{ for } t \in [0, +\infty). \quad (683)$$

Show that $\lim_{t \rightarrow +\infty} w(t) = w^*$.

- (4) Show that the function $\mathcal{L}(x) = x^2 - \cos(x)$ satisfies (682) with suitable d_0, D_0 . Perform numerical experiments in Python for this function and the gradient descent method.

Hint: for (2) use the Banach contraction mapping principle.

Solution. (1) *Unique global minimum.* Since $\mathcal{L}''(x) \geq d_0 > 0$ for all $x \in \mathbb{R}$, the derivative $\mathcal{L}'$ is strictly increasing. Moreover, $\mathcal{L}' \in C(\mathbb{R}, \mathbb{R})$ and for $x \geq 0$,

$$\mathcal{L}'(x) = \mathcal{L}'(0) + \int_0^x \mathcal{L}''(s) \, ds \geq \mathcal{L}'(0) + d_0 x \rightarrow +\infty \text{ as } x \rightarrow +\infty$$

and for $x \leq 0$,

$$\mathcal{L}'(x) = \mathcal{L}'(0) - \int_x^0 \mathcal{L}''(s) \, ds \leq \mathcal{L}'(0) + d_0 x \rightarrow -\infty \text{ as } x \rightarrow -\infty.$$

Thus there exists a unique $w^* \in \mathbb{R}$ such that $\mathcal{L}'(w^*) = 0$. Since $\mathcal{L}$ is strictly convex, this critical point is the unique global minimizer.

(2) *Convergence of GD.* Let

$$T(x) = x - \eta \mathcal{L}'(x).$$

Then, the gradient descent algorithm can be written as $w_{k+1} = T(w_k)$. Since

$$T'(x) = 1 - \eta \mathcal{L}''(x),$$

we have

$$1 - \eta D_0 \leq T'(x) \leq 1 - \eta d_0.$$

Take

$$q \in \left(\frac{D_0 - d_0}{D_0 + d_0}, 1\right), \quad \eta \in \left[\frac{1-q}{d_0}, \frac{1+q}{D_0}\right],$$

then the interval is nonempty because

$$\frac{1-q}{d_0} \leq \frac{1+q}{D_0} \iff q \geq \frac{D_0 - d_0}{D_0 + d_0}.$$

For such η we have

$$1 - \eta d_0 \leq q, \quad 1 - \eta D_0 \geq -q.$$

Hence $|T'(x)| \leq q$ for every $x \in \mathbb{R}$. By the mean-value theorem,

$$|T(x) - T(y)| \leq q|x - y|, \quad x, y \in \mathbb{R}.$$

Thus T is a contraction. Its fixed points satisfy $T(x) = x$, equivalently $\mathcal{L}'(x) = 0$. Therefore, the unique fixed point is w^* . By the Banach contraction principle,

$$w_k = T^k(w_0) \rightarrow w^*$$

for every $w_0 \in \mathbb{R}$.

(3) Convergence of the solution of gradient ODE. Consider

$$w'(t) = -\mathcal{L}'(w(t)), \quad w(0) = w_0.$$

Since $\mathcal{L}'$ is globally Lipschitz with Lipschitz constant D_0 , the solution is unique and global. Using $\mathcal{L}'(w^*) = 0$ and $\mathcal{L}''(x) \geq d_0$ for all $x \in \mathbb{R}$, we obtain by the mean value theorem that for all $w \in \mathbb{R}$

$$(w - w^*)\mathcal{L}'(w) = (w - w^*)(\mathcal{L}'(w) - \mathcal{L}'(w^*)) \geq d_0|w - w^*|^2.$$

Consequently, for all $t \geq 0$

$$\frac{d}{dt}|w(t) - w^*|^2 = 2(w(t) - w^*)w'(t) = -2(w(t) - w^*)\mathcal{L}'(w(t)) \leq -2d_0|w(t) - w^*|^2,$$

and by Gronwall's lemma 10.1

$$|w(t) - w^*|^2 \leq e^{-2d_0t}|w_0 - w^*|^2.$$

Thus $w(t) \rightarrow w^*$ as $t \rightarrow \infty$.

(4) The example $\mathcal{L}(x) = x^2 - \cos x$. Here

$$\mathcal{L}'(x) = 2x + \sin x, \quad \mathcal{L}''(x) = 2 + \cos x.$$

Since $-1 \leq \cos x \leq 1$,

$$1 \leq \mathcal{L}''(x) \leq 3,$$

so one can take $d_0 = 1$ and $D_0 = 3$. The unique minimizer is $w^* = 0$ because $\mathcal{L}'(0) = 0$ and $\mathcal{L}'$ is strictly increasing.

Exercise 4

Let us assume that $\mathcal{L} : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth, and let w^* be a critical point of $\mathcal{L}$. Show that for all $y \in \mathbb{R}^n$

$$|\mathcal{L}(y) - \mathcal{L}(w^*)| \leq \frac{L}{2}\|y - w^*\|^2. \quad (684)$$

Solution. Direct consequence of the Descent Lemma.

Exercise 5

Investigate the behavior of the GD method for the function $\mathcal{L}(x) = x^2$, $x \in \mathbb{R}$.

Solution. Here $\mathcal{L}'(x) = 2x$, so the fixed-step GD recursion is

$$w_{k+1} = w_k - 2\eta w_k = (1 - 2\eta)w_k,$$

and hence

$$w_k = (1 - 2\eta)^k w_0, \quad \mathcal{L}(w_k) = (1 - 2\eta)^{2k} w_0^2.$$

Therefore, we have:

1. if $0 < \eta < 1$, then $|1 - 2\eta| < 1$ and $w_k \rightarrow 0$;
2. if $\eta = 1/2$, then $w_1 = 0$, so the minimizer is reached in one step;
3. if $\eta = 1$, then $w_k = (-1)^k w_0$, hence the sequence oscillates unless $w_0 = 0$;
4. if $\eta > 1$, then $|1 - 2\eta| > 1$, hence $|w_k| \rightarrow \infty$ for $w_0 \neq 0$.

The gradient $\mathcal{L}'(x) = 2x$ is Lipschitz with constant 2, so Theorem 6.1 applies for $\eta \in (0, 1)$.

Exercise 6

Consider the function $\mathcal{L}(x) = \frac{1}{3}x^3$, $x \in \mathbb{R}$.

- (1) For the sequence $\{w^k\}_{k \geq 0}$ given by (209) show that

$$\lim_{k \rightarrow +\infty} w^k = -\infty, \text{ if } w^0 \in (-\infty, 0) \cup (1/\eta, +\infty), \quad (685)$$

$$\lim_{k \rightarrow +\infty} w^k = 0, \text{ if } w^0 \in [0, \frac{1}{\eta}]. \quad (686)$$

- (2) Fix arbitrary $w^0 \in \mathbb{R}$ and consider the following initial value problem (also called the *gradient ODE*)

$$w(0) = w^0, \quad w'(t) = -\mathcal{L}'(w(t)) = -(w(t))^2 \text{ for } t \in [0, +\infty). \quad (687)$$

Solve the above ODE. For $w^0 \in [0, +\infty)$ show that $\lim_{t \rightarrow +\infty} w(t) = 0$. What happens for $w_0 < 0$?

Solution. Since $\mathcal{L}'(x) = x^2$, the GD recursion is

$$w_{k+1} = w_k - \eta w_k^2 = w_k(1 - \eta w_k).$$

Discrete dynamics. If $w_0 \in [0, 1/\eta]$, then $0 \leq w_1 \leq w_0$. By induction,

$$0 \leq w_{k+1} \leq w_k \leq \frac{1}{\eta}.$$

Thus (w_k) is monotone decreasing and bounded below, so $w_k \rightarrow \ell \geq 0$. Passing to the limit in the recursion gives

$$\ell = \ell - \eta \ell^2,$$

so $\ell = 0$. Hence $w_k \rightarrow 0$.

If $w_0 < 0$, then

$$w_{k+1} = w_k - \eta w_k^2 < w_k < 0.$$

Thus (w_k) is strictly decreasing. If it were bounded below, it would converge to a finite limit $\ell < 0$, and the fixed-point equation would again imply $\ell = 0$, a contradiction. Hence $w_k \rightarrow -\infty$.

If $w_0 > 1/\eta$, then $1 - \eta w_0 < 0$, hence $w_1 < 0$, and the previous case implies $w_k \rightarrow -\infty$.

Gradient ODE. The ODE is

$$w'(t) = -w(t)^2, \quad w(0) = w_0.$$

We consider three cases.

- (c1) If $w_0 = 0$, then $w(t) = 0$. If $w_0 \neq 0$, separation of variables gives

$$\frac{1}{w(t)} = \frac{1}{w_0} + t, \quad w(t) = \frac{w_0}{1 + w_0 t},$$

as long as $1 + w_0 t \neq 0$.

(c2) For $w_0 > 0$, the solution is global on $[0, \infty)$ and

$$\lim_{t \rightarrow \infty} w(t) = 0.$$

(c3) For $w_0 < 0$, the maximal interval of existence is

$$0 \leq t < T_* := -\frac{1}{w_0},$$

and

$$\lim_{t \uparrow T_*} w(t) = -\infty.$$

Exercise 7

Let $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$. Show that the following conditions are equivalent:

(i)

$$\exists_{K \in [0, +\infty)} : \forall_{x, y \in \mathbb{R}^n} |\mathcal{L}(x) - \mathcal{L}(y)| \leq K \|x - y\|, \quad (688)$$

(ii)

$$\exists_{K \in [0, +\infty)} : \forall_{x \in \mathbb{R}^n} \|\nabla \mathcal{L}(x)\| \leq K. \quad (689)$$

Moreover, show that if $\mathcal{L}$ is L -smooth then there exists $K \in [0, +\infty)$ such that for all $x \in \mathbb{R}^n$

$$\|\nabla \mathcal{L}(x)\| \leq K(1 + \|x\|). \quad (690)$$

Solution. (i) $\Rightarrow$ (ii) Assume first that $\mathcal{L}$ is globally Lipschitz with constant K . For all $x \in \mathbb{R}^n$ and any unit vector $u \in \mathbb{R}^n$ we have from the properties of directional derivative that

$$|\langle \nabla \mathcal{L}(x), u \rangle| = \left| \lim_{h \rightarrow 0} \frac{\mathcal{L}(x + hu) - \mathcal{L}(x)}{h} \right| \leq K. \quad (691)$$

Fix arbitrary $x \in \mathbb{R}^n$. If $\nabla \mathcal{L}(x) = 0$ then $\|\nabla \mathcal{L}(x)\| \leq K$. So assume that $\|\nabla \mathcal{L}(x)\| > 0$ and take the unit vector $u = \nabla \mathcal{L}(x) / \|\nabla \mathcal{L}(x)\|$. By (691) we get that $\|\nabla \mathcal{L}(x)\| \leq K$.

(ii) $\Rightarrow$ (i) Conversely, assume $\|\nabla \mathcal{L}(z)\| \leq K$ for all $z \in \mathbb{R}^n$. By Theorem 13.1 we get for $x, y \in \mathbb{R}^n$,

$$\mathcal{L}(x) - \mathcal{L}(y) = \int_0^1 \langle \nabla \mathcal{L}(y + t(x - y)), x - y \rangle dt.$$

Hence

$$|\mathcal{L}(x) - \mathcal{L}(y)| \leq \int_0^1 K \|x - y\| dt = K \|x - y\|.$$

Finally, if $\nabla \mathcal{L}$ is Lipschitz with constant L , then

$$\|\nabla \mathcal{L}(x)\| \leq \|\nabla \mathcal{L}(0)\| + L\|x\|.$$

Thus, with

$$K := \max\{\|\nabla \mathcal{L}(0)\|, L\},$$

we get

$$\|\nabla \mathcal{L}(x)\| \leq K(1 + \|x\|).$$

Exercise 8

For the SGD algorithm (318) show that for all $k \geq 0$, $i = 1, \dots, n$

$$\begin{aligned} \text{Var}(w_i^{k+1} \mid \mathcal{F}_k) &= \eta_k^2 \cdot \text{Var}\left(\frac{\partial \mathcal{L}_{J_{k+1}}}{\partial w_i}(w^k) \mid \mathcal{F}_k\right) \\ &= \eta_k^2 \cdot \left[\frac{1}{M} \sum_{j=1}^M \left| \frac{\partial \mathcal{L}_j}{\partial w_i}(w^k) \right|^2 - \left| \frac{\partial \mathcal{L}}{\partial w_i}(w^k) \right|^2 \right]. \end{aligned} \quad (692)$$

Solution. The SGD recursion is as follows

$$w^{k+1} = w^k - \eta_k \nabla \mathcal{L}_{J_{k+1}}(w^k),$$

and for the i -th coordinate we have

$$w_i^{k+1} = w_i^k - \eta_k \frac{\partial \mathcal{L}_{J_{k+1}}}{\partial w_i}(w^k).$$

Since w^k is $\mathcal{F}_k$ -measurable, w_i^k is conditionally deterministic given $\mathcal{F}_k$. Therefore

$$\text{Var}(w_i^{k+1} \mid \mathcal{F}_k) = \eta_k^2 \text{Var}\left(\frac{\partial \mathcal{L}_{J_{k+1}}}{\partial w_i}(w^k) \mid \mathcal{F}_k\right).$$

Conditionally on $\mathcal{F}_k$, the index J_{k+1} is uniform on $\{1, \dots, M\}$ and independent of $\mathcal{F}_k$. Hence

$$\mathbb{E}\left[\frac{\partial \mathcal{L}_{J_{k+1}}}{\partial w_i}(w^k) \mid \mathcal{F}_k\right] = \frac{1}{M} \sum_{j=1}^M \frac{\partial \mathcal{L}_j}{\partial w_i}(w^k) = \frac{\partial \mathcal{L}}{\partial w_i}(w^k),$$

and

$$\mathbb{E}\left[\left|\frac{\partial \mathcal{L}_{J_{k+1}}}{\partial w_i}(w^k)\right|^2 \mid \mathcal{F}_k\right] = \frac{1}{M} \sum_{j=1}^M \left|\frac{\partial \mathcal{L}_j}{\partial w_i}(w^k)\right|^2.$$

Therefore

$$\text{Var}(w_i^{k+1} \mid \mathcal{F}_k) = \eta_k^2 \left[\frac{1}{M} \sum_{j=1}^M \left|\frac{\partial \mathcal{L}_j}{\partial w_i}(w^k)\right|^2 - \left|\frac{\partial \mathcal{L}}{\partial w_i}(w^k)\right|^2 \right].$$

Exercise 9

Find a solution of a gradient system for the function $\mathcal{L}(x) = x^4/4$, $x \in \mathbb{R}$, and discuss the behavior of the solution as $t \rightarrow +\infty$.

Solution. Since $\mathcal{L}'(x) = x^3$, the gradient system is

$$w'(t) = -(w(t))^3, \quad w(0) = w_0.$$

If $w_0 = 0$, then $w(t) = 0$. If $w_0 \neq 0$, separation of variables gives

$$\frac{dw}{w^3} = -dt,$$

so

$$-\frac{1}{2w(t)^2} = -t - \frac{1}{2w_0^2},$$

and equivalently

$$\frac{1}{w(t)^2} = 2t + \frac{1}{w_0^2}.$$

The sign of $w(t)$ is the same as the sign of w_0 , hence

$$w(t) = \frac{w_0}{\sqrt{1 + 2tw_0^2}}.$$

Therefore

$$\lim_{t \rightarrow \infty} w(t) = 0, \quad \mathcal{L}(w(t)) = \frac{w_0^4}{4(1 + 2tw_0^2)^2} \rightarrow 0.$$

The convergence is algebraic rather than exponential, since

$$|w(t)| \sim \frac{1}{\sqrt{2t}}.$$

This reflects the degeneracy $\mathcal{L}''(0) = 0$ at the minimizer.

Exercise 10

For the SGD sequence (318) show, under the assumptions of Theorem 7.1, that $\sum_{k=0}^{+\infty} \|w^{k+1} - w^k\|^2 < +\infty$ almost surely.

Solution. By the SGD recursion,

$$w^{k+1} - w^k = -\eta_k \nabla \mathcal{L}_{J_{k+1}}(w^k).$$

The assumptions of Theorem 7.1 include the uniform bound

$$\|\nabla \mathcal{L}_j(x)\| \leq D, \quad j = 1, \dots, M, \quad x \in \mathbb{R}^n,$$

and the step-size condition

$$\sum_{k=0}^{\infty} \eta_k^2 < \infty.$$

Therefore, with probability one

$$\sum_{k=0}^{\infty} \|w^{k+1} - w^k\|^2 = \sum_{k=0}^{\infty} \eta_k^2 \|\nabla \mathcal{L}_{J_{k+1}}(w^k)\|^2 \leq D^2 \sum_{k=0}^{\infty} \eta_k^2 < \infty.$$

Exercise 11

Show that

$$\begin{aligned} \mathbb{E}(U_k \mid \mathcal{F}_k) &= -\eta_k \left(1 - \frac{L\eta_k}{2}\right) \|\nabla \mathcal{L}(w^k)\|^2 \\ &+ \frac{L}{2} \eta_k^2 \cdot \mathbb{E} \left[\|\nabla \mathcal{L}_{J_{k+1}}(w^k) - \nabla \mathcal{L}(w^k)\|^2 \mid \mathcal{F}_k \right], \end{aligned} \quad (693)$$

and

$$\begin{aligned} &\mathbb{E} \left[\|\nabla \mathcal{L}_{J_{k+1}}(w^k) - \nabla \mathcal{L}(w^k)\|^2 \mid \mathcal{F}_k \right] \\ &= \frac{1}{M} \sum_{j=1}^M \|\nabla \mathcal{L}_j(w^k)\|^2 - \|\nabla \mathcal{L}(w^k)\|^2, \end{aligned} \quad (694)$$

where U_k is defined in (343).

Solution. By definition,

$$U_k = -\eta_k \langle \nabla \mathcal{L}(w^k), \nabla \mathcal{L}_{J_{k+1}}(w^k) \rangle + \frac{L}{2} \eta_k^2 \|\nabla \mathcal{L}_{J_{k+1}}(w^k)\|^2.$$

Set

$$g_k := \nabla \mathcal{L}(w^k), \quad G_k := \nabla \mathcal{L}_{J_{k+1}}(w^k).$$

Conditionally on $\mathcal{F}_k$, w^k and g_k are fixed, while J_{k+1} is uniform and independent of $\mathcal{F}_k$. Hence

$$\mathbb{E}(G_k \mid \mathcal{F}_k) = g_k.$$

Moreover

$$\mathbb{E}(\|G_k\|^2 \mid \mathcal{F}_k) = \|g_k\|^2 + \mathbb{E}(\|G_k - g_k\|^2 \mid \mathcal{F}_k),$$

since

$$\mathbb{E}(\langle G_k - g_k, g_k \rangle \mid \mathcal{F}_k) = 0.$$

Therefore

$$\begin{aligned} \mathbb{E}(U_k \mid \mathcal{F}_k) &= -\eta_k \langle g_k, \mathbb{E}(G_k \mid \mathcal{F}_k) \rangle + \frac{L}{2} \eta_k^2 \mathbb{E}(\|G_k\|^2 \mid \mathcal{F}_k) \\ &= -\eta_k \|g_k\|^2 + \frac{L}{2} \eta_k^2 (\|g_k\|^2 + \mathbb{E}(\|G_k - g_k\|^2 \mid \mathcal{F}_k)) \\ &= -\eta_k \left(1 - \frac{L\eta_k}{2}\right) \|\nabla \mathcal{L}(w^k)\|^2 + \frac{L}{2} \eta_k^2 \mathbb{E}(\|\nabla \mathcal{L}_{J_{k+1}}(w^k) - \nabla \mathcal{L}(w^k)\|^2 \mid \mathcal{F}_k). \end{aligned}$$

This proves the first identity.

For the second identity, conditionally on $\mathcal{F}_k$,

$$\mathbb{E}(\|G_k\|^2 \mid \mathcal{F}_k) = \frac{1}{M} \sum_{j=1}^M \|\nabla \mathcal{L}_j(w^k)\|^2.$$

Since $\mathbb{E}(G_k \mid \mathcal{F}_k) = g_k$,

$$\mathbb{E}(\|G_k - g_k\|^2 \mid \mathcal{F}_k) = \mathbb{E}(\|G_k\|^2 \mid \mathcal{F}_k) - \|g_k\|^2.$$

Thus

$$\mathbb{E}(\|\nabla \mathcal{L}_{J_{k+1}}(w^k) - \nabla \mathcal{L}(w^k)\|^2 \mid \mathcal{F}_k) = \frac{1}{M} \sum_{j=1}^M \|\nabla \mathcal{L}_j(w^k)\|^2 - \|\nabla \mathcal{L}(w^k)\|^2.$$

Exercise 12

Let $A \in \mathbb{R}^{M \times N}$. Show that:

- (i) the matrix $A^T A$ is symmetric and positive semi-definite,
- (ii) for all $\lambda > 0$ the matrix $A^T A + \lambda I$ is symmetric and positive definite.

Solution. Ad. (i) Symmetry follows from

$$(A^T A)^T = A^T A.$$

For every $z \in \mathbb{R}^N$,

$$z^T A^T A z = (Az)^T (Az) = \|Az\|^2 \geq 0,$$

thus $A^T A$ is positive semidefinite.

Ad. (ii) If $\lambda > 0$, then $A^T A + \lambda I$ is symmetric. Moreover, for any $z \neq 0$,

$$z^T (A^T A + \lambda I) z = \|Az\|^2 + \lambda \|z\|^2 > 0,$$

and therefore $A^T A + \lambda I$ is positive definite.

Exercise 13

Consider the function $U(x) = x^4 - x^2$, $x \in \mathbb{R}$. Show that U' is locally Lipschitz and for all $x, y \in \mathbb{R}$

$$(x - y)(U'(x) - U'(y)) \geq -2|x - y|^2. \quad (695)$$

Hence, U is weakly convex. However, it is neither convex nor L -smooth.

Solution. *We have*

$$U'(x) = 4x^3 - 2x, \quad U''(x) = 12x^2 - 2.$$

Since U' is a polynomial, it is locally Lipschitz. For $x, y \in \mathbb{R}$,

$$\begin{aligned} U'(x) - U'(y) &= 4(x^3 - y^3) - 2(x - y) \\ &= (x - y)(4(x^2 + xy + y^2) - 2). \end{aligned}$$

Thus

$$(x - y)(U'(x) - U'(y)) = (x - y)^2(4(x^2 + xy + y^2) - 2).$$

Since

$$x^2 + xy + y^2 = \frac{1}{2}(x^2 + y^2) + \frac{1}{2}(x + y)^2 \geq 0,$$

we obtain

$$4(x^2 + xy + y^2) - 2 \geq -2,$$

and hence

$$(x - y)(U'(x) - U'(y)) \geq -2|x - y|^2.$$

This is the one-dimensional weak convexity inequality with constant 2. The function is not convex because $U''(0) = -2 < 0$. It is not L -smooth because $U''(x) = 12x^2 - 2$ is unbounded, whereas a C^2 one-dimensional function with Lipschitz gradient must have bounded second derivative.

Exercise 14

Consider the functions $\varphi \in C^2(\mathbb{R}^M; \mathbb{R}^N)$ and $F(w) = \|\varphi(w)\|^2$. Show that for all $w \in \mathbb{R}^M$

$$\nabla F(w) = 2(\varphi(w))^T \nabla \varphi(w), \quad (696)$$

$$\nabla^2 F(w) = 2(\nabla \varphi(w))^T \nabla \varphi(w) + 2 \sum_{i=1}^N \varphi_i(w) \nabla^2 \varphi_i(w) \quad (697)$$

where $\nabla \varphi(w)$ is the Jacobian of φ , and $\nabla^2 \varphi_i(w)$ is the Hessian of φ_i .

Solution. *Write*

$$F(w) = \sum_{i=1}^N \varphi_i(w)^2.$$

Assume first that $\varphi \in C^1$. For $j = 1, \dots, M$,

$$\frac{\partial F}{\partial w_j}(w) = 2 \sum_{i=1}^N \varphi_i(w) \frac{\partial \varphi_i}{\partial w_j}(w).$$

In vector form, with gradients treated as column vectors and $\nabla \varphi(w) \in \mathbb{R}^{N \times M}$, this is

$$\nabla F(w) = 2(\nabla \varphi(w))^T \varphi(w).$$

Equivalently, if one writes gradients as row vectors, this becomes

$$(\nabla F(w))^T = 2\varphi(w)^T \nabla \varphi(w).$$

For the Hessian, assume now that $\varphi \in C^2$. Differentiating once more gives, for $j, k = 1, \dots, M$,

$$\frac{\partial^2 F}{\partial w_k \partial w_j}(w) = 2 \sum_{i=1}^N \frac{\partial \varphi_i}{\partial w_k}(w) \frac{\partial \varphi_i}{\partial w_j}(w) + 2 \sum_{i=1}^N \varphi_i(w) \frac{\partial^2 \varphi_i}{\partial w_k \partial w_j}(w),$$

and therefore

$$\nabla^2 F(w) = 2(\nabla \varphi(w))^T \nabla \varphi(w) + 2 \sum_{i=1}^N \varphi_i(w) \nabla^2 \varphi_i(w).$$

Exercise 15

For the SGD algorithm (318) show that the conditional covariance matrix of w^{k+1} satisfies

$$\text{Cov}(w^{k+1} \mid \mathcal{F}_k) = \eta_k^2 \cdot \text{Cov}(Z_k \mid \mathcal{F}_k), \quad k = 0, 1, \dots, \quad (698)$$

where Z_k is defined in (326). Moreover, show that

$$\text{Cov}(Z_k \mid \mathcal{F}_k) = \mathbb{E}(Z_k Z_k^T \mid \mathcal{F}_k) = D(w^k), \quad (699)$$

where for all $w \in \mathbb{R}^n$

$$D(w) = \frac{1}{M} \sum_{j=1}^M \nabla \mathcal{L}_j(w) \nabla^T \mathcal{L}_j(w) - \nabla \mathcal{L}(w) \nabla^T \mathcal{L}(w) \quad (700)$$

is the so-called *diffusion matrix*.

Solution. Recall that

$$w^{k+1} = w^k - \eta_k \nabla \mathcal{L}(w^k) + \eta_k Z_k,$$

where

$$Z_k = \nabla \mathcal{L}(w^k) - \nabla \mathcal{L}_{J_{k+1}}(w^k).$$

The vector $w^k - \eta_k \nabla \mathcal{L}(w^k)$ is $\mathcal{F}_k$ -measurable. Therefore,

$$\text{Cov}(w^{k+1} \mid \mathcal{F}_k) = \eta_k^2 \text{Cov}(Z_k \mid \mathcal{F}_k).$$

Moreover,

$$\mathbb{E}(Z_k \mid \mathcal{F}_k) = 0,$$

so

$$\text{Cov}(Z_k \mid \mathcal{F}_k) = \mathbb{E}(Z_k Z_k^T \mid \mathcal{F}_k).$$

Let

$$G_k := \nabla \mathcal{L}_{J_{k+1}}(w^k), \quad g_k := \nabla \mathcal{L}(w^k) = \mathbb{E}(G_k \mid \mathcal{F}_k).$$

Since $Z_k = g_k - G_k$,

$$\begin{aligned} \mathbb{E}(Z_k Z_k^T \mid \mathcal{F}_k) &= \mathbb{E}((G_k - g_k)(G_k - g_k)^T \mid \mathcal{F}_k) \\ &= \mathbb{E}(G_k G_k^T \mid \mathcal{F}_k) - g_k g_k^T. \end{aligned}$$

Conditionally on $\mathcal{F}_k$, J_{k+1} is uniform on $\{1, \dots, M\}$, hence

$$\mathbb{E}(G_k G_k^T \mid \mathcal{F}_k) = \frac{1}{M} \sum_{j=1}^M \nabla \mathcal{L}_j(w^k) \nabla \mathcal{L}_j(w^k)^T.$$

Therefore

$$\text{Cov}(Z_k | \mathcal{F}_k) = D(w^k),$$

where

$$D(w) = \frac{1}{M} \sum_{j=1}^M \nabla \mathcal{L}_j(w) \nabla \mathcal{L}_j(w)^T - \nabla \mathcal{L}(w) \nabla \mathcal{L}(w)^T.$$

Exercise 16

By using the backpropagation procedure, described in Section 5.2, discuss the problems of *vanishing* and *exploding* gradients in feedforward neural networks. Consider the case of sigmoid and ReLU activation functions.

Solution. For the feedforward neural network N^F we have

$$a^{(0)} = x, \quad A^{(L)}(z) = W^{(L)}z + b^{(L)}, \quad a^{(L)} = (\Phi^{(L)} \circ A^{(L)})(a^{(L-1)}),$$

for $L = 1, 2, \dots, M+1$, where

$$\Phi^{(L)}(s) = [\phi^{(L)}(s_1), \dots, \phi^{(L)}(s_{N_L})]^T.$$

Equivalently,

$$\Sigma^{(L)} = A^{(L)}(a^{(L-1)}) = W^{(L)}a^{(L-1)} + b^{(L)}, \quad a^{(L)} = \Phi^{(L)}(\Sigma^{(L)}).$$

The backpropagation recursion is

$$\delta^{(M+1)} = \frac{\partial L}{\partial a^{(M+1)}}, \quad \delta^{(L)} = \delta^{(L+1)} \frac{\partial a^{(L+1)}}{\partial a^{(L)}}, \quad L = M, M-1, \dots, 1.$$

Here, as in Section 5.2, $\partial L / \partial a^{(L)}$ is treated as the Jacobian of the scalar loss with respect to $a^{(L)}$, so the products are written with $\delta^{(L+1)}$ multiplying the Jacobian from the left. Moreover, by formula 202

$$\frac{\partial a^{(L+1)}}{\partial a^{(L)}} = \left. \frac{\partial \Phi^{(L+1)}}{\partial s}(s) \right|_{s=A^{(L+1)}(a^{(L)})} W^{(L+1)},$$

where

$$\left. \frac{\partial \Phi^{(L+1)}}{\partial s}(s) \right|_{s=A^{(L+1)}(a^{(L)})} = \text{diag}\left((\phi^{(L+1)})'(\Sigma_i^{(L+1)}) \mid i = 1, \dots, N_{L+1}\right).$$

Iterating the recursion gives, for $L = 1, 2, \dots, M$,

$$\delta^{(L)} = \delta^{(M+1)} \frac{\partial a^{(M+1)}}{\partial a^{(M)}} \frac{\partial a^{(M)}}{\partial a^{(M-1)}} \dots \frac{\partial a^{(L+1)}}{\partial a^{(L)}}.$$

Hence,

$$\|\delta^{(L)}\| \leq \|\delta^{(M+1)}\| \prod_{r=L}^M \left\| \frac{\partial a^{(r+1)}}{\partial a^{(r)}} \right\|,$$

and therefore

$$\|\delta^{(L)}\| \leq \|\delta^{(M+1)}\| \prod_{r=L}^M \left\| \left. \frac{\partial \Phi^{(r+1)}}{\partial s}(s) \right|_{s=A^{(r+1)}(a^{(r)})} \right\| \|W^{(r+1)}\|.$$

This product is the source of the vanishing and exploding gradient phenomena. If many factors in the product are smaller than one, then the norm of the backpropagated quantity $\delta^{(L)}$ may become very small for early

layers L . This is the vanishing-gradient problem. If many factors are larger than one, then $\delta^{(L)}$ may become very large. This is the exploding-gradient problem. The parameter gradients are

$$\frac{\partial L}{\partial W^{(L)}} = \delta^{(L)} \cdot \frac{\partial a^{(L)}}{\partial W^{(L)}}, \quad \frac{\partial L}{\partial b^{(L)}} = \delta^{(L)} \cdot \frac{\partial a^{(L)}}{\partial b^{(L)}}.$$

Thus, if $\delta^{(L)}$ is very small, the update of $W^{(L)}$ and $b^{(L)}$ is negligible. On the other hand, if $\delta^{(L)}$ is very large, the update may be unstable.

Consider first the sigmoid activation function

$$\phi(x) = \frac{1}{1 + e^{-x}}.$$

Then

$$\phi'(x) = \phi(x)(1 - \phi(x)) \leq \frac{1}{4}, \quad x \in \mathbb{R}.$$

Consequently, for a sigmoid layer,

$$\left. \frac{\partial \Phi^{(L)}}{\partial s}(s) \right|_{s=A^{(L)}(a^{(L-1)})} = \text{diag}\left(\phi'(\Sigma_i^{(L)}) \mid i = 1, \dots, N_L\right),$$

where every diagonal entry is at most $1/4$. Hence, for the Frobenius norm

$$\left\| \left. \frac{\partial \Phi^{(L)}}{\partial s}(s) \right|_{s=A^{(L)}(a^{(L-1)})} \right\| \leq \frac{\sqrt{N_L}}{4}.$$

In particular, if the layer widths are fixed and the quantities

$$\frac{\sqrt{N_{r+1}}}{4} \|W^{(r+1)}\|$$

are frequently smaller than one, then the product controlling $\|\delta^{(L)}\|$ decays rapidly with depth. Moreover, when $|\Sigma_i^{(L)}|$ is large, the sigmoid is saturated and $\phi'(\Sigma_i^{(L)})$ is much smaller than $1/4$, which makes the decay even stronger. Hence sigmoid networks are especially prone to vanishing gradients. Exploding gradients can still occur if the matrices $W^{(L)}$ have sufficiently large norms, so that the weight factors dominate the small derivative factors.

Now consider the ReLU activation

$$\phi(x) = \max\{0, x\}.$$

At points where $x \neq 0$,

$$\phi'(x) = \begin{cases} 1, & x > 0, \\ 0, & x < 0. \end{cases}$$

Therefore, away from the non-differentiability points $\Sigma_i^{(L)} = 0$,

$$\left. \frac{\partial \Phi^{(L)}}{\partial s}(s) \right|_{s=A^{(L)}(a^{(L-1)})} = \text{diag}\left(\mathbf{1}_{\{\Sigma_i^{(L)} > 0\}} \mid i = 1, \dots, N_L\right).$$

At the points $\Sigma_i^{(L)} = 0$ the ReLU derivative does not exist. In implementations, one usually chooses a subgradient convention, for example, 0. For active ReLU neurons, the derivative factor is 1, so ReLU does not introduce the systematic factor $1/4$ that appears for the sigmoid activation. This is why ReLU often

mitigates the classical sigmoid-type vanishing-gradient mechanism. However, inactive ReLU neurons have derivative factor 0. If a component satisfies $\Sigma_i^{(L)} < 0$, then the corresponding diagonal entry in

$$\frac{\partial \Phi^{(L)}}{\partial s}$$

is zero, and the corresponding part of the backpropagated gradient is killed. If this happens persistently, the corresponding neuron may stop learning; this is the dead-ReLU phenomenon. Exploding gradients may also occur for ReLU networks, because on active paths the derivative factors equal 1 and the product is controlled mainly by the norms of the matrices $W^{(L)}$. Thus, vanishing and exploding gradients are consequences of repeated multiplication in the recursion

$$\delta^{(L)} = \delta^{(L+1)} \frac{\partial a^{(L+1)}}{\partial a^{(L)}}.$$

Sigmoid activations tend to cause vanishing gradients through small diagonal factors in $\partial \Phi^{(L)} / \partial s$, especially in saturated regimes. ReLU activations avoid this systematic shrinkage on active neurons, but they can create exactly zero gradients on inactive neurons and can still produce exploding gradients when the matrices $W^{(L)}$ are large.

Exercise 17

Minimize the loss (106) and compute the estimators $\hat{\kappa}_N, \hat{\mu}_N$.

Solution. Denote by $X_i := X(t_i)$ for $i = 0, \dots, N-1$. The least-squares problem is the ordinary linear regression

$$Z_i = aX_i + b + \varepsilon_i.$$

Define the sample means

$$\bar{X} := \frac{1}{N} \sum_{i=0}^{N-1} X_i, \quad \bar{Z} := \frac{1}{N} \sum_{i=0}^{N-1} Z_i,$$

and assume

$$S_{XX} := \sum_{i=0}^{N-1} (X_i - \bar{X})^2 > 0.$$

The normal equations are

$$\begin{aligned} \sum_{i=0}^{N-1} (Z_i - aX_i - b) &= 0, \\ \sum_{i=0}^{N-1} X_i (Z_i - aX_i - b) &= 0. \end{aligned}$$

The first equation gives

$$b = \bar{Z} - a\bar{X}.$$

Substituting this into the second equation yields

$$\hat{a}_N = \frac{\sum_{i=0}^{N-1} (X_i - \bar{X})(Z_i - \bar{Z})}{\sum_{i=0}^{N-1} (X_i - \bar{X})^2}.$$

Then

$$\hat{b}_N = \bar{Z} - \hat{a}_N \bar{X}.$$

Equivalently, with design matrix

$$A = \begin{bmatrix} 1 & X_0 \\ 1 & X_1 \\ \vdots & \vdots \\ 1 & X_{N-1} \end{bmatrix}, \quad \theta = \begin{bmatrix} b \\ a \end{bmatrix}, \quad Z = \begin{bmatrix} Z_0 \\ \vdots \\ Z_{N-1} \end{bmatrix},$$

one has

$$\hat{\theta} = (A^T A)^{-1} A^T Z$$

provided $A^T A$ is invertible. From the Euler-Maruyama parametrization

$$a = -\kappa\sqrt{h}, \quad b = \kappa\sqrt{h}\mu = -a\mu,$$

where $h = T/N$. Thus, if $\hat{a}_N \neq 0$,

$$\hat{\kappa}_N = -\frac{\hat{a}_N}{\sqrt{h}}, \quad \hat{\mu}_N = -\frac{\hat{b}_N}{\hat{a}_N}.$$

The assumptions $S_{XX} > 0$ and $\hat{a}_N \neq 0$ are finite-sample identifiability conditions. If $S_{XX} = 0$, the regression slope is not identifiable. If $\hat{a}_N = 0$, then the relation $b = -a\mu$ cannot identify μ .

Exercise 18

For $a > 25/3$ consider the function

$$f(x) = x^2 + a \ln(1 + x^2), \quad x \in \mathbb{R}. \quad (701)$$

Show that it is L -smooth and of PL-type, but it is not a convex function.

Solution. First,

$$f'(x) = 2x + \frac{2ax}{1+x^2} = 2x \left(1 + \frac{a}{1+x^2} \right),$$

and

$$f''(x) = 2 + 2a \frac{1-x^2}{(1+x^2)^2}.$$

Since

$$\left| \frac{1-x^2}{(1+x^2)^2} \right| \leq 1,$$

we get

$$|f''(x)| \leq 2 + 2a.$$

Hence f' is globally Lipschitz, so f is L -smooth, for instance with $L = 2 + 2a$. The function has a global minimum $f^* = 0$ at $x = 0$, because both terms x^2 and $a \log(1 + x^2)$ are nonnegative and vanish only at 0. Moreover,

$$\log(1 + x^2) \leq x^2,$$

so

$$f(x) \leq (1 + a)x^2.$$

On the other hand,

$$|f'(x)|^2 = 4x^2 \left(1 + \frac{a}{1+x^2} \right)^2 \geq 4x^2.$$

Therefore

$$f(x) - f^* = f(x) \leq \frac{1+a}{4} |f'(x)|^2.$$

Equivalently,

$$\frac{1}{2} |f'(x)|^2 \geq \frac{2}{1+a} (f(x) - f^*).$$

Thus f satisfies the PL inequality with PL constant $\mu = 2/(1+a)$. It remains to show nonconvexity. Let $s = x^2$. For

$$g(s) := \frac{1-s}{(1+s)^2}, \quad s \geq 0,$$

we have

$$g'(s) = \frac{s-3}{(1+s)^3},$$

so its minimum is attained at $s = 3$ and equals

$$g(3) = \frac{1-3}{(1+3)^2} = -\frac{1}{8}.$$

Hence

$$f''(\sqrt{3}) = 2 + 2a \left(-\frac{1}{8} \right) = 2 - \frac{a}{4}.$$

Since $a > 25/3 > 8$, we have $2 - a/4 < 0$. Thus $f''(\sqrt{3}) < 0$, so f is not convex.

Exercise 19

Show that in Theorem 10.5 (ii) the assumption $\mathcal{L} \in C^2(\mathbb{R}^n; \mathbb{R})$ can be weakened to $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ and the limit $\lim_{t \rightarrow +\infty} \nabla \mathcal{L}(w(t)) = 0$ still holds.

Hint. Prove that the function $\phi_h(t) = \|w(t+h) - w(t)\|^2$ is nonincreasing for all $h \in \mathbb{R}$ with respect to the variable t .

Solution. Let w be a C^1 solution of the gradient flow

$$w'(t) = -\nabla \mathcal{L}(w(t)), \quad t \geq 0,$$

and assume that $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ is convex and that w^* is a critical point. Then w^* is a global minimizer. We have that

$$\frac{d}{dt} \mathcal{L}(w(t)) = \langle \nabla \mathcal{L}(w(t)), w'(t) \rangle = -\|\nabla \mathcal{L}(w(t))\|^2.$$

Since $\mathcal{L}(w(t)) \geq \mathcal{L}(w^*)$, integration over $[0, T]$ gives

$$\int_0^T \|\nabla \mathcal{L}(w(t))\|^2 dt = \mathcal{L}(w(0)) - \mathcal{L}(w(T)) \leq \mathcal{L}(w(0)) - \mathcal{L}(w^*).$$

Letting $T \rightarrow \infty$ yields

$$\int_0^\infty \|\nabla \mathcal{L}(w(t))\|^2 dt < \infty.$$

For any fixed $h > 0$, define

$$\varphi_h(t) := \|w(t+h) - w(t)\|^2, \quad t \geq 0.$$

Then

$$\begin{aligned} \varphi'_h(t) &= 2\langle w(t+h) - w(t), w'(t+h) - w'(t) \rangle \\ &= -2\langle w(t+h) - w(t), \nabla \mathcal{L}(w(t+h)) - \nabla \mathcal{L}(w(t)) \rangle. \end{aligned}$$

For a differentiable convex function, the gradient is monotone, i.e.

$$\langle x - y, \nabla \mathcal{L}(x) - \nabla \mathcal{L}(y) \rangle \geq 0,$$

and therefore, for all $t \geq 0$

$$\varphi'_h(t) \leq 0.$$

Thus φ_h is nonincreasing in t for every $h > 0$. Hence, if $0 \leq s \leq t$, then

$$\left\| \frac{w(t+h) - w(t)}{h} \right\|^2 \leq \left\| \frac{w(s+h) - w(s)}{h} \right\|^2.$$

Letting $h \downarrow 0$ and using the continuity of $x \rightarrow \|x\|$, and the fact that $w \in C^1$, we get

$$\|w'(t)\|^2 \leq \|w'(s)\|^2, \quad 0 \leq s \leq t.$$

Hence, $t \rightarrow \|w'(t)\|$ is nonnegative and nonincreasing. Since $w'(t) = -\nabla \mathcal{L}(w(t))$, the function

$$t \rightarrow \|\nabla \mathcal{L}(w(t))\| = \|w'(t)\|$$

is nonincreasing and nonnegative. Hence, by Theorem 10.4 (C2) we get

$$\lim_{t \rightarrow \infty} \|\nabla \mathcal{L}(w(t))\| = \inf_{t \geq 0} \|\nabla \mathcal{L}(w(t))\| = 0,$$

which also implies that

$$\lim_{t \rightarrow \infty} \nabla \mathcal{L}(w(t)) = 0.$$

Exercise 20

Show that the loss function (119) can be written as in (132).

Hint. Use the fact that for the sigmoid activation φ , the following holds

$$-\ln(\varphi(z)) = \ln(1 + e^{-z}), \quad (702)$$

$$-\ln(1 - \varphi(z)) = \ln(1 + e^z), \quad (703)$$

for all $z \in \mathbb{R}$.

Solution. Let

$$z_j := \langle x_j, w \rangle, \quad \phi(z) = \frac{1}{1 + e^{-z}}.$$

The binary cross-entropy loss is

$$\mathcal{L}(w) = -\frac{1}{M} \sum_{j=1}^M [y_j \log \phi(z_j) + (1 - y_j) \log(1 - \phi(z_j))].$$

Using

$$-\log \phi(z) = \log(1 + e^{-z}), \quad -\log(1 - \phi(z)) = \log(1 + e^z),$$

we distinguish the two possible labels. If $y_j = 1$, then $t_j = 1$, and the j -th contribution is

$$-\log \phi(z_j) = \log(1 + e^{-z_j}) = \log(1 + e^{-t_j z_j}).$$

If $y_j = 0$, then $t_j = -1$, and the j -th contribution is

$$-\log(1 - \phi(z_j)) = \log(1 + e^{z_j}) = \log(1 + e^{-t_j z_j}).$$

Thus, for both cases,

$$-y_j \log \phi(z_j) - (1 - y_j) \log(1 - \phi(z_j)) = \log(1 + e^{-t_j z_j}).$$

Substituting $z_j = \langle x_j, w \rangle$ gives

$$L(w) = \frac{1}{M} \sum_{j=1}^M \log \left(1 + e^{-t_j \langle x_j, w \rangle} \right).$$

Exercise 21

Assume that $\mathcal{L} \in C^1(\mathbb{R}^n; \mathbb{R})$ is L -smooth and consider the gradient descent algorithm with clipping, i.e., $\eta > 0$, $w^0 \in \mathbb{R}^n$ and

$$w^{k+1} = w^k - \eta \cdot R_D(\nabla \mathcal{L}(w^k)), \quad k = 0, 1, \dots, \quad (704)$$

where R_D is defined in (388) for a fixed $D \in (0, +\infty)$. Moreover, assume that $\eta \in (0, 1/L)$ and that there exists $\mathcal{L}^* \in (-\infty, \mathcal{L}(w^0)]$ such that $\mathcal{L}(w) \geq \mathcal{L}^*$ for all $w \in \mathbb{R}^n$. Show that the sequence $\{\mathcal{L}(w^k)\}_{k \geq 0}$ converges and

$$\lim_{k \rightarrow +\infty} \nabla \mathcal{L}(w^k) = 0. \quad (705)$$

Solution. From Lemma 13.2 we have for all $k \geq 0$ that

$$\begin{aligned} \mathcal{L}(w^{k+1}) - \mathcal{L}(w^k) &\leq \langle \nabla \mathcal{L}(w^k), w^{k+1} - w^k \rangle + \frac{L}{2} \|w^{k+1} - w^k\|^2 \\ &= -\eta \langle \nabla \mathcal{L}(w^k), R_D(\nabla \mathcal{L}(w^k)) \rangle + \frac{L\eta}{2} \|R_D(\nabla \mathcal{L}(w^k))\|^2. \end{aligned} \quad (706)$$

Since, by Proposition 7.1 we get that

$$-\eta \langle \nabla \mathcal{L}(w^k), R_D(\nabla \mathcal{L}(w^k)) \rangle \leq -\eta (\min\{\|\nabla \mathcal{L}(w^k)\|, D\})^2, \quad (707)$$

$$\|R_D(\nabla \mathcal{L}(w^k))\|^2 = (\min\{\|\nabla \mathcal{L}(w^k)\|, D\})^2, \quad (708)$$

we arrive at

$$\mathcal{L}(w^{k+1}) - \mathcal{L}(w^k) \leq -\eta \left(1 - \frac{L\eta}{2}\right) (\min\{\|\nabla \mathcal{L}(w^k)\|, D\})^2. \quad (709)$$

We assumed that $\eta \in (0, 1/L)$, so

$$-\eta < -\eta \left(1 - \frac{L\eta}{2}\right) < -\frac{\eta}{2}, \quad (710)$$

and hence for all $k \geq 0$

$$\mathcal{L}(w^{k+1}) - \mathcal{L}(w^k) \leq -\frac{\eta}{2} Q_k^2 \leq 0, \quad (711)$$

where we denote by $Q_k = \min\{\|\nabla \mathcal{L}(w^k)\|, D\}$. Therefore, the sequence $\{\mathcal{L}(w^k)\}_{k \geq 0}$ is nonincreasing and bounded from below by $\mathcal{L}^*$. Hence, there exists $\lim_{k \rightarrow +\infty} \mathcal{L}(w^k) \in [\mathcal{L}^*, \mathcal{L}(w^0)]$. Moreover, for all $M \geq 1$

$$0 \leq \mathcal{L}(w^M) - \mathcal{L}(w^0) = \sum_{k=0}^{M-1} (\mathcal{L}(w^{k+1}) - \mathcal{L}(w^k)) \leq -\frac{\eta}{2} \sum_{k=0}^{M-1} Q_k^2, \quad (712)$$

which implies that

$$\sum_{k=0}^{M-1} Q_k^2 \leq 2(\mathcal{L}(w^0) - \mathcal{L}(w^M))/\eta \leq 2(\mathcal{L}(w^0) - \mathcal{L}^*)/\eta. \quad (713)$$

The sequence $\left\{ \sum_{k=0}^{M-1} Q_k^2 \right\}_{M \geq 1}$ is nonnegative, nondecreasing and bounded from below. Therefore,

$$\sum_{k=0}^{+\infty} Q_k^2 = \lim_{M \rightarrow +\infty} \sum_{k=0}^{M-1} Q_k^2 \leq 2(\mathcal{L}(w^0) - \mathcal{L}^*)/\eta < +\infty, \quad (714)$$

and this implies that $\lim_{k \rightarrow +\infty} Q_k = \lim_{k \rightarrow +\infty} \min\{\|\nabla \mathcal{L}(w^k)\|, D\} = 0$. Since $D > 0$, it holds that $\lim_{k \rightarrow +\infty} \|\nabla \mathcal{L}(w^k)\| = 0$.

16 Bibliography

- [1] M. V. BALASHOV, M. O. GOLUBEV, About the Lipschitz property of the metric projection in the Hilbert space, *J. Math. Anal. Appl.* 394 (2012), 545–551
- [2] M. BARANEK, P. PRZYBYŁOWICZ, SPINNs – Deep learning framework for approximation of stochastic differential equations, <https://arxiv.org/abs/2512.14258>
- [3] I. BARBĀLAT, Systèmes d’équations différentielles d’oscillations non Linéaires, *Rev. Math. Pures Appl.* 4 (1959), 267–270
- [4] V. BARBU, *Differential Equations*, Springer, 2016.
- [5] A. BARBU, S-C. ZHU, *Monte Carlo Methods*, Springer, 2020
- [6] D. P. BERTSEKAS, J. N. TSITSIKLIS, Gradient Convergence in Gradient methods with Errors, *SIAM J. OPTIM.*, Vol. 10, No. 3, pp. 627–642
- [7] P. BŁASZCZYK, K. KLIMCZAK, A. MAHDI, P. OPROCHA, P. POTORSKI, P. PRZYBYŁOWICZ, M. SOBIERAJ, On automatic calibration of the SIRD epidemiological model for COVID-19 data in Poland, <https://arxiv.org/abs/2204.12346>
- [8] R. T. Q. CHEN, Y. RUBANOVA, J. BETTENCOURT, AND D. K. DUVENAUD, Neural Ordinary Differential Equations. *Advances in Neural Information Processing Systems*. 31. 2018, 6572–6583
- [9] M. P. DEISENROTH, A. A. FAISAL, C. S. ONG, *Mathematics for Machine Learning*, Cambridge University Press, 2020, <https://mml-book.com>
- [10] O. DÜRR, B. SICK, E. MURINA, *Probabilistic Deep Learning with Python, Keras and TensorFlow Probability*, Manning, 2020.
- [11] A. F. FILIPPOV, *Differential Equations with Discontinuous Righthand Sides*, Kluwer Academic Publishers, 1988
- [12] G. GARRIGOS, R. M. GOWER, *Handbook of Convergence Theorems for (Stochastic) Gradient Methods*, arXiv:2301.11235
- [13] A. GÉRON, *Uczenie maszynowe z użyciem Scikit-Learn, Keras i TensorFlow. Wydanie III*, Helion, 2023
- [14] C. GRAHAM, D. TALAY, *Stochastic Simulation and Monte Carlo Methods*, Springer, 2013
- [15] L. V. HIEN, Some Results on the Strict Fréchet Differentiability of the Metric Projection Operator in Hilbert Spaces, <https://arxiv.org/abs/2403.14512>
- [16] N. ISOBE, M. OKUMURA, Variational Formulations of ODE-Net as a Mean-Field Optimal Control Problem and Existence Results, to appear in *Journal of Machine Learning*
- [17] S. JASTRZĘBSKI ET AL., Width of Minima Reached by Stochastic Gradient Descent is Influenced by Learning Rate to Batch Size Ratio. In: Kůrková, V., Manolopoulos, Y., Hammer, B., Iliadis, L., Maglogiannis, I. (eds) *Artificial Neural Networks and Machine Learning – ICANN 2018*. ICANN 2018. *Lecture Notes in Computer Science()*, vol 11141. Springer, 2018
- [18] A. JENTZEN, B. KUCKUCK, P. VON WURSTEMBERGER, *Mathematical Introduction to Deep Learning: Methods, Implementations, and Theory*, <https://arxiv.org/abs/2310.20360>
- [19] A. KAŁUŻA, P. M. MORKISZ, B. MULEWICZ, P. PRZYBYŁOWICZ, M. WIĄCEK, Deep learning-based estimation of time-dependent parameters in Markov models with application to nonlinear regression and SDEs, *Appl. Math. and Comp.* 480C (2024), 128906

- [20] H. KARIMI, J. NUTINI, M. SCHMIDT, Linear Convergence of Gradient and Proximal-Gradient Methods Under the Polyak-Łojasiewicz Condition. In: Frasconi, P., Landwehr, N., Manco, G., Vreeken, J. (eds) Machine Learning and Knowledge Discovery in Databases. ECML PKDD 2016. Lecture Notes in Computer Science(), vol 9851. Springer, <https://arxiv.org/abs/1608.04636>
- [21] J. KIEFER, J. WOLFOWITZ, Stochastic Estimation of the Maximum of a Regression Function, *Ann. Math. Statist.* 23(3): 462-466 (September, 1952). DOI: 10.1214/aoms/1177729392
- [22] P. MERTIKOPOULOS, N. HALLAK, A. KAVIS, V. CEVHER, On the almost sure convergence of stochastic gradient descent in non-convex problems, <https://inria.hal.science/hal-03043771v1>
- [23] J. MEI, C. XIAO, C. SZEPESVÁRI, D. SCHUURMANS, On the Global Convergence Rates of Softmax Policy Gradient Methods, *Proceedings of the 37th International Conference on Machine Learning*, Online, PMLR 119, 2020.
- [24] S. MISRA, S. THAKUR, M. GHOSH, S. K. SAHA, An Autoencoder Based Model for Detecting Fraudulent Credit Card Transaction. *Procedia Computer Science*, Volume 167, 2020, 254–262, <https://doi.org/10.1016/j.procs.2020.03.219>.
- [25] Y. NESTEROV, *Lectures on Convex Optimization - Second Edition*, Springer, 2018
- [26] G. PAGÉS, *Numerical Probability - An Introductory with Application to Finance*, Springer, 2018
- [27] E. PARDOUX, A. RASCANU, *Stochastic Differential Equations, Backward SDEs, Partial Differential Equations*, Springer, 2014.
- [28] J. PEYPOUQUET, *Convex Optimization in Normed Spaces - Theory, Methods and Examples*, Springer, 2015.
- [29] E. PLAUT, From Principal Subspaces to Principal Components with Linear Autoencoders, <https://arxiv.org/abs/1804.10253>
- [30] SIMON J. D. PRINCE, *Understanding Deep Learning*, The MIT Press, 2023.
- [31] P. PRZYBYŁOWICZ, *Foundations of Monte Carlo methods and stochastic simulations – From Monte Carlo Lebesgue integration to weak approximation of SDEs*, <https://arxiv.org/abs/2208.05531>
- [32] P. PRZYBYŁOWICZ, M. SOBIERAJ, On the randomized Euler scheme for SDEs with integral-form drift, <https://arxiv.org/abs/2405.20481>
- [33] D. E. RUMELHART, G. E. HINTON, R. J. WILLIAMS, Learning representations by back-propagating errors. *Nature* 323, 533–536 (1986). <https://doi.org/10.1038/323533a0>
- [34] M. RAISSI, P. PERDIKARIS, G. E. KARNIADAKIS, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J. Comput. Phys.* 378 (2019), pp. 686–707. url: doi.org/10.1016/j.jcp.2018.10.045.
- [35] H. ROBBINS, D. SIEGMUND, A convergence theorem for non negative almost supermartingales and some applications, *Optimizing Methods in Statistics*, Academic Press, 1971, 233–257,
- [36] R. RUDNICKI, *Wykłady z analizy matematycznej*, Wydanie II, Wydawnictwo Naukowe PWN, 2020
- [37] D. E. STEWART, *Numerical Analysis: A Graduate Course*, Springer, 2022
- [38] J. STOER, R. BURLISCH, *Introduction to Numerical Analysis*, 3rd Ed., Springer-Verlag, 2002

- [39] S. H. STROGATZ, *Nonlinear Dynamics and Chaos - with applications to physics, biology, chemistry and engineering*, CRC Press, 2024
- [40] M. SZÖLGYENYI, Stochastic differential equations with irregular coefficients: mind the gap! Internationale Mathematische Nachrichten, 246:43-56, 2021
- [41] T. TAO, *An Introduction to Measure Theory*, American Mathematical Society, 2011.
- [42] D. WACHSMUTH, G. WACHSMUTH, A simple proof of the Baillon–Haddad theorem on open subsets of Hilbert spaces, <https://arxiv.org/abs/2204.00282>
- [43] <https://medium.com/ai-enthusiast/gradient-clipping-preventing-exploding-gradients-in-deep-models-d2ed5fbda157>
- [44] <https://medium.com/@stacymacbrains/anomaly-detection-using-autoencoders-a-deep-dive-into-fraud-detection-9f59bcb5ab32>

Optimization algorithm	Main advantages	Main disadvantages
Full-batch gradient descent	- Deterministic, exact gradient; very stable updates - Monotonic decrease of the training loss - Easy to debug and reproduce	- Needs the entire dataset in memory for each step - High risk of getting trapped in sharp local minima (little inherent noise)
Mini-batch SGD ($1 < b \ll M$)	- Many updates per epoch - Natural noise in gradients helps jump out of shallow local minima - Efficient GPU/TPU utilisation	- Gradient still noisy and may require learning-rate schedules for stability - Batch size b must be tuned (too small = unstable, too large = memory heavy) - Convergence path is non-deterministic
Online SGD ($b = 1$)	- <i>Maximal</i> gradient noise results in a strong ability to escape poor local minima - Minimal memory footprint; supports on-line learning - Very frequent parameter updates	- Highly erratic loss curve; needs small learning rates and many epochs - Inefficient use of vectorised hardware (little parallelism) - The most unstable convergence, can wander for a long time before settling